

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\124-DopeRev 02.pvdb

Date of Analysis : Jul 30, 2026 3:35pm

PV Elite 28 U2, April 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec :	762.002 x 826.002 x 762.002 mm ID
Vessel Design Length, Tangent to Tangent	14686.00 mm
Distance of Bottom Tangent above Grade	3500.00 mm
Distance of Base above Grade	300.00 mm
Specified Datum Line Distance	0.00 mm
Internal Design Temperature	82 °C
Internal Design Pressure	9.403 MPa
External Design Temperature	82 °C
External Design Pressure	0.103 MPa
Maximum Allowable Working Pressure	9.56 MPa
External Max. Allowable Working Pressure	4.23 MPa
Shop Test Pressure	13.776 MPa
Field Test Pressure	12.224 MPa
Required Minimum Design Metal Temperature	-28.0 °C
Warmest Computed Minimum Design Metal Temperature	4.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Flange	SA-266 2	...	...	K03506	
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-105	...	...	K03504	+IT
Nozzle Flg	SA-105	...	...	K03504	
Basering	SA-516 70	...	...	K02700	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Base Bolting	SA-193 B7	...	<= 64	G41400	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt-2	...	...	...	1.6000	N/A	No
Skirt-1	...	...	...	1.6000	N/A	No
Bottom head	9.412	0.10	9.730	3.0000	No	No
Shell-1	9.410	0.10	9.960	3.0000	N/A	No
Flange-1	9.403	0.10	9.970	3.0000	N/A	No
Flange-2	9.403	0.10	9.970	3.0000	N/A	No
Shell-2	9.403	0.10	9.970	3.0000	N/A	No
Shell-3	9.403	0.10	9.970	3.0000	N/A	No
Flange-3	9.403	0.10	9.970	3.0000	N/A	No
Flange-4	9.403	0.10	9.970	3.0000	N/A	No
Shell-4	9.403	0.10	9.970	3.0000	N/A	No
Shell-5	9.403	0.10	9.970	3.0000	N/A	No
Flange-5	9.403	0.10	9.970	3.0000	N/A	No



Flange-6	9.403	0.10	9.970	3.0000	N/A	No
Shell-6	9.403	0.10	9.970	3.0000	N/A	No
Flange-7	9.403	0.10	9.970	3.0000	N/A	No
Flange-8	9.403	0.10	11.300	3.0000	N/A	No
Top head	9.403	0.10	9.560	3.0000	No	No

Liquid Level: 1040.50 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 0.920

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Skirt	2200.0	2200.0	16.0	16.0	...	...	0.70	0.70
Skirt	3200.0	1000.0	16.0	16.0	...	...	0.70	0.50
Ellipse	3250.0	50.0	36.0	30.0	29.1	5.1	1.00	1.00
Cylinder	5250.0	2000.0	32.0	32.0	30.3	6.9	1.00	1.00
Body Flg	5490.5	236.0	178.0	176.0	161.5	161.5	1.00	1.00
Body Flg	5726.5	236.0	178.0	176.0	161.5	161.5	1.00	1.00
Cylinder	8726.5	3000.0	32.0	32.0	30.3	8.2	1.00	1.00
Cylinder	9876.5	1150.0	32.0	32.0	30.3	8.2	1.00	1.00
Body Flg	10117.0	236.0	178.0	176.0	161.5	161.5	1.00	1.00
Body Flg	10353.0	236.0	178.0	176.0	161.5	161.5	1.00	1.00
Cylinder	13353.0	3000.0	32.0	32.0	30.4	8.2	1.00	1.00
Cylinder	14481.0	1128.0	32.0	32.0	30.3	8.2	1.00	1.00
Body Flg	14721.5	236.0	178.0	176.0	161.5	161.5	1.00	1.00
Body Flg	14957.5	236.0	178.0	176.0	161.5	161.5	1.00	1.00
Cylinder	17357.5	2400.0	32.0	32.0	30.3	7.1	1.00	1.00
Body Flg	17600.0	238.0	180.0	178.0	166.2	166.2	1.00	1.00
Body Flg	17836.0	236.0	178.0	176.0	158.0	158.0	1.00	1.00
Ellipse	17886.0	50.0	36.0	29.5	29.1	5.1	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N4_80	WRC-107/537	0.871	n/a	Passed
N1_150	WRC-297	0.935	Shell	Passed
N2_80	WRC-297	0.935	Shell	Passed
N3_150	WRC-107/537	0.886	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1100	13510	19830.2	124878792	255876320
20	30	2700	12106.2	19732.7	96689520	212339440
30	40	3225	11405.3	19668.9	84928960	192630704
40	50	4250	11366.8	19583.8	84359432	191648976
50	60	5372.5	9819.07	18917.3	63120756	153047072
60	70	5608.5	9652.06	18586.9	60822232	148619792
70	80	7226.5	9485.05	18239.5	58563152	144272512
80	90	9301.5	7163.44	16682.3	33580292	91868600
90	100	9999	6273.49	15984	25822684	73005888
100	110	10235	6106.48	15302.5	24361258	69312584
110	120	11853	5939.47	14602.3	22939264	65782384
120	130	13917	3617.86	10089.1	8597463	28730226
130	140	14603.5	2744.93	8994.11	4995039	17922432
140	150	14839.5	2577.92	7934.53	4366689	15924049
150	160	16157.5	2410.91	6854.96	3777769	14178185
160	170	17481	553.624	3815.8	216393	1350892
170	180	17718	385.416	2487.54	104603	600492
180	190	17861	219.912	1141.67	33145.2	172073

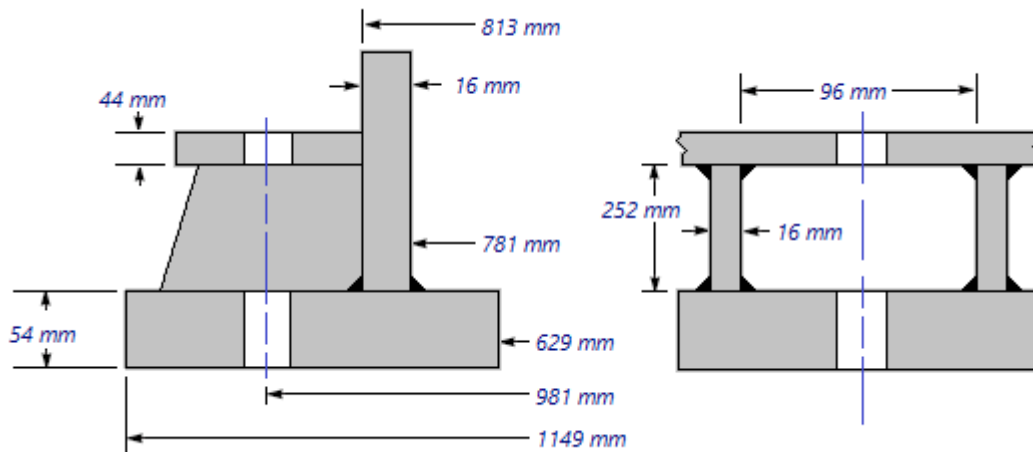


Abs Max of the all of the Stress Ratio's : 0.7684

Basering Data : Continuous Top Ring W/Gussets

Thickness of Basering	54.0000	mm
Inside Diameter of Basering	629.0000	mm
Outside Diameter of Basering	1149.0000	mm
Nominal Diameter of Bolts	48.0000	mm
Diameter of Bolt Circle	981.0000	mm
Number of Bolts	16	
Thickness of Gusset Plates	16.0000	mm
Average Width of Gusset Plates	158.0000	mm
Height of Gussets	252.0000	mm
Distance between Gussets	96.0000	mm
Thickness of Top Plate or Ring	44.0000	mm
Radial Width of the Top Plate	168.0000	mm
Number of Gussets	32	

Basering Sketch for Cylindrical Skirts:



Basering Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	53901.	N
Total Earthquake Shear on Support	60221.	N
Wind Moment on Support	481291456.	N-mm
Earthquake Moment on Support	612289024.	N-mm
Maximum Initial Bolt Load	141626.	N

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	19452.7	kgm
Shop Test - Fabricated + water (full)	26272.4	kgm
Shipping - Fab. + rem. intls.+ shipping app.	21502.4	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	21502.4	kgm
Empty - Fab. + intls. + details + wghts.	21502.4	kgm
Operating - Empty + operating liquid (no ca)	22606.1	kgm
Field Test - Empty weight + test medium (full)	27565.2	kgm

Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N4_80	9.73	OK	...	OK	7.80	OK	Passed
N1_150	9.96	OK	...	OK	9.22	OK	Passed
K1_50	9.96	...	...	OK	8.26	OK	No Calc[*]
K2_50	9.96	...	...	OK	8.26	OK	No Calc[*]
K3_50	9.96	...	...	OK	8.26	OK	No Calc[*]
H2_200	9.97	OK	...	...		OK	Passed
H1_200	9.97	OK	...	...		OK	Passed
N2_80	9.97	OK	...	OK	7.80	OK	Passed
K4_50	9.97	...	...	OK	8.26	OK	No Calc[*]
N3_150	9.56	OK	...	OK	9.22	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 9.56 Nozzle : N3150
 Minimum MAWP Shells/Flanges : 9.56 Element : Top head
 Minimum MAPnc Shells/Flanges : 10.6 Element : Top head

Computed Vessel M.A.W.P. : 9.56 MPa

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N4_80	0.000	0.000	145.299
40	N1_150	1225.000	180.000	310.145
40	K1_50	250.000	0.000	162.648
40	K2_50	1250.000	0.000	162.648
40	K3_50	600.000	180.000	162.648
70	H2_200	2870.491	0.000	386.231
110	H1_200	7826.198	0.000	377.100
150	N2_80	12502.198	180.000	150.862
150	K4_50	13263.998	0.000	162.648
180	N3_150	0.000	0.000	310.145

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
K1_50	50 mm	Actual	LWN	104.6	26.924	...	...	222	900
K2_50	50 mm	Actual	LWN	104.6	26.924	...	...	222	900
K3_50	50 mm	Actual	LWN	104.6	26.924	...	...	222	900
K4_50	50 mm	Actual	LWN	104.6	26.924	...	...	222	900
N4_80	80 mm	160	WNF	88.9	11.125	...	...	231	900
N2_80	80 mm	160	WNF	88.9	11.125	...	...	234	900
N1_150	150 mm	80	WNF	168.3	10.973	...	...	241	900
N3_150	150 mm	80	WNF	168.3	10.973	...	...	234	900
H2_200	200 mm	120	WNF	219.1	18.263	...	...	248	900
H1_200	200 mm	120	WNF	219.1	18.263	...	...	248	900

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K1_50	SA-105	30.800	10.000	...	...	...
K2_50	SA-105	30.800	10.000	...	...	...
K3_50	SA-105	30.800	10.000	...	...	...
K4_50	SA-105	32.000	10.000	...	...	...
N4_80	SA-105	30.000	10.000	...	...	...
N2_80	SA-105	32.000	10.000	...	...	...
N1_150	SA-105	30.800	10.000	...	...	...
N3_150	SA-105	29.400	10.000	...	...	...
H2_200	SA-105	32.000	10.000	...	...	...
H1_200	SA-105	32.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
K1_50	3450.000	0.0	187.32	0.00	Shell-1
K2_50	4450.000	0.0	187.32	0.00	Shell-1
K3_50	3800.000	180.0	187.32	0.00	Shell-1
K4_50	16463.998	0.0	187.32	0.00	Shell-6
N4_80	...	0.0	200.00	0.00	Bottom head
N2_80	15702.198	180.0	200.00	0.00	Shell-6
N1_150	4425.000	180.0	200.00	0.00	Shell-1
N3_150	...	0.0	200.00	0.00	Top head
H2_200	6070.491	0.0	200.00	0.00	Shell-2
H1_200	11026.197	0.0	200.00	0.00	Shell-4



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	SKIRT: 16.0mm THK X 781.0mm ID X 2200.0mm	SA-516 70
1	SKIRT: 16.0mm THK X 781.0mm ID X 1000.0mm	SA-516 70
2	ELLIPTICAL HEAD: 2.0 X 1, 36.0mm THK X 762.0mm ID X 50.0mm	SA-516 70
6	WELD NECK FLANGE: 178.0mm THK X 1126.0mm OD	SA-266 2
2	CYLINDER: 32.0mm THK X 762.0mm ID X 3000.0mm	SA-516 70
1	CYLINDER: 32.0mm THK X 762.0mm ID X 1150.0mm	SA-516 70
1	CYLINDER: 32.0mm THK X 762.0mm ID X 1128.0mm	SA-516 70
1	CYLINDER: 32.0mm THK X 762.0mm ID X 2400.0mm	SA-516 70
1	WELD NECK FLANGE: 180.0mm THK X 1128.0mm OD	SA-266 2
1	WELD NECK FLANGE: 178.0mm THK X 1128.0mm OD	SA-266 2
1	INSULATION: 500mm X 80mm THK	
2	INSULATION: 240mm X 80mm THK	
1	INSULATION: 2000mm X 80mm THK	
7	INSULATION: 206mm X 80mm THK	
1	PACKING: 2900.000mm	...
2	INSULATION: 3000mm X 80mm THK	
1	PACKING: 800.000mm	...
1	INSULATION: 1150mm X 80mm THK	
1	PACKING: 2922.000mm	...
1	PACKING: 778.000mm	...
1	INSULATION: 1128mm X 80mm THK	
1	INSULATION: 2400mm X 80mm THK	
1	INSULATION: 194mm X 80mm THK	
2	CLASS 900 GR 1.1, 200.0mm BLIND FLANGE(S)	SA-105
1	BASERING: 1149.0mm OD X 629.0 ID X 54.0mm	SA-516 70
16	BASERING BOLTS: 48.0mm DIA	SA-193 B7
16	BASERING BOLT NUTS: 48.0mm DIA	...
1	BASERING TOP RING: 1149.0mm OD X 813.0 ID X 44.0mm	...
32	BASERING GUSSET PLATES: 158.0mm X 158.0mm X 252.0mm X 16.0mm	...
4	GASKET: 826.0mm OD X 794.0mm ID	...
96	BODY FLANGE BOLTS: 56.0mm DIA	SA-193 B7
192	NUTS FOR BODY FLANGE BOLTS: 56.0mm DIA	...
1	CYLINDRICAL SEGMENT 32.0mm THK X 997.4mm X 2000.0mm	SA-516 70



1	CYLINDRICAL SEGMENT 32.0mm THK X 704.9mm X 2000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 32.0mm THK X 691.6mm X 2000.0mm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Flange-1	[11]	B	6	0		0.892	32.000	1.00	No
Flange-2	[11]	B	6	0		0.888	32.000	1.00	No
Flange-3	[11]	B	6	-2		0.866	32.000	1.00	No
Flange-4	[11]	B	6	-2		0.866	32.000	1.00	No
Flange-5	[11]	B	6	-2		0.866	32.000	1.00	No
Flange-6	[11]	B	6	-2		0.866	32.000	1.00	No
Flange-7	[11]	B	6	-2		0.866	32.000	1.00	No
Flange-8	[11]	B	-8	-20		0.779	36.000	1.00	Yes
Bottom head	[10]	B	4	3		0.982	30.000	1.00	No
Bottom head	[7]	B	9	0		0.842	36.000	1.00	No
Shell-1	[8]	B	6	3		0.957	32.000	1.00	No
Shell-2	[8]	B	6	3		0.957	32.000	1.00	No
Shell-3	[8]	B	6	3		0.957	32.000	1.00	No
Shell-4	[8]	B	6	3		0.960	32.000	1.00	No
Shell-5	[8]	B	6	3		0.957	32.000	1.00	No
Shell-6	[8]	B	6	3		0.957	32.000	1.00	No
Top head	[10]	B	4	4		1.000	29.500	1.00	No
Top head	[7]	B	9	0		0.841	36.000	1.00	No
N480	[1]	B	-8	-48	-29	0.438	9.735	1.00	No
Nozzle Flg	[5]	A	-18	-39					
N1150	[1]	B	6	2		0.941	32.000	1.00	No
Nozzle Flg	[5]	!	-18	-39					
K150	[1]	B	1	-104		0.084	26.924	1.00	No
Nozzle Flg	[5]	A	-18	-39					
K250	[1]	B	1	-104		0.084	26.924	1.00	No
Nozzle Flg	[5]	A	-18	-39					
K350	[1]	B	1	-104		0.084	26.924	1.00	No
Nozzle Flg	[5]	A	-18	-39					
H2200	[1]	B	6	2		0.941	32.000	1.00	No
Nozzle Flg	[5]	A	-18	-39					
H1200	[1]	B	6	2		0.941	32.000	1.00	No
Nozzle Flg	[5]	A	-18	-39					
N280	[1]	B	6	2		0.941	32.000	1.00	No
Nozzle Flg	[5]	A	-18	-39					
K450	[1]	B	1	-104		0.084	26.924	1.00	No
Nozzle Flg	[5]	A	-18	-39					
N3150	[1]	B	4	-3		0.888	29.500	1.00	No
Nozzle Flg	[5]	A	-18	-39					
Nozl. Bolting	[22]		-48			1.000	64.000	1.00	No
Bolting	[21]		-48						
Bolting	[21]		-48						
Bolting	[21]		-48						
Bolting	[21]		-48						

Warmest MDMT: 9 4

Required Minimum Design Metal Temperature -28.0 °C
 Warmest Computed Minimum Design Metal Temperature 4.0 °C

Warning: Computed overall MDMT was higher than the required value !

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3



- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note	20	40	There is a high jump in the Joint Eff.
Note	10	20	The wind diameter multiplier is very big.
Note	20	30	The wind diameter multiplier is very big.
Note	30	40	The wind diameter multiplier is very big.
Note	40	50	The wind diameter multiplier is very big.
Note	50	60	The wind diameter multiplier is very big.
Note	60	70	The wind diameter multiplier is very big.
Note	70	80	The wind diameter multiplier is very big.
Note	80	90	The wind diameter multiplier is very big.
Note	90	100	The wind diameter multiplier is very big.
Note	100	110	The wind diameter multiplier is very big.
Note	110	120	The wind diameter multiplier is very big.
Note	120	130	The wind diameter multiplier is very big.
Note	130	140	The wind diameter multiplier is very big.
Note	140	150	The wind diameter multiplier is very big.
Note	150	160	The wind diameter multiplier is very big.
Note	160	170	The wind diameter multiplier is very big.
Note	170	180	The wind diameter multiplier is very big.
Note	180	190	The wind diameter multiplier is very big.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.



Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	9.4029	MPa
Design Internal Temperature	82.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-28.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	Y	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16	IP+VE+EW	
Load Case 17	NP+VO+OW	
Load Case 18	FS+BS+IP+OW	
Load Case 19	FS+BS+EP+OW	
Load Case 20	BL+IP+OW	
Wind Design Code	ASCE/SEI 7-16	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 97.2	km/hr
Surface Roughness Category	C: Open Terrain	
Type of Surface	Moderately Smooth	
Base Elevation	300	mm
Percent Wind for Hydrotest	50.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	H or Hh	0 mm
Distance Upwind of Crest	Lh	0 mm
Distance from Crest to the Vessel	x	0 mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration	[Ss] 0.147
Max. Eff. Ground Acceleration	[S] 0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/18
Element From Node	10



Element To Node	20
Element Type	Skirt Support
Description	Skirt-2
Distance "FROM" to "TO"	2200 mm
Skirt Outside Diameter	813 mm
Diameter of Skirt at Base	813 mm
Skirt Thickness	16 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	16 mm
External Corrosion Allowance	0.8 mm
Design Temperature Internal Pressure	82 °C
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.64
Material Name	SA-516 70
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	138 MPa
Allowable Stress, Hydrotest	234 MPa
Material Density	0.00775 kg/cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	242.6 MPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No
Element From Node	10
Detail Type	Weight
Detail ID	Tailing lug
Dist. from "FROM" Node / Offset dist	0 mm
Miscellaneous Weight	1961.2 N
Offset from Element Centerline	0 mm

Element#	2/18
Element From Node	20
Element To Node	30
Element Type	Skirt Support
Description	Skirt-1
Distance "FROM" to "TO"	1000 mm
Skirt Outside Diameter	813 mm
Diameter of Skirt at Base	813 mm
Skirt Thickness	16 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	16 mm
External Corrosion Allowance	0.8 mm
Design Temperature Internal Pressure	82 °C
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.64
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.5
Weld is pre-Heated	No
Element From Node	20
Detail Type	Insulation
Detail ID	FP
Dist. from "FROM" Node / Offset dist	500 mm
Height/Length of Insulation	500 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm ³

Element#	3/18
Element From Node	30
Element To Node	40



Element Type	Elliptical
Description	Bottom head
Distance "FROM" to "TO"	50 mm
Inside Diameter	762 mm
Element Thickness	30 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	36 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	9.4029 MPa
Design Temperature Internal Pressure	82 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.64
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	30
Detail Type	Liquid
Detail ID	Liquid 30
Dist. from "FROM" Node / Offset dist	-190.5 mm
Height/Length of Liquid	240.5 mm
Liquid Density	0.0009185 kg/cm³
Element From Node	30
Detail Type	Insulation
Detail ID	Ins: 30
Dist. from "FROM" Node / Offset dist	-190.5 mm
Height/Length of Insulation	240.5 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm³
Element From Node	30
Detail Type	Nozzle
Detail ID	N480
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	80 mm
Nozzle Schedule	160
Nozzle Class	900
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT
Element From Node	30
Detail Type	For./Mom.
Detail ID	Guide sup later
Dist. from "FROM" Node / Offset dist	-190 mm
Force in X Direction	-20791 N
Force in Y Direction	0 N
Force in Z Direction	14291 N
Moment about X Axis	0 N-mm
Moment about Y Axis	0 N-mm
Moment about Z Axis	0 N-mm
Force/Moment Combination Method	SRSS
Element#	4/18
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Shell-1
Distance "FROM" to "TO"	2000 mm
Inside Diameter	762 mm
Element Thickness	32 mm
Internal Corrosion Allowance	3 mm



Nominal Thickness	32	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.4029	MPa
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	800	mm
Liquid Density	0.0009185	kg/cm³
Element From Node	40	
Detail Type	Insulation	
Detail ID	Ins: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N1150	
Dist. from "FROM" Node / Offset dist	1175	mm
Nozzle Diameter	150	mm
Nozzle Schedule	80	
Nozzle Class	900	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	K150	
Dist. from "FROM" Node / Offset dist	200	mm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	900	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	231.38	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	K250	
Dist. from "FROM" Node / Offset dist	1200	mm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	900	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	231.38	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	K350	
Dist. from "FROM" Node / Offset dist	550	mm



Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	900	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	231.38	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT
Element From Node	40	
Detail Type	Weight	
Detail ID	N1FY	
Dist. from "FROM" Node / Offset dist	1175	mm
Miscellaneous Weight	2539.8	N
Offset from Element Centerline	1649	mm

Element#	5/18	
Element From Node	50	
Element To Node	60	
Element Type	Flange	
Description	Flange-1	
Distance "FROM" to "TO"	236	mm
Flange Inside Diameter	762	mm
Element Thickness	176	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	178	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.4029	MPa
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	138	MPa
Allowable Stress, Hydrotest	225	MPa
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	230.08	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	206	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element#	6/18	
Element From Node	60	
Element To Node	70	
Element Type	Flange	
Description	Flange-2	
Distance "FROM" to "TO"	236	mm
Flange Inside Diameter	762	mm
Element Thickness	176	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	178	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.4029	MPa



Design Temperature Internal Pressure	82	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-266 2	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	60	
Detail Type	Insulation	
Detail ID	Ins: 60	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	206	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element#	7/18	
Element From Node	70	
Element To Node	80	
Element Type	Cylinder	
Description	Shell-2	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	762	mm
Element Thickness	32	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	32	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.4029	MPa
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	138	MPa
Allowable Stress, Hydrotest	234	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	242.6	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	70	
Detail Type	Packing	
Detail ID	Bed 2	
Dist. from "FROM" Node / Offset dist	100	mm
Height of Packed Section	2900	mm
Density	0.000224	kg/cm³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.92000002	

Element From Node	70	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element From Node	70	
Detail Type	Nozzle	
Detail ID	H2200	
Dist. from "FROM" Node / Offset dist	343.99	mm



Nozzle Diameter	200	mm
Nozzle Schedule	120	
Nozzle Class	900	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT

Element From Node	70	
Detail Type	Weight	
Detail ID	H2Davit	
Dist. from "FROM" Node / Offset dist	343.99	mm
Miscellaneous Weight	490.3	N
Offset from Element Centerline	350	mm

Element From Node	70	
Detail Type	Weight	
Detail ID	Bed support 2	
Dist. from "FROM" Node / Offset dist	50	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element From Node	70	
Detail Type	Weight	
Detail ID	Bed sup grid 2	
Dist. from "FROM" Node / Offset dist	80	mm
Miscellaneous Weight	343.21	N
Offset from Element Centerline	0	mm

Element#	8/18	
Element From Node	80	
Element To Node	90	
Element Type	Cylinder	
Description	Shell-3	
Distance "FROM" to "TO"	1150	mm
Inside Diameter	762	mm
Element Thickness	32	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	32	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.4029	MPa
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	80	
Detail Type	Packing	
Detail ID	Bed2	
Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	800	mm
Density	0.000224	kg/cm³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.9200002	

Element From Node	80	
Detail Type	Insulation	
Detail ID	Ins: 80	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1150	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³



Element From Node	80	
Detail Type	Weight	
Detail ID	Bed limiter sup	
Dist. from "FROM" Node / Offset dist	820	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element From Node	80	
Detail Type	Weight	
Detail ID	Bed limiter	
Dist. from "FROM" Node / Offset dist	850	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element#	9/18	
Element From Node	90	
Element To Node	100	
Element Type	Flange	
Description	Flange-3	
Distance "FROM" to "TO"	236	mm
Flange Inside Diameter	762	mm
Element Thickness	176	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	178	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.4029	MPa
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	138	MPa
Allowable Stress, Hydrotest	225	MPa
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	230.08	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	90	
Detail Type	Insulation	
Detail ID	Ins: 90	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	206	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³

Element#	10/18	
Element From Node	100	
Element To Node	110	
Element Type	Flange	
Description	Flange-4	
Distance "FROM" to "TO"	236	mm
Flange Inside Diameter	762	mm
Element Thickness	176	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	178	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.4029	MPa
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.1034	MPa



Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-266 2	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	100	
Detail Type	Insulation	
Detail ID	Ins: 100	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	206	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element#	11/18	
Element From Node	110	
Element To Node	120	
Element Type	Cylinder	
Description	Shell-4	
Distance "FROM" to "TO"	3000	mm
Element Outside Diameter	826	mm
Element Thickness	32	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	32	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.4029	MPa
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	138	MPa
Allowable Stress, Hydrotest	234	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	242.6	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	110	
Detail Type	Packing	
Detail ID	Bed-1	
Dist. from "FROM" Node / Offset dist	78	mm
Height of Packed Section	2922	mm
Density	0.000224	kg/cm³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.92000002	

Element From Node	110	
Detail Type	Insulation	
Detail ID	Ins: 110	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element From Node	110	
Detail Type	Nozzle	
Detail ID	H1200	
Dist. from "FROM" Node / Offset dist	673.2	mm
Nozzle Diameter	200	mm
Nozzle Schedule	120	



Nozzle Class 900
 Layout Angle 0.0
 Blind Flange (Y/N) Y
 Weight of Nozzle (Used if > 0) 0 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-105 +IT

Element From Node 110
 Detail Type Weight
 Detail ID H1Davit
 Dist. from "FROM" Node / Offset dist 673.2 mm
 Miscellaneous Weight 490.3 N
 Offset from Element Centerline 350 mm

Element From Node 110
 Detail Type Weight
 Detail ID Clip weight
 Dist. from "FROM" Node / Offset dist 665 mm
 Miscellaneous Weight 3922.4 N
 Offset from Element Centerline 0 mm

Element From Node 110
 Detail Type Weight
 Detail ID Bed support
 Dist. from "FROM" Node / Offset dist 49.999 mm
 Miscellaneous Weight 98.06 N
 Offset from Element Centerline 0 mm

Element From Node 110
 Detail Type Weight
 Detail ID Bed sup grid
 Dist. from "FROM" Node / Offset dist 60 mm
 Miscellaneous Weight 343.21 N
 Offset from Element Centerline 0 mm

Element From Node 110
 Detail Type Weight
 Detail ID Misc wt 7%
 Dist. from "FROM" Node / Offset dist 607 mm
 Miscellaneous Weight 12748 N
 Offset from Element Centerline 0 mm

Element# 12/18
 Element From Node 120
 Element To Node 130
 Element Type Cylinder
 Description Shell-5
 Distance "FROM" to "TO" 1128 mm
 Inside Diameter 762 mm
 Element Thickness 32 mm
 Internal Corrosion Allowance 3 mm
 Nominal Thickness 32 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 9.4029 MPa
 Design Temperature Internal Pressure 82 °C
 Design External Pressure 0.1034 MPa
 Design Temperature External Pressure 82 °C
 Effective Diameter Multiplier 1.64
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 120
 Detail Type Packing
 Detail ID Bed1
 Dist. from "FROM" Node / Offset dist 0 mm
 Height of Packed Section 778 mm



Density	0.000224	kg/cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.9200002	
Element From Node	120	
Detail Type	Insulation	
Detail ID	Ins: 120	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1128	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm ³
Element From Node	120	
Detail Type	Weight	
Detail ID	Bed limiter sup	
Dist. from "FROM" Node / Offset dist	800	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm
Element From Node	120	
Detail Type	Weight	
Detail ID	Bed limiter 1	
Dist. from "FROM" Node / Offset dist	820	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element#	13/18
Element From Node	130
Element To Node	140
Element Type	Flange
Description	Flange-5
Distance "FROM" to "TO"	236 mm
Flange Inside Diameter	762 mm
Element Thickness	176 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	178 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	9.4029 MPa
Design Temperature Internal Pressure	82 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.64
Material Name	SA-266 2
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	138 MPa
Allowable Stress, Hydrotest	225 MPa
Material Density	0.00775 kg/cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	230.08 MPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	130
Detail Type	Insulation
Detail ID	Ins: 130
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	206 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm ³

Element#	14/18
Element From Node	140



Element To Node	150
Element Type	Flange
Description	Flange-6
Distance "FROM" to "TO"	236 mm
Flange Inside Diameter	762 mm
Element Thickness	176 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	178 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	9.4029 MPa
Design Temperature Internal Pressure	82 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.64
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	140
Detail Type	Insulation
Detail ID	Ins: 140
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	206 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm³

Element#	15/18
Element From Node	150
Element To Node	160
Element Type	Cylinder
Description	Shell-6
Distance "FROM" to "TO"	2400 mm
Inside Diameter	762 mm
Element Thickness	32 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	32 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	9.4029 MPa
Design Temperature Internal Pressure	82 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.64
Material Name	SA-516 70
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	138 MPa
Allowable Stress, Hydrotest	234 MPa
Material Density	0.00775 kg/cm³
P Number Thickness	32.004 mm
Yield Stress, Operating	242.6 MPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	150
Detail Type	Insulation
Detail ID	Ins: 150
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	2400 mm
Thickness of Insulation	80 mm
Density	0.000352 kg/cm³
Element From Node	150
Detail Type	Nozzle
Detail ID	N280



Dist. from "FROM" Node / Offset dist	744.7	mm
Nozzle Diameter	80	mm
Nozzle Schedule	160	
Nozzle Class	900	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT

Element From Node	150	
Detail Type	Nozzle	
Detail ID	K450	
Dist. from "FROM" Node / Offset dist	1506.5	mm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	900	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	231.98	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT

Element From Node	150	
Detail Type	Weight	
Detail ID	N2Pipe	
Dist. from "FROM" Node / Offset dist	744.7	mm
Miscellaneous Weight	49.03	N
Offset from Element Centerline	0	mm

Element From Node	150	
Detail Type	Weight	
Detail ID	Demister sup	
Dist. from "FROM" Node / Offset dist	2250	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element From Node	150	
Detail Type	Weight	
Detail ID	Demister	
Dist. from "FROM" Node / Offset dist	2300	mm
Miscellaneous Weight	196.12	N
Offset from Element Centerline	0	mm

Element From Node	150	
Detail Type	Weight	
Detail ID	Lifting lug1	
Dist. from "FROM" Node / Offset dist	2300	mm
Miscellaneous Weight	1470.9	N
Offset from Element Centerline	0	mm

Element From Node	150	
Detail Type	Weight	
Detail ID	Lifting lug 2	
Dist. from "FROM" Node / Offset dist	2300	mm
Miscellaneous Weight	1470.9	N
Offset from Element Centerline	0	mm

Element#	16/18	
Element From Node	160	
Element To Node	170	
Element Type	Flange	
Description	Flange-7	
Distance "FROM" to "TO"	238	mm
Flange Inside Diameter	762	mm
Element Thickness	178	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	180	mm



External Corrosion Allowance	0	mm
Design Internal Pressure	9.4029	MPa
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	138	MPa
Allowable Stress, Hydrotest	225	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	230.08	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	160	
Detail Type	Insulation	
Detail ID	Ins: 160	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	206	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element#	17/18	
Element From Node	170	
Element To Node	180	
Element Type	Flange	
Description	Flange-8	
Distance "FROM" to "TO"	236	mm
Flange Inside Diameter	762	mm
Element Thickness	176	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	178	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	9.4029	MPa
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-266 2	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	170	
Detail Type	Insulation	
Detail ID	Ins: 170	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	194	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³

Element#	18/18	
Element From Node	180	
Element To Node	190	
Element Type	Elliptical	
Description	Top head	
Distance "FROM" to "TO"	50	mm
Inside Diameter	762	mm
Element Thickness	29.5	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	36	mm



External Corrosion Allowance	0	mm
Design Internal Pressure	9.4029	MPa
Design Temperature Internal Pressure	82	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.64	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	138	MPa
Allowable Stress, Hydrotest	234	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	242.6	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	180	
Detail Type	Insulation	
Detail ID	Ins: 180	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	240.5	mm
Thickness of Insulation	80	mm
Density	0.000352	kg/cm³
Element From Node	180	
Detail Type	Nozzle	
Detail ID	N3150	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	150	mm
Nozzle Schedule	80	
Nozzle Class	900	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	180	
Detail Type	Weight	
Detail ID	N3Wt	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	3137.9	N
Offset from Element Centerline	1649	mm
Element From Node	180	
Detail Type	For./Mom.	
Detail ID	N3F/M	
Dist. from "FROM" Node / Offset dist	170	mm
Force in X Direction	10721	N
Force in Y Direction	-14845	N
Force in Z Direction	10721	N
Moment about X Axis	4950120	N-mm
Moment about Y Axis	0	N-mm
Moment about Z Axis	4950120	N-mm
Force/Moment Combination Method	SRSS	



XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Skirt-2	...	...	2200	...	2200
Skirt-1	...	...	3200	...	1000
Bottom head	...	...	3250	...	50
Shell-1	...	...	5250	...	2000
Flange-1	...	...	5490.5	...	236
Flange-2	...	...	5726.5	...	236
Shell-2	...	...	8726.5	...	3000
Shell-3	...	...	9876.5	...	1150
Flange-3	...	...	10117	...	236
Flange-4	...	...	10353	...	236
Shell-4	...	...	13353	...	3000
Shell-5	...	...	14481	...	1128
Flange-5	...	...	14721.5	...	236
Flange-6	...	...	14957.5	...	236
Shell-6	...	...	17357.5	...	2400
Flange-7	...	...	17600	...	238
Flange-8	...	...	17836	...	236
Top head	...	...	17886	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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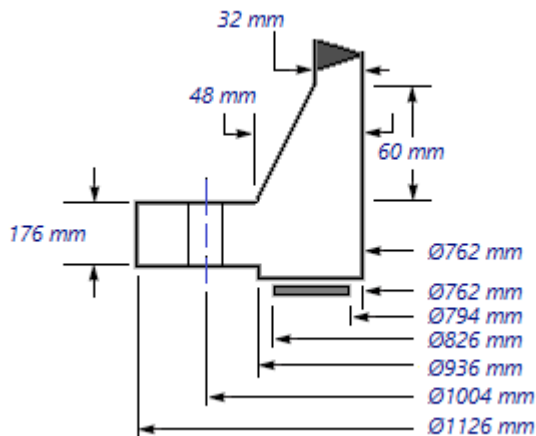


Flange Input Data Values

Description: FLANGE-1 :

Flange-1

Flange Type	Integral Weld Neck		
Design Pressure	P	9.40	MPa
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	762.002	mm
Flange Outside Diameter	A	1126.000	mm
Flange Thickness	t	176.0000	mm
Thickness of Hub at Small End	go	32.0000	mm
Thickness of Hub at Large End	gl	48.0000	mm
Length of Hub	h	60.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	1004.000	mm
Nominal Bolt Diameter	a	56.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	936.000	mm
Flange Face Inside Diameter	Fid	762.002	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	826.000	mm
Gasket Inside Diameter	Gi	794.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		115933.00	N
Bending Moment on Flange		175200992.00	N-mm





Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (9.4 \cdot (762/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 9.4) + Ca$$

$$= 30.2804 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 29) / (384 + 0.6 \cdot 29)$$

$$= 9.970 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 32) / (381 + 0.6 \cdot 32)$$

$$= 11.034 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	768.002 mm
Corroded Large Hub,	g1Cor = g1-ci	45.000 mm
Corroded Small Hub,	g0Cor = go-ci	29.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	72.999 mm
Gasket Contact Width,	N = (Go - Gi)/2	16.000 mm
Basic Gasket Width,	bo = N/2	8.000 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	7.127 mm
Gasket Reaction Diameter,	G = Go - 2 * b	811.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 812^2 \cdot 9.4$$

$$= 4865799.500 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 812 \cdot 3 \cdot 9.4$$

$$= 1025361.688 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 768^2 \cdot 9.4$$

$$= 4355515.000 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 4865799 - 4355515$$

$$= 510284.281 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(4865799 + 1025361 + 0, 0)$$

$$= 5891161.500 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot Ip$$

$$= 68.9 \cdot 7.13 \cdot \pi \cdot 812 + 0 \cdot 0 \cdot 0$$

$$= 1253136.250 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((5891161 + 115933 + 4 \cdot 175200992/812) / 172, 1253136/172)$$

$$= 39946.410 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 56 + 6 \cdot 176/(3 + 0.5)$$

$$= 413.714 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1004 \cdot \sin(\pi / 24)$$



= 131.048 mm

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max\left(\sqrt{\frac{Bs}{(2a + t)}}, 1\right)$$

$$= \max\left(\sqrt{\frac{131}{(2 \cdot 56 + 176)}}, 1\right)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	39946.410	44705.398	
Radial Distance between Hub and Bolts:	71.500	72.999	
Radial Distance between Bolts and Edge:	56.500	61.000	
Circ. Spacing between the Bolts:	124.700	131.048	413.714

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 44705 \cdot 172 / (68.9 \cdot \pi(826 + 794))$$

$$= 21.912 \text{ mm [Note: Exceeds actual gasket width, 16.000]}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 44705$$

$$= 7688536.50 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 5891161 - 4865799$$

$$= 1025361.88 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1004 - 812) / 2$$

$$= 96.1274 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1004 - 768) / 2 + 96.1)$$

$$= 107.0632 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1004 - 768 - 45) / 2$$

$$= 95.4990 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 179, Bb = 176, Cc = 60 \text{ and } Ddg = 37 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 179 \cdot 176^3 \cdot (1/3 - 0.210 \cdot (176/179) \cdot (1 - 1/12 \cdot (176/179)^4))$$

$$= 139483744.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 60 \cdot 37^3 \cdot (1/3 - 0.105 \cdot (37/60) \cdot (1 - 1/192 \cdot (37/60)^4))$$

$$= 816420.562 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 8512 + 49.8$$

$$= 140300160.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 \left(\frac{1}{3} - 0.21 \left(\frac{t}{AR} \right) \left(1 - \frac{1}{12} \left(\frac{t}{AR} \right)^4 \right) \right)$$

$$= 179 \cdot 176^3 \left(\frac{1}{3} - 0.21 \left(\frac{176}{179} \right) \left(1 - \frac{1}{12} \left(\frac{176}{179} \right)^4 \right) \right)$$

$$= 139483744.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2)$$

$$= \max(140300160, 139483744)$$

$$= 140300160.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 3.63 \cdot 29^2 \cdot 149 \cdot 768 / 0.33$$

$$= 93579992.000 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	4355515.	95.4990	1.0000	416116000.
Face Pressure, Mt	510284.	107.0632	1.0000	54654832.
Gasket Load, Mg	1025362.	96.1274	1.0000	98605376.
Gasket Seating, Matm	7688536.	96.1274	1.0000	739379008.
Moment Load, Moe	...	...	...	63318776.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me \left[\frac{1}{(0.384 \cdot IP + I)} \right] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$$

$$= 4 \cdot 175200992 [93579992 / (0.384 \cdot 140300160 + 93579992)] \cdot (95.5 / (1004 - 2 \cdot 95.5)) +$$

$$115933 \cdot 95.5$$

$$= 63318776.000 \text{ N-mm}$$

Total Moment for Operation, Mo 632694912. N-mm
 Total Moment for Gasket seating, Mg 739379008. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 149.238 mm
 Hub Ratio, h/h0 = HL / H0 0.402
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.552

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.860
 Factor V per 4.16.5 0.327
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1126 / 768$$

$$= 1.466$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.47^2 \cdot \log_{10}(1.47))) / (1.47^2 - 1)) / (1.47 - 1)$$

$$= 5.245$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.47^2 (1 + 8.55246 \cdot \log_{10}(1.47)) - 1) / ((1.0472 + 1.9448 \cdot 1.47^2) \cdot (1.47 - 1))$$

$$= 1.725$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.47^2 (1 + 8.55246 \cdot \log_{10}(1.47)) - 1) / (1.36136 (1.47^2 - 1) \cdot (1.47 - 1))$$

$$= 5.764$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.47^2 + 1) / (1.47^2 - 1)$$

$$= 2.740$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned} &= U \cdot g_0^2 \cdot h_0 / V \\ &= 5.76 \cdot 29^2 \cdot 149 / 0.33 \\ &= 2213042.750 \text{ mm}^3 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned} &= F / h_0 \\ &= 0.86 / 149 \\ &= 0.006 \text{ 1/mm} \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned} &= (t \cdot e + 1) / T + t^3 / d \\ &= (176 \cdot 0.0058 + 1) / 1.73 + 176^3 / 2213042 \\ &= 3.631 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned} &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\ &= (1 \cdot 632694912 / 768) / (3.63 \cdot 45^2) \\ &= 112.01 \text{ MPa} \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned} &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\ &= (1 \cdot 739379008 / 768) / (3.63 \cdot 45^2) \\ &= 130.90 \text{ MPa} \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned} &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\ &= (2.35 \cdot 632694912 / 768) / (3.63 \cdot 176^2) \\ &= 17.23 \text{ MPa} \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned} &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\ &= (2.35 \cdot 739379008 / 768) / (3.63 \cdot 176^2) \\ &= 20.13 \text{ MPa} \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned} &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\ &= (5.24 \cdot 632694912 / (176^2 \cdot 768)) - 2.74 \cdot 17.2 \\ &= 92.25 \text{ MPa} \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned} &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\ &= (5.24 \cdot 739379008 / (176^2 \cdot 768)) - 2.74 \cdot 20.1 \\ &= 107.81 \text{ MPa} \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned} &= (SHo + \max(SRo, STo)) / 2 \\ &= (112 + \max(17.2, 92.3)) / 2 \\ &= 102.13 \text{ MPa} \end{aligned}$$

Average Flange Stress, Seating [SAa]:

$$\begin{aligned} &= (SHa + \max(SRa, STa)) / 2 \\ &= (131 + \max(20.1, 108)) / 2 \\ &= 119.35 \text{ MPa} \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned} &= (Wm_1 + Fa + 4Me/G) / Ab \\ &= 6870074 / 44705 \\ &= 153.69 \text{ MPa} \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned} &= (Wm_2 / Ab) \\ &= (1253136 / 44705) \\ &= 28.03 \text{ MPa} \end{aligned}$$

Flange Stress Analysis Results: MPa



	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	112.01	207.00	130.90	207.00
Radial Flange	17.23	138.00	20.13	138.00
Tangential Flange	92.25	138.00	107.81	138.00
Maximum Average	102.13	138.00	119.35	138.00
Bolting	153.69	172.00	28.03	172.00

Minimum Required Flange Thickness 161.468 mm

Estimated M.A.W.P. (Operating) 9.97 MPa

Estimated Finished Weight of Flange at given Thk. 783.1 kgm

Estimated Unfinished Weight of Forging at given Thk 987.3 kgm

Flange Stress results per WRC-538 at computed S_{fmax} (MPa):

		Actual	Operating Allowed	stress ratio	Actual	Gasket Seating Allowed	stress ratio
Longitudinal Hub	Sh	411.26	499.99	0.823	411.26	499.99	0.823
Tangential Flange	St	338.72	374.99	0.903	338.72	374.99	0.903
Tangential Flange	Sto	148.50	250.00	0.594	148.50	250.00	0.594
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	401.97	749.99	0.536	401.97	749.99	0.536
Combined	Sh + Sto	559.76	749.99	0.746	559.76	749.99	0.746
Radial Flange	Sr	63.25	499.99	0.126	63.25	499.99	0.126
Maximum Stress Ratio				1.000			1.000

Coefficients for *f* and *f2*

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:



$$= \max(1; F1num / F1den)$$

$$= \max(1; 1.37/1.44)$$

$$= 1.0000$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; fn2) \text{ if } h/ho < 0.35, \text{ else } \max(0.25; fn3)$$

$$= \max(0.25; 0.53/1.45) \text{ if } h/ho < 0.35, \text{ else } \max(0.25; 0.27/0.76)$$

$$= 0.3611$$

In this case, the hub ratio h/ho , 0.4020 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, $Sfmax$, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 MPa
Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	540.4 MPa
Target assembly gasket stress	SgT	172.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	1862.7250 mm ²
Max. permissible single flange rotation for gasket	Thetamax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	230.1 MPa
Single flange rotation at $Sfmax$	Thetaamax	0.336 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (826^2 - 794^2)$$

$$= 40714.9 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 172 \cdot 40715 / (24 \cdot 1863)$$

$$= 156.6 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(157; 508)$$

$$= 156.6 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(157; 290)$$

$$= 290.0 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(290; 540)$$

$$= 290.0 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$\therefore Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$\therefore 290 \geq 140 (40715 / (24 \cdot 1863))$$

$$\therefore 290.0 \geq 127.5 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$\therefore Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$\therefore 290 \geq (97 \cdot 40715 + \pi/4 \cdot 9.4 \cdot 794^2) / (0.7 \cdot 1863 \cdot 24)$$

$$\therefore 290.0 \geq 275.0 \text{ MPa} \text{ Check Passes}$$



Check if Gasket Maximum Stress is achieved. (Eq. O-9)

:: Sbsel $\leq$ Sgmax(Ag / (nb • Abs))
 :: 290 $\leq$ 380 (40715/(24 • 1863))
 :: 290.0 $\leq$ 346.1 MPa Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

:: Sbsel $\leq$ Sfmax(thetagmax / thetafmax)
 :: 290 $\leq$ 540 (1/0.34)
 :: 290.0 $\leq$ 1606.9 MPa Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

= Sbsel • K • Abs • ϕ b
 = 290 • 0.2 • 1863 • 56
 = 6052070.0 N-mm

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 MPa
 Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since Sb $\geq$ SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 = 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
 = 52.14•739379008/1•1•0.33/(3.02•202713•29²•149•0.3)
 = 0.546 (should be $\leq$ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 = 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
 = 52.14•632694912/1•1•0.33/(3.02•199226•29²•149•0.3)
 = 0.475 (should be $\leq$ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 = 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
 = 52.14•739379008/1•1•0.33/(3.63•202713•29²•149•0.3)
 = 0.454 (should be $\leq$ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 = 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
 = 52.14•632694912/1•1•0.33/(3.63•199226•29²•149•0.3)
 = 0.395 (should be $\leq$ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.89, Temperature Reduction per Fig. UCS 66.1 = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 0 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	113.66	207.00	130.78	207.00
Radial Flange	19.45	138.00	22.38	138.00
Tangential Flange	87.66	138.00	100.86	138.00
Maximum Average	100.66	138.00	115.82	138.00
Bolting	153.69	172.00	28.03	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 157.632 mm

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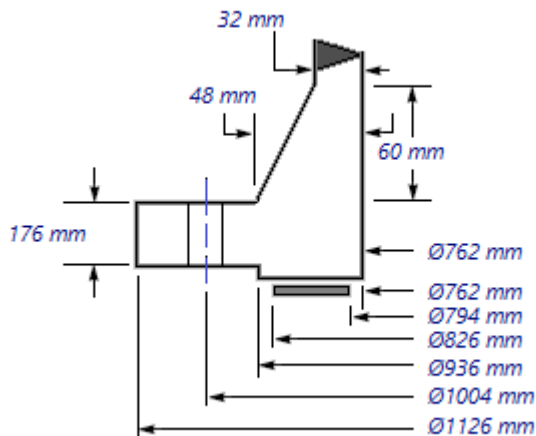


Flange Input Data Values

Description: FLANGE-2 :

Flange-2

Flange Type	Integral Weld Neck		
Design Pressure	P	9.40	MPa
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	762.002	mm
Flange Outside Diameter	A	1126.000	mm
Flange Thickness	t	176.0000	mm
Thickness of Hub at Small End	go	32.0000	mm
Thickness of Hub at Large End	gl	48.0000	mm
Length of Hub	h	60.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	1004.000	mm
Nominal Bolt Diameter	a	56.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	936.000	mm
Flange Face Inside Diameter	Fid	762.002	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	826.000	mm
Gasket Inside Diameter	Gi	794.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		108484.00	N
Bending Moment on Flange		170335008.00	N-mm





Hub Small End Required Thickness due to Internal Pressure:

$$\begin{aligned} &= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= (9.4 \cdot (762/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 9.4) + Ca \\ &= 30.2804 \text{ mm} \end{aligned}$$

Small End Hub MAWP:

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\ &= (138 \cdot 1 \cdot 29) / (384 + 0.6 \cdot 29) \\ &= 9.970 \text{ MPa} \end{aligned}$$

Hub Small End Hub MAPnc:

$$\begin{aligned} &= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\ &= (138 \cdot 1 \cdot 32) / (381 + 0.6 \cdot 32) \\ &= 11.034 \text{ MPa} \end{aligned}$$

Corroded Flange ID,	Bcor = B+2*ci	768.002 mm
Corroded Large Hub,	g1Cor = g1-ci	45.000 mm
Corroded Small Hub,	g0Cor = go-ci	29.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	72.999 mm
Gasket Contact Width,	N = (Go - Gi)/2	16.000 mm
Basic Gasket Width,	bo = N/2	8.000 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	7.127 mm
Gasket Reaction Diameter,	G = Go - 2 * b	811.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$\begin{aligned} &= \pi/4 \cdot G^2 \cdot P \\ &= \pi/4 \cdot 812^2 \cdot 9.4 \\ &= 4865799.500 \text{ N} \end{aligned}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$\begin{aligned} &= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P \\ &= 2 \cdot 7.13 \cdot \pi \cdot 812 \cdot 3 \cdot 9.4 \\ &= 1025361.688 \text{ N} \end{aligned}$$

Hydrostatic End Load at Flange ID [Hd]:

$$\begin{aligned} &= \pi/4 \cdot Bcor^2 \cdot P \\ &= \pi/4 \cdot 768^2 \cdot 9.4 \\ &= 4355515.000 \text{ N} \end{aligned}$$

Pressure Force on Flange Face [Ht]:

$$\begin{aligned} &= H - Hd \\ &= 4865799 - 4355515 \\ &= 510284.281 \text{ N} \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned} &= \max(Wo,1 + Wo,2 + H'p, 0) \\ &= \max(4865799 + 1025361 + 0, 0) \\ &= 5891161.500 \text{ N} \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned} &= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp \\ &= 68.9 \cdot 7.13 \cdot \pi \cdot 812 + 0 \cdot 0 \cdot 0 \\ &= 1253136.250 \text{ N} \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned} &= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa) \\ &= \max((5891161 + 108484 + 4 \cdot 170335008/812) / 172, 1253136/172) \\ &= 39763.734 \text{ mm}^2 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$\begin{aligned} &= 2a + 6t/(m + 0.5) \\ &= 2 \cdot 56 + 6 \cdot 176/(3 + 0.5) \\ &= 413.714 \text{ mm} \end{aligned}$$

Actual Circumferential Bolt Spacing [Bs]:

$$\begin{aligned} &= C \cdot \sin(\pi / n) \\ &= 1004 \cdot \sin(\pi/24) \end{aligned}$$



= 131.048 mm

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max\left(\sqrt{\frac{Bs}{(2a + t)}}, 1\right)$$

$$= \max\left(\sqrt{\frac{131}{(2 \cdot 56 + 176)}}, 1\right)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	39763.734	44705.398	
Radial Distance between Hub and Bolts:	71.500	72.999	
Radial Distance between Bolts and Edge:	56.500	61.000	
Circ. Spacing between the Bolts:	124.700	131.048	413.714

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 44705 \cdot 172 / (68.9 \cdot \pi(826 + 794))$$

$$= 21.912 \text{ mm [Note: Exceeds actual gasket width, 16.000]}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 44705$$

$$= 7688536.50 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 5891161 - 4865799$$

$$= 1025361.88 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1004 - 812) / 2$$

$$= 96.1274 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 \cdot (C - B) / 2 + hg$$

$$= 0.5 \cdot (1004 - 768) / 2 + 96.1$$

$$= 107.0632 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1004 - 768 - 45) / 2$$

$$= 95.4990 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 179, Bb = 176, Cc = 60 \text{ and } Ddg = 37 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 179 \cdot 176^3 \cdot (1/3 - 0.210 \cdot (176/179) \cdot (1 - 1/12 \cdot (176/179)^4))$$

$$= 139483744.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 60 \cdot 37^3 \cdot (1/3 - 0.105 \cdot (37/60) \cdot (1 - 1/192 \cdot (37/60)^4))$$

$$= 816420.562 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 8512 + 49.8$$



$$= 140300160.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 \left(\frac{1}{3} - 0.21 \left(\frac{t}{AR} \right) \left(1 - \frac{1}{12} \left(\frac{t}{AR} \right)^4 \right) \right)$$

$$= 179 \cdot 176^3 \left(\frac{1}{3} - 0.21 \left(\frac{176}{179} \right) \left(1 - \frac{1}{12} \left(\frac{176}{179} \right)^4 \right) \right)$$

$$= 139483744.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2)$$

$$= \max(140300160, 139483744)$$

$$= 140300160.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 3.63 \cdot 29^2 \cdot 149 \cdot 768 / 0.33$$

$$= 93579992.000 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	4355515.	95.4990	1.0000	416116000.
Face Pressure, Mt	510284.	107.0632	1.0000	54654832.
Gasket Load, Mg	1025362.	96.1274	1.0000	98605376.
Gasket Seating, Matm	7688536.	96.1274	1.0000	739379008.
Moment Load, Moe	...	...	...	61156140.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me \left[\frac{1}{(0.384 \cdot IP + I)} \right] \cdot \left(\frac{hD}{(C-2 \cdot hD)} \right) + FA \cdot hD$$

$$= 4 \cdot 170335008 [93579992 / (0.384 \cdot 140300160 + 93579992)] \cdot (95.5 / (1004 - 2 \cdot 95.5)) +$$

$$108484 \cdot 95.5$$

$$= 61156140.000 \text{ N-mm}$$

Total Moment for Operation, Mo 630532352. N-mm
 Total Moment for Gasket seating, Mg 739379008. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 149.238 mm
 Hub Ratio, h/h0 = HL / H0 0.402
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.552

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.860
 Factor V per 4.16.5 0.327
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1126/768$$

$$= 1.466$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.47^2 \cdot \log_{10}(1.47))) / (1.47^2 - 1)) / (1.47 - 1)$$

$$= 5.245$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.47^2 (1 + 8.55246 \cdot \log_{10}(1.47)) - 1) / ((1.0472 + 1.9448 \cdot 1.47^2) \cdot (1.47 - 1))$$

$$= 1.725$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.47^2 (1 + 8.55246 \cdot \log_{10}(1.47)) - 1) / (1.36136 (1.47^2 - 1) \cdot (1.47 - 1))$$

$$= 5.764$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.47^2 + 1) / (1.47^2 - 1)$$

$$= 2.740$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned} &= U \cdot g_0^2 \cdot h_0 / V \\ &= 5.76 \cdot 29^2 \cdot 149 / 0.33 \\ &= 2213042.750 \text{ mm}^3 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned} &= F / h_0 \\ &= 0.86 / 149 \\ &= 0.006 \text{ 1/mm} \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned} &= (t \cdot e + 1) / T + t^3 / d \\ &= (176 \cdot 0.0058 + 1) / 1.73 + 176^3 / 2213042 \\ &= 3.631 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned} &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\ &= (1 \cdot 630532352 / 768) / (3.63 \cdot 45^2) \\ &= 111.63 \text{ MPa} \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned} &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\ &= (1 \cdot 739379008 / 768) / (3.63 \cdot 45^2) \\ &= 130.90 \text{ MPa} \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned} &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\ &= (2.35 \cdot 630532352 / 768) / (3.63 \cdot 176^2) \\ &= 17.17 \text{ MPa} \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned} &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\ &= (2.35 \cdot 739379008 / 768) / (3.63 \cdot 176^2) \\ &= 20.13 \text{ MPa} \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned} &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\ &= (5.24 \cdot 630532352 / (176^2 \cdot 768)) - 2.74 \cdot 17.2 \\ &= 91.94 \text{ MPa} \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned} &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\ &= (5.24 \cdot 739379008 / (176^2 \cdot 768)) - 2.74 \cdot 20.1 \\ &= 107.81 \text{ MPa} \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned} &= (SHo + \max(SRo, STo)) / 2 \\ &= (112 + \max(17.2, 91.9)) / 2 \\ &= 101.78 \text{ MPa} \end{aligned}$$

Average Flange Stress, Seating [SAa]:

$$\begin{aligned} &= (SHa + \max(SRa, STa)) / 2 \\ &= (131 + \max(20.1, 108)) / 2 \\ &= 119.35 \text{ MPa} \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned} &= (Wm_1 + Fa + 4Me/G) / Ab \\ &= 6838657 / 44705 \\ &= 152.98 \text{ MPa} \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned} &= (Wm_2 / Ab) \\ &= (1253136 / 44705) \\ &= 28.03 \text{ MPa} \end{aligned}$$

Flange Stress Analysis Results: MPa



	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	111.63	207.00	130.90	207.00
Radial Flange	17.17	138.00	20.13	138.00
Tangential Flange	91.94	138.00	107.81	138.00
Maximum Average	101.78	138.00	119.35	138.00
Bolting	152.98	172.00	28.03	172.00

Minimum Required Flange Thickness 161.468 mm

Estimated M.A.W.P. (Operating) 9.97 MPa

Estimated Finished Weight of Flange at given Thk. 783.1 kgm

Estimated Unfinished Weight of Forging at given Thk 987.3 kgm

Flange Stress results per WRC-538 at computed Sfmax (MPa):

		Actual	Operating Allowed	stress ratio	Actual	Gasket Seating Allowed	stress ratio
Longitudinal Hub	Sh	411.26	499.99	0.823	411.26	499.99	0.823
Tangential Flange	St	338.72	374.99	0.903	338.72	374.99	0.903
Tangential Flange	Sto	148.50	250.00	0.594	148.50	250.00	0.594
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	401.97	749.99	0.536	401.97	749.99	0.536
Combined	Sh + Sto	559.76	749.99	0.746	559.76	749.99	0.746
Radial Flange	Sr	63.25	499.99	0.126	63.25	499.99	0.126
Maximum Stress Ratio				1.000			1.000

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:



$$= \max(1; F1num / F1den)$$

$$= \max(1; 1.37/1.44)$$

$$= 1.0000$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; fn2) \text{ if } h/ho < 0.35, \text{ else } \max(0.25; fn3)$$

$$= \max(0.25; 0.53/1.45) \text{ if } h/ho < 0.35, \text{ else } \max(0.25; 0.27/0.76)$$

$$= 0.3611$$

In this case, the hub ratio h/ho , 0.4020 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, $Sfmax$, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.9 MPa
Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	540.4 MPa
Target assembly gasket stress	SgT	172.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	1862.7250 mm ²
Max. permissible single flange rotation for gasket	Thetamax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	230.1 MPa
Single flange rotation at $Sfmax$	Thetaamax	0.336 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (826^2 - 794^2)$$

$$= 40714.9 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 172 \cdot 40715 / (24 \cdot 1863)$$

$$= 156.6 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(157; 508)$$

$$= 156.6 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(157; 290)$$

$$= 290.0 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(290; 540)$$

$$= 290.0 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$\therefore Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$\therefore 290 \geq 140 (40715 / (24 \cdot 1863))$$

$$\therefore 290.0 \geq 127.5 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$\therefore Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$\therefore 290 \geq (97 \cdot 40715 + \pi/4 \cdot 9.4 \cdot 794^2) / (0.7 \cdot 1863 \cdot 24)$$

$$\therefore 290.0 \geq 275.0 \text{ MPa} \text{ Check Passes}$$



Check if Gasket Maximum Stress is achieved. (Eq. O-9)

:: Sbsel $\leq$ Sgmax(Ag / (nb • Abs))
 :: 290 $\leq$ 380 (40715/(24 • 1863))
 :: 290.0 $\leq$ 346.1 MPa Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

:: Sbsel $\leq$ Sfmax(thetagmax / thetafmax)
 :: 290 $\leq$ 540 (1/0.34)
 :: 290.0 $\leq$ 1606.9 MPa Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

= Sbsel • K • Abs • ϕ b
 = 290 • 0.2 • 1863 • 56
 = 6052070.0 N-mm

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 MPa
 Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since Sb $\geq$ SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 = 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
 = 52.14•739379008/1•1•0.33/(3.02•202713•29²•149•0.3)
 = 0.546 (should be $\leq$ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 = 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
 = 52.14•630532352/1•1•0.33/(3.02•199226•29²•149•0.3)
 = 0.474 (should be $\leq$ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 = 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
 = 52.14•739379008/1•1•0.33/(3.63•202713•29²•149•0.3)
 = 0.454 (should be $\leq$ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 = 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
 = 52.14•630532352/1•1•0.33/(3.63•199226•29²•149•0.3)
 = 0.394 (should be $\leq$ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.89, Temperature Reduction per Fig. UCS 66.1 = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 0 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	113.27	207.00	130.78	207.00
Radial Flange	19.39	138.00	22.38	138.00
Tangential Flange	87.36	138.00	100.86	138.00
Maximum Average	100.31	138.00	115.82	138.00
Bolting	152.98	172.00	28.03	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 157.632 mm

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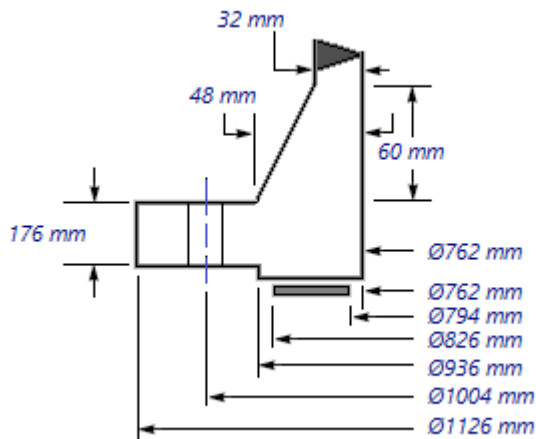


Flange Input Data Values

Description: FLANGE-3 :

Flange-3

Flange Type	Integral Weld Neck		
Design Pressure	P	9.40	MPa
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	762.002	mm
Flange Outside Diameter	A	1126.000	mm
Flange Thickness	t	176.0000	mm
Thickness of Hub at Small End	go	32.0000	mm
Thickness of Hub at Large End	gl	48.0000	mm
Length of Hub	h	60.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	1004.000	mm
Nominal Bolt Diameter	a	56.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	936.000	mm
Flange Face Inside Diameter	Fid	762.002	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	826.000	mm
Gasket Inside Diameter	Gi	794.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		76109.61	N
Bending Moment on Flange		83714096.00	N-mm





Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (9.4 \cdot (762/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 9.4) + Ca$$

$$= 30.2804 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 29) / (384 + 0.6 \cdot 29)$$

$$= 9.970 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 32) / (381 + 0.6 \cdot 32)$$

$$= 11.034 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	768.002 mm
Corroded Large Hub,	g1Cor = g1-ci	45.000 mm
Corroded Small Hub,	g0Cor = go-ci	29.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	72.999 mm
Gasket Contact Width,	N = (Go - Gi)/2	16.000 mm
Basic Gasket Width,	bo = N/2	8.000 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	7.127 mm
Gasket Reaction Diameter,	G = Go - 2 * b	811.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 812^2 \cdot 9.4$$

$$= 4865799.500 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 812 \cdot 3 \cdot 9.4$$

$$= 1025361.688 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 768^2 \cdot 9.4$$

$$= 4355515.000 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 4865799 - 4355515$$

$$= 510284.281 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(4865799 + 1025361 + 0, 0)$$

$$= 5891161.500 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot Ip$$

$$= 68.9 \cdot 7.13 \cdot \pi \cdot 812 + 0 \cdot 0 \cdot 0$$

$$= 1253136.250 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((5891161 + 76110 + 4 \cdot 83714096/812) / 172, 1253136/172)$$

$$= 37094.625 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 56 + 6 \cdot 176/(3 + 0.5)$$

$$= 413.714 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1004 \cdot \sin(\pi/24)$$



= 131.048 mm

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{\frac{Bs}{(2a + t)}}, 1)$$

$$= \max(\sqrt{\frac{131}{(2 \cdot 56 + 176)}}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	37094.625	44705.398	
Radial Distance between Hub and Bolts:	71.500	72.999	
Radial Distance between Bolts and Edge:	56.500	61.000	
Circ. Spacing between the Bolts:	124.700	131.048	413.714

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 44705 \cdot 172 / (68.9 \cdot \pi(826 + 794))$$

$$= 21.912 \text{ mm [Note: Exceeds actual gasket width, 16.000]}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 44705$$

$$= 7688536.50 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 5891161 - 4865799$$

$$= 1025361.88 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1004 - 812) / 2$$

$$= 96.1274 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1004 - 768) / 2 + 96.1)$$

$$= 107.0632 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1004 - 768 - 45) / 2$$

$$= 95.4990 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 179, Bb = 176, Cc = 60 \text{ and } Ddg = 37 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 179 \cdot 176^3 \cdot (1/3 - 0.210 \cdot (176/179) \cdot (1 - 1/12 \cdot (176/179)^4))$$

$$= 139483744.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 60 \cdot 37^3 \cdot (1/3 - 0.105 \cdot (37/60) \cdot (1 - 1/192 \cdot (37/60)^4))$$

$$= 816420.562 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 8512 + 49.8$$



$$= 140300160.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 \left(\frac{1}{3} - 0.21 \left(\frac{t}{AR} \right) \left(1 - \frac{1}{12} \left(\frac{t}{AR} \right)^4 \right) \right)$$

$$= 179 \cdot 176^3 \left(\frac{1}{3} - 0.21 \left(\frac{176}{179} \right) \left(1 - \frac{1}{12} \left(\frac{176}{179} \right)^4 \right) \right)$$

$$= 139483744.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2)$$

$$= \max(140300160, 139483744)$$

$$= 140300160.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 3.63 \cdot 29^2 \cdot 149 \cdot 768 / 0.33$$

$$= 93579992.000 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	4355515.	95.4990	1.0000	416116000.
Face Pressure, Mt	510284.	107.0632	1.0000	54654832.
Gasket Load, Mg	1025362.	96.1274	1.0000	98605376.
Gasket Seating, Matm	7688536.	96.1274	1.0000	739379008.
Moment Load, Moe	...	...	...	32233864.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me \left[\frac{1}{(0.384 \cdot IP + I)} \right] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$$

$$= 4 \cdot 83714096 [93579992 / (0.384 \cdot 140300160 + 93579992)] \cdot (95.5 / (1004 - 2 \cdot 95.5)) +$$

$$76110 \cdot 95.5$$

$$= 32233864.000 \text{ N-mm}$$

Total Moment for Operation, Mo 601610048. N-mm
 Total Moment for Gasket seating, Mg 739379008. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 149.238 mm
 Hub Ratio, h/h0 = HL / H0 0.402
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.552

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.860
 Factor V per 4.16.5 0.327
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1126/768$$

$$= 1.466$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.47^2 \cdot \log_{10}(1.47))) / (1.47^2 - 1)) / (1.47 - 1)$$

$$= 5.245$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.47^2 (1 + 8.55246 \cdot \log_{10}(1.47)) - 1) / ((1.0472 + 1.9448 \cdot 1.47^2) \cdot (1.47 - 1))$$

$$= 1.725$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.47^2 (1 + 8.55246 \cdot \log_{10}(1.47)) - 1) / (1.36136 (1.47^2 - 1) \cdot (1.47 - 1))$$

$$= 5.764$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.47^2 + 1) / (1.47^2 - 1)$$

$$= 2.740$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned} &= U \cdot g_0^2 \cdot h_0 / V \\ &= 5.76 \cdot 29^2 \cdot 149 / 0.33 \\ &= 2213042.750 \text{ mm}^3 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned} &= F / h_0 \\ &= 0.86 / 149 \\ &= 0.006 \text{ 1/mm} \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned} &= (t \cdot e + 1) / T + t^3/d \\ &= (176 \cdot 0.0058 + 1) / 1.73 + 176^3 / 2213042 \\ &= 3.631 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned} &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\ &= (1 \cdot 601610048 / 768) / (3.63 \cdot 45^2) \\ &= 106.51 \text{ MPa} \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned} &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\ &= (1 \cdot 739379008 / 768) / (3.63 \cdot 45^2) \\ &= 130.90 \text{ MPa} \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned} &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\ &= (2.35 \cdot 601610048 / 768) / (3.63 \cdot 176^2) \\ &= 16.38 \text{ MPa} \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned} &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\ &= (2.35 \cdot 739379008 / 768) / (3.63 \cdot 176^2) \\ &= 20.13 \text{ MPa} \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned} &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\ &= (5.24 \cdot 601610048 / (176^2 \cdot 768)) - 2.74 \cdot 16.4 \\ &= 87.72 \text{ MPa} \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned} &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\ &= (5.24 \cdot 739379008 / (176^2 \cdot 768)) - 2.74 \cdot 20.1 \\ &= 107.81 \text{ MPa} \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned} &= (SHo + \max(SRo, STo)) / 2 \\ &= (107 + \max(16.4, 87.7)) / 2 \\ &= 97.11 \text{ MPa} \end{aligned}$$

Average Flange Stress, Seating [SAa]:

$$\begin{aligned} &= (SHa + \max(SRa, STa)) / 2 \\ &= (131 + \max(20.1, 108)) / 2 \\ &= 119.35 \text{ MPa} \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned} &= (Wm_1 + Fa + 4Me/G) / Ab \\ &= 6379618 / 44705 \\ &= 142.72 \text{ MPa} \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned} &= (Wm_2 / Ab) \\ &= (1253136 / 44705) \\ &= 28.03 \text{ MPa} \end{aligned}$$

Flange Stress Analysis Results: MPa



	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	106.51	207.00	130.90	207.00
Radial Flange	16.38	138.00	20.13	138.00
Tangential Flange	87.72	138.00	107.81	138.00
Maximum Average	97.11	138.00	119.35	138.00
Bolting	142.72	172.00	28.03	172.00

Minimum Required Flange Thickness 161.468 mm

Estimated M.A.W.P. (Operating) 9.97 MPa

Estimated Finished Weight of Flange at given Thk. 783.1 kgm

Estimated Unfinished Weight of Forging at given Thk 987.3 kgm

Flange Stress results per WRC-538 at computed S_{fmax} (MPa):

		Actual	Operating Allowed	stress ratio	Actual	Gasket Seating Allowed	stress ratio
Longitudinal Hub	Sh	411.26	499.99	0.823	411.26	499.99	0.823
Tangential Flange	St	338.72	374.99	0.903	338.72	374.99	0.903
Tangential Flange	Sto	148.50	250.00	0.594	148.50	250.00	0.594
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	401.97	749.99	0.536	401.97	749.99	0.536
Combined	Sh + Sto	559.76	749.99	0.746	559.76	749.99	0.746
Radial Flange	Sr	63.25	499.99	0.126	63.25	499.99	0.126
Maximum Stress Ratio				1.000			1.000

Coefficients for f and f_2

	f	f_2	f_3
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g_1}{g_0} \right) + e \left(\frac{h}{h_0} \right) + g \left(\frac{g_1}{g_0} \right)^2 + i \left(\frac{h}{h_0} \right)^2 + k \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}{\left[1 + b \left(\frac{g_1}{g_0} \right) + d \left(\frac{h}{h_0} \right) + f \left(\frac{g_1}{g_0} \right)^2 + h \left(\frac{h}{h_0} \right)^2 + j \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}$$

Hub Stress Correction Factor [f]:



= max(1; F1num / F1den)
 = max(1; 1.37/1.44)
 = 1.0000

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)
 = max(0.25; 0.53/1.45) if h/ho < 0.35, else max (0.25; 0.27/0.76)
 = 0.3611

In this case, the hub ratio h/ho, 0.4020 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 MPa
Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	540.4 MPa
Target assembly gasket stress	SgT	172.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	1862.7250 mm ²
Max. permissible single flange rotation for gasket	Thetamax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	230.1 MPa
Single flange rotation at Sfmax	Thetaamax	0.336 deg

Gasket Contact Area [Ag]:

= $\pi/4 (GO.D.^2 - GI.D.^2)$
 = $\pi/4 (826^2 - 794^2)$
 = 40714.9 mm²

Target Bolt Stress [Sbsel]: (Eq. O-1)

= $SgT \cdot Ag / (nb \cdot Ab)$
 = $172 \cdot 40715 / (24 \cdot 1863)$
 = 156.6 MPa

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

= min(Sbsel; Sbmax)
 = min(157 ; 508)
 = 156.6 MPa

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

= max(Sbsel; Sbmin)
 = max(157 ; 290)
 = 290.0 MPa

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

= min(Sbsel; Sfmax)
 = min(290 ; 540)
 = 290.0 MPa

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

:: Sbsel $\geq$ Sgmin-S (Ag / (nb • Abs))
 :: 290 $\geq$ 140 (40715 / (24 • 1863))
 :: 290.0 $\geq$ 127.5 MPa Check Passes

Check if Gasket Operating Stress is maintained. (Eq. O-8)

:: Sbsel $\geq$ (Sgmin-O • Ag + $\pi/4 \cdot Pmax \cdot GI.D.^2$) / ($\phi G \cdot Abs \cdot nb$)
 :: 290 $\geq$ (97 • 40715 + $\pi/4 \cdot 9.4 \cdot 794^2$) / (0.7 • 1863 • 24)
 :: 290.0 $\geq$ 275.0 MPa Check Passes



Check if Gasket Maximum Stress is achieved. (Eq. O-9)

:: Sbsel $\leq$ Sgmax(Ag / (nb • Abs))
 :: 290 $\leq$ 380 (40715/(24 • 1863))
 :: 290.0 $\leq$ 346.1 MPa Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

:: Sbsel $\leq$ Sfmax(thetagmax / thetafmax)
 :: 290 $\leq$ 540 (1/0.34)
 :: 290.0 $\leq$ 1606.9 MPa Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

= Sbsel • K • Abs • ϕ b
 = 290 • 0.2 • 1863 • 56
 = 6052070.0 N-mm

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 MPa
 Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since Sb $\geq$ SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 = 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
 = 52.14•739379008/1•1•0.33/(3.02•202713•29²•149•0.3)
 = 0.546 (should be $\leq$ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 = 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
 = 52.14•601610048/1•1•0.33/(3.02•199226•29²•149•0.3)
 = 0.452 (should be $\leq$ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 = 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
 = 52.14•739379008/1•1•0.33/(3.63•202713•29²•149•0.3)
 = 0.454 (should be $\leq$ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 = 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
 = 52.14•601610048/1•1•0.33/(3.63•199226•29²•149•0.3)
 = 0.376 (should be $\leq$ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.87, Temperature Reduction per Fig. UCS 66.1 = 7 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -2 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	108.00	207.00	130.78	207.00
Radial Flange	18.49	138.00	22.38	138.00
Tangential Flange	83.30	138.00	100.86	138.00
Maximum Average	95.65	138.00	115.82	138.00
Bolting	142.72	172.00	28.03	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 157.632 mm

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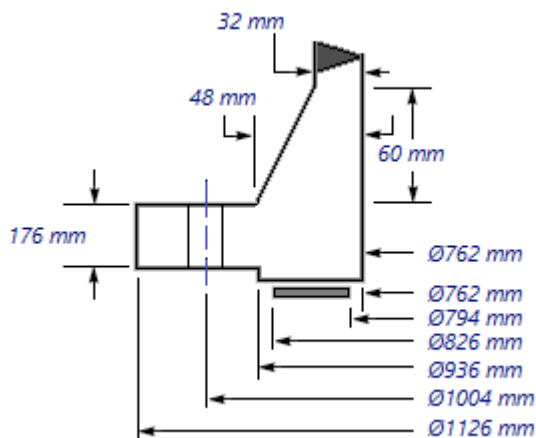


Flange Input Data Values

Description: FLANGE-4 :

Flange-4

Flange Type	Integral Weld Neck		
Design Pressure	P	9.40	MPa
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	762.002	mm
Flange Outside Diameter	A	1126.000	mm
Flange Thickness	t	176.0000	mm
Thickness of Hub at Small End	go	32.0000	mm
Thickness of Hub at Large End	gl	48.0000	mm
Length of Hub	h	60.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	1004.000	mm
Nominal Bolt Diameter	a	56.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	936.000	mm
Flange Face Inside Diameter	Fid	762.002	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	826.000	mm
Gasket Inside Diameter	Gi	794.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		68913.40	N
Bending Moment on Flange		79887400.00	N-mm





Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (9.4 \cdot (762/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 9.4) + Ca$$

$$= 30.2804 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 29) / (384 + 0.6 \cdot 29)$$

$$= 9.970 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 32) / (381 + 0.6 \cdot 32)$$

$$= 11.034 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	768.002 mm
Corroded Large Hub,	g1Cor = g1-ci	45.000 mm
Corroded Small Hub,	g0Cor = go-ci	29.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	72.999 mm
Gasket Contact Width,	N = (Go - Gi)/2	16.000 mm
Basic Gasket Width,	bo = N/2	8.000 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	7.127 mm
Gasket Reaction Diameter,	G = Go - 2 * b	811.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 812^2 \cdot 9.4$$

$$= 4865799.500 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 812 \cdot 3 \cdot 9.4$$

$$= 1025361.688 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 768^2 \cdot 9.4$$

$$= 4355515.000 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 4865799 - 4355515$$

$$= 510284.281 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(4865799 + 1025361 + 0, 0)$$

$$= 5891161.500 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.9 \cdot 7.13 \cdot \pi \cdot 812 + 0 \cdot 0 \cdot 0$$

$$= 1253136.250 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((5891161 + 68913 + 4 \cdot 79887400/812) / 172, 1253136/172)$$

$$= 36943.184 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 56 + 6 \cdot 176/(3 + 0.5)$$

$$= 413.714 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1004 \cdot \sin(\pi/24)$$



= 131.048 mm

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{\frac{Bs}{(2a + t)}}, 1)$$

$$= \max(\sqrt{\frac{131}{(2 \cdot 56 + 176)}}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	36943.184	44705.398	
Radial Distance between Hub and Bolts:	71.500	72.999	
Radial Distance between Bolts and Edge:	56.500	61.000	
Circ. Spacing between the Bolts:	124.700	131.048	413.714

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 44705 \cdot 172 / (68.9 \cdot \pi(826 + 794))$$

$$= 21.912 \text{ mm [Note: Exceeds actual gasket width, 16.000]}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 44705$$

$$= 7688536.50 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 5891161 - 4865799$$

$$= 1025361.88 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1004 - 812) / 2$$

$$= 96.1274 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1004 - 768) / 2 + 96.1)$$

$$= 107.0632 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1004 - 768 - 45) / 2$$

$$= 95.4990 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 179, Bb = 176, Cc = 60 \text{ and } Ddg = 37 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 179 \cdot 176^3 \cdot (1/3 - 0.210 \cdot (176/179) \cdot (1 - 1/12 \cdot (176/179)^4))$$

$$= 139483744.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 60 \cdot 37^3 \cdot (1/3 - 0.105 \cdot (37/60) \cdot (1 - 1/192 \cdot (37/60)^4))$$

$$= 816420.562 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 8512 + 49.8$$



$$= 140300160.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 \left(\frac{1}{3} - 0.21 \left(\frac{t}{AR} \right) \left(1 - \frac{1}{12} \left(\frac{t}{AR} \right)^4 \right) \right)$$

$$= 179 \cdot 176^3 \left(\frac{1}{3} - 0.21 \left(\frac{176}{179} \right) \left(1 - \frac{1}{12} \left(\frac{176}{179} \right)^4 \right) \right)$$

$$= 139483744.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2)$$

$$= \max(140300160, 139483744)$$

$$= 140300160.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 3.63 \cdot 29^2 \cdot 149 \cdot 768 / 0.33$$

$$= 93579992.000 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	4355515.	95.4990	1.0000	416116000.
Face Pressure, Mt	510284.	107.0632	1.0000	54654832.
Gasket Load, Mg	1025362.	96.1274	1.0000	98605376.
Gasket Seating, Matm	7688536.	96.1274	1.0000	739379008.
Moment Load, Moe	...	...	...	30405280.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me \left[\frac{1}{(0.384 \cdot IP + I)} \right] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$$

$$= 4 \cdot 79887400 [93579992 / (0.384 \cdot 140300160 + 93579992)] \cdot (95.5 / (1004 - 2 \cdot 95.5)) +$$

$$68913 \cdot 95.5$$

$$= 30405280.000 \text{ N-mm}$$

Total Moment for Operation, Mo 599781440. N-mm
 Total Moment for Gasket seating, Mg 739379008. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 149.238 mm
 Hub Ratio, h/h0 = HL / H0 0.402
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.552

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.860
 Factor V per 4.16.5 0.327
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1126/768$$

$$= 1.466$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.47^2 \cdot \log_{10}(1.47))) / (1.47^2 - 1)) / (1.47 - 1)$$

$$= 5.245$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.47^2 (1 + 8.55246 \cdot \log_{10}(1.47)) - 1) / ((1.0472 + 1.9448 \cdot 1.47^2) \cdot (1.47 - 1))$$

$$= 1.725$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.47^2 (1 + 8.55246 \cdot \log_{10}(1.47)) - 1) / (1.36136 (1.47^2 - 1) \cdot (1.47 - 1))$$

$$= 5.764$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.47^2 + 1) / (1.47^2 - 1)$$

$$= 2.740$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned} &= U \cdot g_0^2 \cdot h_0 / V \\ &= 5.76 \cdot 29^2 \cdot 149 / 0.33 \\ &= 2213042.750 \text{ mm}^3 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned} &= F / h_0 \\ &= 0.86 / 149 \\ &= 0.006 \text{ 1/mm} \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned} &= (t \cdot e + 1) / T + t^3/d \\ &= (176 \cdot 0.0058 + 1) / 1.73 + 176^3 / 2213042 \\ &= 3.631 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned} &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\ &= (1 \cdot 599781440 / 768) / (3.63 \cdot 45^2) \\ &= 106.18 \text{ MPa} \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned} &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\ &= (1 \cdot 739379008 / 768) / (3.63 \cdot 45^2) \\ &= 130.90 \text{ MPa} \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned} &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\ &= (2.35 \cdot 599781440 / 768) / (3.63 \cdot 176^2) \\ &= 16.33 \text{ MPa} \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned} &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\ &= (2.35 \cdot 739379008 / 768) / (3.63 \cdot 176^2) \\ &= 20.13 \text{ MPa} \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned} &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\ &= (5.24 \cdot 599781440 / (176^2 \cdot 768)) - 2.74 \cdot 16.3 \\ &= 87.45 \text{ MPa} \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned} &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\ &= (5.24 \cdot 739379008 / (176^2 \cdot 768)) - 2.74 \cdot 20.1 \\ &= 107.81 \text{ MPa} \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned} &= (SHo + \max(SRo, STo)) / 2 \\ &= (106 + \max(16.3, 87.5)) / 2 \\ &= 96.82 \text{ MPa} \end{aligned}$$

Average Flange Stress, Seating [SAa]:

$$\begin{aligned} &= (SHa + \max(SRa, STa)) / 2 \\ &= (131 + \max(20.1, 108)) / 2 \\ &= 119.35 \text{ MPa} \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned} &= (Wm_1 + Fa + 4Me/G) / Ab \\ &= 6353572 / 44705 \\ &= 142.13 \text{ MPa} \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned} &= (Wm_2 / Ab) \\ &= (1253136 / 44705) \\ &= 28.03 \text{ MPa} \end{aligned}$$

Flange Stress Analysis Results: MPa



	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	106.18	207.00	130.90	207.00
Radial Flange	16.33	138.00	20.13	138.00
Tangential Flange	87.45	138.00	107.81	138.00
Maximum Average	96.82	138.00	119.35	138.00
Bolting	142.13	172.00	28.03	172.00

Minimum Required Flange Thickness 161.468 mm

Estimated M.A.W.P. (Operating) 9.97 MPa

Estimated Finished Weight of Flange at given Thk. 783.1 kgm

Estimated Unfinished Weight of Forging at given Thk 987.3 kgm

Flange Stress results per WRC-538 at computed S_{fmax} (MPa):

		Actual	Operating Allowed	stress ratio	Actual	Gasket Seating Allowed	stress ratio
Longitudinal Hub	Sh	411.26	499.99	0.823	411.26	499.99	0.823
Tangential Flange	St	338.72	374.99	0.903	338.72	374.99	0.903
Tangential Flange	Sto	148.50	250.00	0.594	148.50	250.00	0.594
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	401.97	749.99	0.536	401.97	749.99	0.536
Combined	Sh + Sto	559.76	749.99	0.746	559.76	749.99	0.746
Radial Flange	Sr	63.25	499.99	0.126	63.25	499.99	0.126
Maximum Stress Ratio				1.000			1.000

Coefficients for f and f_2

	f	f_2	f_3
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g_1}{g_0} \right) + e \left(\frac{h}{h_0} \right) + g \left(\frac{g_1}{g_0} \right)^2 + i \left(\frac{h}{h_0} \right)^2 + k \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}{\left[1 + b \left(\frac{g_1}{g_0} \right) + d \left(\frac{h}{h_0} \right) + f \left(\frac{g_1}{g_0} \right)^2 + h \left(\frac{h}{h_0} \right)^2 + j \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}$$

Hub Stress Correction Factor [f]:



$$= \max(1; F1num / F1den)$$

$$= \max(1; 1.37/1.44)$$

$$= 1.0000$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; fn2) \text{ if } h/ho < 0.35, \text{ else } \max(0.25; fn3)$$

$$= \max(0.25; 0.53/1.45) \text{ if } h/ho < 0.35, \text{ else } \max(0.25; 0.27/0.76)$$

$$= 0.3611$$

In this case, the hub ratio h/ho , 0.4020 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sf_{max} , until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 MPa
Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	540.4 MPa
Target assembly gasket stress	SgT	172.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	1862.7250 mm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	230.1 MPa
Single flange rotation at Sf_{max}	Thetaamax	0.336 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (826^2 - 794^2)$$

$$= 40714.9 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 172 \cdot 40715 / (24 \cdot 1863)$$

$$= 156.6 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(157; 508)$$

$$= 156.6 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(157; 290)$$

$$= 290.0 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(290; 540)$$

$$= 290.0 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$\therefore Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$\therefore 290 \geq 140 (40715 / (24 \cdot 1863))$$

$$\therefore 290.0 \geq 127.5 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$\therefore Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$\therefore 290 \geq (97 \cdot 40715 + \pi/4 \cdot 9.4 \cdot 794^2) / (0.7 \cdot 1863 \cdot 24)$$

$$\therefore 290.0 \geq 275.0 \text{ MPa} \text{ Check Passes}$$



Check if Gasket Maximum Stress is achieved. (Eq. O-9)

:: Sbsel $\leq$ Sgmax(Ag / (nb • Abs))
 :: 290 $\leq$ 380 (40715/(24 • 1863))
 :: 290.0 $\leq$ 346.1 MPa Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

:: Sbsel $\leq$ Sfmax(thetagmax / thetafmax)
 :: 290 $\leq$ 540 (1/0.34)
 :: 290.0 $\leq$ 1606.9 MPa Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

= Sbsel • K • Abs • ϕ b
 = 290 • 0.2 • 1863 • 56
 = 6052070.0 N-mm

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 MPa
 Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since Sb $\geq$ SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 = 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
 = 52.14•739379008/1•1•0.33/(3.02•202713•29²•149•0.3)
 = 0.546 (should be $\leq$ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 = 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
 = 52.14•599781440/1•1•0.33/(3.02•199226•29²•149•0.3)
 = 0.451 (should be $\leq$ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 = 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
 = 52.14•739379008/1•1•0.33/(3.63•202713•29²•149•0.3)
 = 0.454 (should be $\leq$ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 = 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
 = 52.14•599781440/1•1•0.33/(3.63•199226•29²•149•0.3)
 = 0.375 (should be $\leq$ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.87, Temperature Reduction per Fig. UCS 66.1 = 7 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -2 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	107.67	207.00	130.78	207.00
Radial Flange	18.43	138.00	22.38	138.00
Tangential Flange	83.04	138.00	100.86	138.00
Maximum Average	95.35	138.00	115.82	138.00
Bolting	142.13	172.00	28.03	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 157.632 mm

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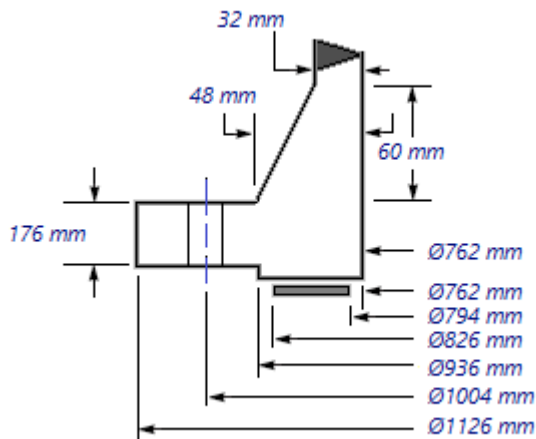


Flange Input Data Values

Description: FLANGE-5 :

Flange-5

Flange Type	Integral Weld Neck		
Design Pressure	P	9.40	MPa
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	762.002	mm
Flange Outside Diameter	A	1126.000	mm
Flange Thickness	t	176.0000	mm
Thickness of Hub at Small End	go	32.0000	mm
Thickness of Hub at Large End	gl	48.0000	mm
Length of Hub	h	60.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	1004.000	mm
Nominal Bolt Diameter	a	56.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	936.000	mm
Flange Face Inside Diameter	Fid	762.002	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	826.000	mm
Gasket Inside Diameter	Gi	794.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		37417.30	N
Bending Moment on Flange		20375300.00	N-mm



Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (9.4 \cdot (762/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 9.4) + Ca$$

$$= 30.2804 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 29) / (384 + 0.6 \cdot 29)$$

$$= 9.970 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 32) / (381 + 0.6 \cdot 32)$$

$$= 11.034 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	768.002 mm
Corroded Large Hub,	g1Cor = g1-ci	45.000 mm
Corroded Small Hub,	g0Cor = go-ci	29.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	72.999 mm
Gasket Contact Width,	N = (Go - Gi)/2	16.000 mm
Basic Gasket Width,	bo = N/2	8.000 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	7.127 mm
Gasket Reaction Diameter,	G = Go - 2 * b	811.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 812^2 \cdot 9.4$$

$$= 4865799.500 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 812 \cdot 3 \cdot 9.4$$

$$= 1025361.688 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 768^2 \cdot 9.4$$

$$= 4355515.000 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 4865799 - 4355515$$

$$= 510284.281 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(4865799 + 1025361 + 0, 0)$$

$$= 5891161.500 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.9 \cdot 7.13 \cdot \pi \cdot 812 + 0 \cdot 0 \cdot 0$$

$$= 1253136.250 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((5891161 + 37417 + 4 \cdot 20375300/812) / 172, 1253136/172)$$

$$= 35055.590 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 56 + 6 \cdot 176/(3 + 0.5)$$

$$= 413.714 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1004 \cdot \sin(\pi/24)$$



= 131.048 mm

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max\left(\sqrt{\frac{Bs}{(2a + t)}}, 1\right)$$

$$= \max\left(\sqrt{\frac{131}{(2 \cdot 56 + 176)}}, 1\right)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	35055.590	44705.398	
Radial Distance between Hub and Bolts:	71.500	72.999	
Radial Distance between Bolts and Edge:	56.500	61.000	
Circ. Spacing between the Bolts:	124.700	131.048	413.714

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 44705 \cdot 172 / (68.9 \cdot \pi(826 + 794))$$

$$= 21.912 \text{ mm [Note: Exceeds actual gasket width, 16.000]}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 44705$$

$$= 7688536.50 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 5891161 - 4865799$$

$$= 1025361.88 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1004 - 812) / 2$$

$$= 96.1274 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1004 - 768) / 2 + 96.1)$$

$$= 107.0632 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1004 - 768 - 45) / 2$$

$$= 95.4990 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 179, Bb = 176, Cc = 60 \text{ and } Ddg = 37 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 179 \cdot 176^3 \cdot (1/3 - 0.210 \cdot (176/179) \cdot (1 - 1/12 \cdot (176/179)^4))$$

$$= 139483744.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 60 \cdot 37^3 \cdot (1/3 - 0.105 \cdot (37/60) \cdot (1 - 1/192 \cdot (37/60)^4))$$

$$= 816420.562 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 8512 + 49.8$$



$$= 140300160.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 \left(\frac{1}{3} - 0.21 \left(\frac{t}{AR} \right) \left(1 - \frac{1}{12} \left(\frac{t}{AR} \right)^4 \right) \right)$$

$$= 179 \cdot 176^3 \left(\frac{1}{3} - 0.21 \left(\frac{176}{179} \right) \left(1 - \frac{1}{12} \left(\frac{176}{179} \right)^4 \right) \right)$$

$$= 139483744.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2)$$

$$= \max(140300160, 139483744)$$

$$= 140300160.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 3.63 \cdot 29^2 \cdot 149 \cdot 768 / 0.33$$

$$= 93579992.000 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	4355515.	95.4990	1.0000	416116000.
Face Pressure, Mt	510284.	107.0632	1.0000	54654832.
Gasket Load, Mg	1025362.	96.1274	1.0000	98605376.
Gasket Seating, Matm	7688536.	96.1274	1.0000	739379008.
Moment Load, Moe	...	...	...	9650430.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me \left[\frac{1}{(0.384 \cdot IP + I)} \right] \cdot (hD / (C - 2 \cdot hD)) + FA \cdot hD$$

$$= 4 \cdot 20375300 [93579992 / (0.384 \cdot 140300160 + 93579992)] \cdot (95.5 / (1004 - 2 \cdot 95.5)) +$$

$$37417 \cdot 95.5$$

$$= 9650430.000 \text{ N-mm}$$

Total Moment for Operation, Mo 579026624. N-mm
 Total Moment for Gasket seating, Mg 739379008. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 149.238 mm
 Hub Ratio, h/h0 = HL / H0 0.402
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.552

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.860
 Factor V per 4.16.5 0.327
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1126/768$$

$$= 1.466$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.47^2 \cdot \log_{10}(1.47))) / (1.47^2 - 1)) / (1.47 - 1)$$

$$= 5.245$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.47^2 (1 + 8.55246 \cdot \log_{10}(1.47)) - 1) / ((1.0472 + 1.9448 \cdot 1.47^2) \cdot (1.47 - 1))$$

$$= 1.725$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.47^2 (1 + 8.55246 \cdot \log_{10}(1.47)) - 1) / (1.36136 (1.47^2 - 1) \cdot (1.47 - 1))$$

$$= 5.764$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.47^2 + 1) / (1.47^2 - 1)$$

$$= 2.740$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned} &= U \cdot g_0^2 \cdot h_0 / V \\ &= 5.76 \cdot 29^2 \cdot 149 / 0.33 \\ &= 2213042.750 \text{ mm}^3 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned} &= F / h_0 \\ &= 0.86 / 149 \\ &= 0.006 \text{ 1/mm} \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned} &= (t \cdot e + 1) / T + t^3 / d \\ &= (176 \cdot 0.0058 + 1) / 1.73 + 176^3 / 2213042 \\ &= 3.631 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned} &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\ &= (1 \cdot 579026560 / 768) / (3.63 \cdot 45^2) \\ &= 102.51 \text{ MPa} \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned} &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\ &= (1 \cdot 739379008 / 768) / (3.63 \cdot 45^2) \\ &= 130.90 \text{ MPa} \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned} &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\ &= (2.35 \cdot 579026560 / 768) / (3.63 \cdot 176^2) \\ &= 15.76 \text{ MPa} \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned} &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\ &= (2.35 \cdot 739379008 / 768) / (3.63 \cdot 176^2) \\ &= 20.13 \text{ MPa} \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned} &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\ &= (5.24 \cdot 579026560 / (176^2 \cdot 768)) - 2.74 \cdot 15.8 \\ &= 84.43 \text{ MPa} \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned} &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\ &= (5.24 \cdot 739379008 / (176^2 \cdot 768)) - 2.74 \cdot 20.1 \\ &= 107.81 \text{ MPa} \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned} &= (SHo + \max(SRo, STo)) / 2 \\ &= (103 + \max(15.8, 84.4)) / 2 \\ &= 93.47 \text{ MPa} \end{aligned}$$

Average Flange Stress, Seating [SAa]:

$$\begin{aligned} &= (SHa + \max(SRa, STa)) / 2 \\ &= (131 + \max(20.1, 108)) / 2 \\ &= 119.35 \text{ MPa} \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned} &= (Wm_1 + Fa + 4Me/G) / Ab \\ &= 6028940 / 44705 \\ &= 134.87 \text{ MPa} \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned} &= (Wm_2 / Ab) \\ &= (1253136 / 44705) \\ &= 28.03 \text{ MPa} \end{aligned}$$

Flange Stress Analysis Results: MPa



	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	102.51	207.00	130.90	207.00
Radial Flange	15.76	138.00	20.13	138.00
Tangential Flange	84.43	138.00	107.81	138.00
Maximum Average	93.47	138.00	119.35	138.00
Bolting	134.87	172.00	28.03	172.00

Minimum Required Flange Thickness 161.468 mm

Estimated M.A.W.P. (Operating) 9.97 MPa

Estimated Finished Weight of Flange at given Thk. 783.1 kgm

Estimated Unfinished Weight of Forging at given Thk 987.3 kgm

Flange Stress results per WRC-538 at computed Sfmax (MPa):

		Actual	Operating Allowed	stress ratio	Actual	Gasket Seating Allowed	stress ratio
Longitudinal Hub	Sh	411.26	499.99	0.823	411.26	499.99	0.823
Tangential Flange	St	338.72	374.99	0.903	338.72	374.99	0.903
Tangential Flange	Sto	148.50	250.00	0.594	148.50	250.00	0.594
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	401.97	749.99	0.536	401.97	749.99	0.536
Combined	Sh + Sto	559.76	749.99	0.746	559.76	749.99	0.746
Radial Flange	Sr	63.25	499.99	0.126	63.25	499.99	0.126
Maximum Stress Ratio				1.000			1.000

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:



$$= \max(1; F1num / F1den)$$

$$= \max(1; 1.37/1.44)$$

$$= 1.0000$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; fn2) \text{ if } h/ho < 0.35, \text{ else } \max(0.25; fn3)$$

$$= \max(0.25; 0.53/1.45) \text{ if } h/ho < 0.35, \text{ else } \max(0.25; 0.27/0.76)$$

$$= 0.3611$$

In this case, the hub ratio h/ho , 0.4020 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, $Sfmax$, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 MPa
Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	540.4 MPa
Target assembly gasket stress	SgT	172.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	1862.7250 mm ²
Max. permissible single flange rotation for gasket	Thetamax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	230.1 MPa
Single flange rotation at $Sfmax$	Thetafmax	0.336 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (826^2 - 794^2)$$

$$= 40714.9 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 172 \cdot 40715 / (24 \cdot 1863)$$

$$= 156.6 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(157; 508)$$

$$= 156.6 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(157; 290)$$

$$= 290.0 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(290; 540)$$

$$= 290.0 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$\therefore Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$\therefore 290 \geq 140 (40715 / (24 \cdot 1863))$$

$$\therefore 290.0 \geq 127.5 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$\therefore Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$\therefore 290 \geq (97 \cdot 40715 + \pi/4 \cdot 9.4 \cdot 794^2) / (0.7 \cdot 1863 \cdot 24)$$

$$\therefore 290.0 \geq 275.0 \text{ MPa} \text{ Check Passes}$$



Check if Gasket Maximum Stress is achieved. (Eq. O-9)

:: Sbsel $\leq$ Sgmax(Ag / (nb • Abs))
 :: 290 $\leq$ 380 (40715/(24 • 1863))
 :: 290.0 $\leq$ 346.1 MPa Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

:: Sbsel $\leq$ Sfmax(thetagmax / thetafmax)
 :: 290 $\leq$ 540 (1/0.34)
 :: 290.0 $\leq$ 1606.9 MPa Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

= Sbsel • K • Abs • ϕ b
 = 290 • 0.2 • 1863 • 56
 = 6052070.0 N-mm

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 MPa
 Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since Sb $\geq$ SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 = 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
 = 52.14•739379008/1•1•0.33/(3.02•202713•29²•149•0.3)
 = 0.546 (should be $\leq$ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 = 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
 = 52.14•579026560/1•1•0.33/(3.02•199226•29²•149•0.3)
 = 0.435 (should be $\leq$ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 = 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
 = 52.14•739379008/1•1•0.33/(3.63•202713•29²•149•0.3)
 = 0.454 (should be $\leq$ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 = 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
 = 52.14•579026560/1•1•0.33/(3.63•199226•29²•149•0.3)
 = 0.362 (should be $\leq$ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.87, Temperature Reduction per Fig. UCS 66.1 = 7 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -2 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	103.89	207.00	130.78	207.00
Radial Flange	17.78	138.00	22.38	138.00
Tangential Flange	80.13	138.00	100.86	138.00
Maximum Average	92.01	138.00	115.82	138.00
Bolting	134.87	172.00	28.03	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 157.632 mm

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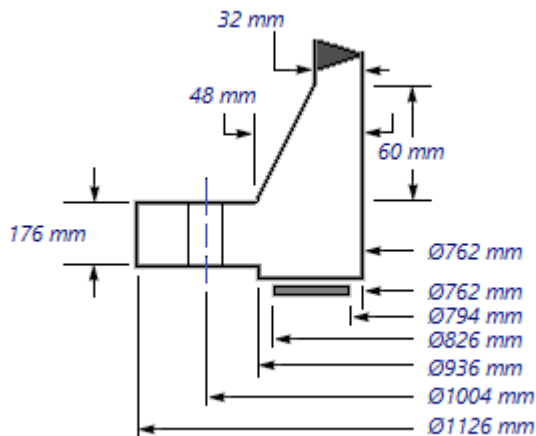


Flange Input Data Values

Description: FLANGE-6 :

Flange-6

Flange Type	Integral Weld Neck		
Design Pressure	P	9.40	MPa
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	762.002	mm
Flange Outside Diameter	A	1126.000	mm
Flange Thickness	t	176.0000	mm
Thickness of Hub at Small End	go	32.0000	mm
Thickness of Hub at Large End	gl	48.0000	mm
Length of Hub	h	60.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	1004.000	mm
Nominal Bolt Diameter	a	56.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	936.000	mm
Flange Face Inside Diameter	Fid	762.002	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	826.000	mm
Gasket Inside Diameter	Gi	794.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		30965.80	N
Bending Moment on Flange		18257800.00	N-mm





Hub Small End Required Thickness due to Internal Pressure:

$$\begin{aligned} &= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= (9.4 \cdot (762/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 9.4) + Ca \\ &= 30.2804 \text{ mm} \end{aligned}$$

Small End Hub MAWP:

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\ &= (138 \cdot 1 \cdot 29) / (384 + 0.6 \cdot 29) \\ &= 9.970 \text{ MPa} \end{aligned}$$

Hub Small End Hub MAPnc:

$$\begin{aligned} &= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\ &= (138 \cdot 1 \cdot 32) / (381 + 0.6 \cdot 32) \\ &= 11.034 \text{ MPa} \end{aligned}$$

Corroded Flange ID,	Bcor = B+2*ci	768.002 mm
Corroded Large Hub,	g1Cor = g1-ci	45.000 mm
Corroded Small Hub,	g0Cor = go-ci	29.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	72.999 mm
Gasket Contact Width,	N = (Go - Gi)/2	16.000 mm
Basic Gasket Width,	bo = N/2	8.000 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	7.127 mm
Gasket Reaction Diameter,	G = Go - 2 * b	811.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$\begin{aligned} &= \pi/4 \cdot G^2 \cdot P \\ &= \pi/4 \cdot 812^2 \cdot 9.4 \\ &= 4865799.500 \text{ N} \end{aligned}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$\begin{aligned} &= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P \\ &= 2 \cdot 7.13 \cdot \pi \cdot 812 \cdot 3 \cdot 9.4 \\ &= 1025361.688 \text{ N} \end{aligned}$$

Hydrostatic End Load at Flange ID [Hd]:

$$\begin{aligned} &= \pi/4 \cdot Bcor^2 \cdot P \\ &= \pi/4 \cdot 768^2 \cdot 9.4 \\ &= 4355515.000 \text{ N} \end{aligned}$$

Pressure Force on Flange Face [Ht]:

$$\begin{aligned} &= H - Hd \\ &= 4865799 - 4355515 \\ &= 510284.281 \text{ N} \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned} &= \max(Wo,1 + Wo,2 + H'p, 0) \\ &= \max(4865799 + 1025361 + 0, 0) \\ &= 5891161.500 \text{ N} \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned} &= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp \\ &= 68.9 \cdot 7.13 \cdot \pi \cdot 812 + 0 \cdot 0 \cdot 0 \\ &= 1253136.250 \text{ N} \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned} &= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa) \\ &= \max((5891161 + 30966 + 4 \cdot 18257800/812) / 172, 1253136/172) \\ &= 34957.434 \text{ mm}^2 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$\begin{aligned} &= 2a + 6t/(m + 0.5) \\ &= 2 \cdot 56 + 6 \cdot 176/(3 + 0.5) \\ &= 413.714 \text{ mm} \end{aligned}$$

Actual Circumferential Bolt Spacing [Bs]:

$$\begin{aligned} &= C \cdot \sin(\pi / n) \\ &= 1004 \cdot \sin(\pi/24) \end{aligned}$$

= 131.048 mm

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{\frac{Bs}{(2a + t)}}, 1)$$

$$= \max(\sqrt{\frac{131}{(2 \cdot 56 + 176)}}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	34957.434	44705.398	
Radial Distance between Hub and Bolts:	71.500	72.999	
Radial Distance between Bolts and Edge:	56.500	61.000	
Circ. Spacing between the Bolts:	124.700	131.048	413.714

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 44705 \cdot 172 / (68.9 \cdot \pi(826 + 794))$$

$$= 21.912 \text{ mm [Note: Exceeds actual gasket width, 16.000]}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 44705$$

$$= 7688536.50 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 5891161 - 4865799$$

$$= 1025361.88 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1004 - 812) / 2$$

$$= 96.1274 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1004 - 768) / 2 + 96.1)$$

$$= 107.0632 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1004 - 768 - 45) / 2$$

$$= 95.4990 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 179, Bb = 176, Cc = 60 \text{ and } Ddg = 37 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 179 \cdot 176^3 \cdot (1/3 - 0.210 \cdot (176/179) \cdot (1 - 1/12 \cdot (176/179)^4))$$

$$= 139483744.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 60 \cdot 37^3 \cdot (1/3 - 0.105 \cdot (37/60) \cdot (1 - 1/192 \cdot (37/60)^4))$$

$$= 816420.562 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 8512 + 49.8$$



$$= 140300160.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 \left(\frac{1}{3} - 0.21 \left(\frac{t}{AR} \right) \left(1 - \frac{1}{12} \left(\frac{t}{AR} \right)^4 \right) \right)$$

$$= 179 \cdot 176^3 \left(\frac{1}{3} - 0.21 \left(\frac{176}{179} \right) \left(1 - \frac{1}{12} \left(\frac{176}{179} \right)^4 \right) \right)$$

$$= 139483744.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2)$$

$$= \max(140300160, 139483744)$$

$$= 140300160.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 3.63 \cdot 29^2 \cdot 149 \cdot 768 / 0.33$$

$$= 93579992.000 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	4355515.	95.4990	1.0000	416116000.
Face Pressure, Mt	510284.	107.0632	1.0000	54654832.
Gasket Load, Mg	1025362.	96.1274	1.0000	98605376.
Gasket Seating, Matm	7688536.	96.1274	1.0000	739379008.
Moment Load, Moe	...	...	...	8402656.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me \left[\frac{1}{(0.384 \cdot IP + I)} \right] \cdot \left(\frac{hD}{(C-2 \cdot hD)} \right) + FA \cdot hD$$

$$= 4 \cdot 18257800 \left[\frac{93579992}{(0.384 \cdot 140300160 + 93579992)} \right] \cdot \left(\frac{95.5}{(1004 - 2 \cdot 95.5)} \right) +$$

$$30966 \cdot 95.5$$

$$= 8402656.000 \text{ N-mm}$$

Total Moment for Operation, Mo 577778880. N-mm
 Total Moment for Gasket seating, Mg 739379008. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 149.238 mm
 Hub Ratio, h/h0 = HL / H0 0.402
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.552

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.860
 Factor V per 4.16.5 0.327
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1126/768$$

$$= 1.466$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 \cdot K^2 \cdot \log_{10}(K))) / (K^2 - 1) / (K - 1)$$

$$= (0.66845 + (5.7169 \cdot 1.47^2 \cdot \log_{10}(1.47))) / (1.47^2 - 1) / (1.47 - 1)$$

$$= 5.245$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.47^2(1 + 8.55246 \cdot \log_{10}(1.47)) - 1) / ((1.0472 + 1.9448 \cdot 1.47^2) \cdot (1.47 - 1))$$

$$= 1.725$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.47^2(1 + 8.55246 \cdot \log_{10}(1.47)) - 1) / (1.36136(1.47^2 - 1) \cdot (1.47 - 1))$$

$$= 5.764$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.47^2 + 1) / (1.47^2 - 1)$$

$$= 2.740$$



Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned} &= U \cdot g_0^2 \cdot h_0 / V \\ &= 5.76 \cdot 29^2 \cdot 149 / 0.33 \\ &= 2213042.750 \text{ mm}^3 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned} &= F / h_0 \\ &= 0.86 / 149 \\ &= 0.006 \text{ 1/mm} \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned} &= (t \cdot e + 1) / T + t^3/d \\ &= (176 \cdot 0.0058 + 1) / 1.73 + 176^3 / 2213042 \\ &= 3.631 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned} &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\ &= (1 \cdot 577778880 / 768) / (3.63 \cdot 45^2) \\ &= 102.29 \text{ MPa} \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned} &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\ &= (1 \cdot 739379008 / 768) / (3.63 \cdot 45^2) \\ &= 130.90 \text{ MPa} \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned} &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\ &= (2.35 \cdot 577778880 / 768) / (3.63 \cdot 176^2) \\ &= 15.73 \text{ MPa} \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned} &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\ &= (2.35 \cdot 739379008 / 768) / (3.63 \cdot 176^2) \\ &= 20.13 \text{ MPa} \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned} &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\ &= (5.24 \cdot 577778880 / (176^2 \cdot 768)) - 2.74 \cdot 15.7 \\ &= 84.25 \text{ MPa} \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned} &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\ &= (5.24 \cdot 739379008 / (176^2 \cdot 768)) - 2.74 \cdot 20.1 \\ &= 107.81 \text{ MPa} \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned} &= (SHo + \max(SRo, STo)) / 2 \\ &= (102 + \max(15.7, 84.2)) / 2 \\ &= 93.27 \text{ MPa} \end{aligned}$$

Average Flange Stress, Seating [SAa]:

$$\begin{aligned} &= (SHa + \max(SRa, STa)) / 2 \\ &= (131 + \max(20.1, 108)) / 2 \\ &= 119.35 \text{ MPa} \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned} &= (Wm_1 + Fa + 4Me/G) / Ab \\ &= 6012059 / 44705 \\ &= 134.49 \text{ MPa} \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned} &= (Wm_2 / Ab) \\ &= (1253136 / 44705) \\ &= 28.03 \text{ MPa} \end{aligned}$$

Flange Stress Analysis Results: MPa



	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	102.29	207.00	130.90	207.00
Radial Flange	15.73	138.00	20.13	138.00
Tangential Flange	84.25	138.00	107.81	138.00
Maximum Average	93.27	138.00	119.35	138.00
Bolting	134.49	172.00	28.03	172.00

Minimum Required Flange Thickness 161.468 mm

Estimated M.A.W.P. (Operating) 9.97 MPa

Estimated Finished Weight of Flange at given Thk. 783.1 kgm

Estimated Unfinished Weight of Forging at given Thk 987.3 kgm

Flange Stress results per WRC-538 at computed S_{fmax} (MPa):

		Actual	Operating Allowed	stress ratio	Actual	Gasket Seating Allowed	stress ratio
Longitudinal Hub	Sh	411.26	499.99	0.823	411.26	499.99	0.823
Tangential Flange	St	338.72	374.99	0.903	338.72	374.99	0.903
Tangential Flange	Sto	148.50	250.00	0.594	148.50	250.00	0.594
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	401.97	749.99	0.536	401.97	749.99	0.536
Combined	Sh + Sto	559.76	749.99	0.746	559.76	749.99	0.746
Radial Flange	Sr	63.25	499.99	0.126	63.25	499.99	0.126
Maximum Stress Ratio				1.000			1.000

Coefficients for *f* and *f*₂

	<i>f</i>	<i>f</i> ₂	<i>f</i> ₃
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:



= max(1; F1num / F1den)
 = max(1; 1.37/1.44)
 = 1.0000

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)
 = max(0.25; 0.53/1.45) if h/ho < 0.35, else max (0.25; 0.27/0.76)
 = 0.3611

In this case, the hub ratio h/ho, 0.4020 was $\geq$ 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 MPa
Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	540.4 MPa
Target assembly gasket stress	SgT	172.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	1862.7250 mm ²
Max. permissible single flange rotation for gasket	Thetamax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	230.1 MPa
Single flange rotation at Sfmax	Thetaamax	0.336 deg

Gasket Contact Area [Ag]:

= $\pi/4 (GO.D.^2 - GI.D.^2)$
 = $\pi/4 (826^2 - 794^2)$
 = 40714.9 mm²

Target Bolt Stress [Sbsel]: (Eq. O-1)

= $SgT \cdot Ag / (nb \cdot Ab)$
 = $172 \cdot 40715 / (24 \cdot 1863)$
 = 156.6 MPa

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

= min(Sbsel; Sbmax)
 = min(157 ; 508)
 = 156.6 MPa

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

= max(Sbsel; Sbmin)
 = max(157 ; 290)
 = 290.0 MPa

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

= min(Sbsel; Sfmax)
 = min(290 ; 540)
 = 290.0 MPa

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

:: Sbsel $\geq$ Sgmin-S (Ag / (nb • Abs))
 :: 290 $\geq$ 140 (40715 / (24 • 1863))
 :: 290.0 $\geq$ 127.5 MPa Check Passes

Check if Gasket Operating Stress is maintained. (Eq. O-8)

:: Sbsel $\geq$ (Sgmin-O • Ag + $\pi/4 \cdot Pmax \cdot GI.D.^2$) / ($\phi G \cdot Abs \cdot nb$)
 :: 290 $\geq$ (97 • 40715 + $\pi/4 \cdot 9.4 \cdot 794^2$) / (0.7 • 1863 • 24)
 :: 290.0 $\geq$ 275.0 MPa Check Passes



Check if Gasket Maximum Stress is achieved. (Eq. O-9)

:: Sbsel $\leq$ Sgmax(Ag / (nb • Abs))
 :: 290 $\leq$ 380 (40715/(24 • 1863))
 :: 290.0 $\leq$ 346.1 MPa Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

:: Sbsel $\leq$ Sfmax(thetagmax / thetafmax)
 :: 290 $\leq$ 540 (1/0.34)
 :: 290.0 $\leq$ 1606.9 MPa Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

= Sbsel • K • Abs • ϕ b
 = 290 • 0.2 • 1863 • 56
 = 6052070.0 N-mm

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 MPa
 Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since Sb $\geq$ SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 = 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
 = 52.14•739379008/1•1•0.33/(3.02•202713•29²•149•0.3)
 = 0.546 (should be $\leq$ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 = 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
 = 52.14•577778880/1•1•0.33/(3.02•199226•29²•149•0.3)
 = 0.434 (should be $\leq$ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 = 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
 = 52.14•739379008/1•1•0.33/(3.63•202713•29²•149•0.3)
 = 0.454 (should be $\leq$ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 = 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
 = 52.14•577778880/1•1•0.33/(3.63•199226•29²•149•0.3)
 = 0.361 (should be $\leq$ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.87, Temperature Reduction per Fig. UCS 66.1 = 7 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -2 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	103.66	207.00	130.78	207.00
Radial Flange	17.74	138.00	22.38	138.00
Tangential Flange	79.95	138.00	100.86	138.00
Maximum Average	91.81	138.00	115.82	138.00
Bolting	134.49	172.00	28.03	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 157.632 mm

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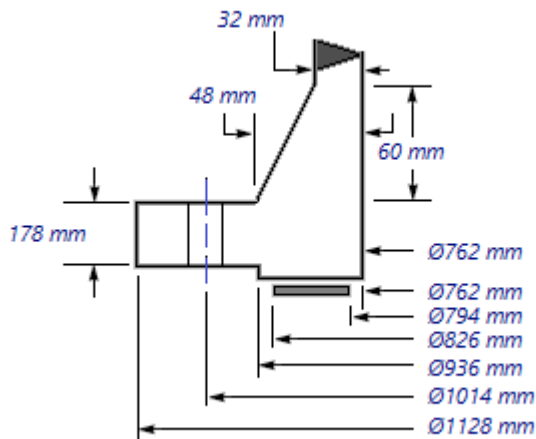


Flange Input Data Values

Description: FLANGE-7 :

Flange-7

Flange Type	Integral Weld Neck		
Design Pressure	P	9.40	MPa
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	762.002	mm
Flange Outside Diameter	A	1128.000	mm
Flange Thickness	t	178.0000	mm
Thickness of Hub at Small End	go	32.0000	mm
Thickness of Hub at Large End	gl	48.0000	mm
Length of Hub	h	60.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	1014.000	mm
Nominal Bolt Diameter	a	56.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	936.000	mm
Flange Face Inside Diameter	Fid	762.002	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	826.000	mm
Gasket Inside Diameter	Gi	794.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		9228.11	N
Bending Moment on Flange		1188450.00	N-mm



Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (9.4 \cdot (762/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 9.4) + Ca$$

$$= 30.2804 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 29) / (384 + 0.6 \cdot 29)$$

$$= 9.970 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 32) / (381 + 0.6 \cdot 32)$$

$$= 11.034 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	768.002 mm
Corroded Large Hub,	g1Cor = g1-ci	45.000 mm
Corroded Small Hub,	g0Cor = go-ci	29.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	77.999 mm
Gasket Contact Width,	N = (Go - Gi)/2	16.000 mm
Basic Gasket Width,	bo = N/2	8.000 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	7.127 mm
Gasket Reaction Diameter,	G = Go - 2 * b	811.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 812^2 \cdot 9.4$$

$$= 4865799.500 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 812 \cdot 3 \cdot 9.4$$

$$= 1025361.688 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 768^2 \cdot 9.4$$

$$= 4355515.000 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 4865799 - 4355515$$

$$= 510284.281 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(4865799 + 1025361 + 0, 0)$$

$$= 5891161.500 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot Ip$$

$$= 68.9 \cdot 7.13 \cdot \pi \cdot 812 + 0 \cdot 0$$

$$= 1253136.250 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((5891161 + 9228 + 4 \cdot 1188450/812) / 172, 1253136/172)$$

$$= 34342.164 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 56 + 6 \cdot 178/(3 + 0.5)$$

$$= 417.143 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1014 \cdot \sin(\pi/24)$$

= 132.354 mm

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{\frac{Bs}{(2a + t)}}, 1)$$

$$= \max(\sqrt{\frac{132}{(2 \cdot 56 + 178)}}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	34342.164	44705.398	
Radial Distance between Hub and Bolts:	71.500	77.999	
Radial Distance between Bolts and Edge:	56.500	57.000	
Circ. Spacing between the Bolts:	124.700	132.354	417.143

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 44705 \cdot 172 / (68.9 \cdot \pi(826 + 794))$$

$$= 21.912 \text{ mm [Note: Exceeds actual gasket width, 16.000]}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 44705$$

$$= 7688536.50 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 5891161 - 4865799$$

$$= 1025361.88 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1014 - 812) / 2$$

$$= 101.1274 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1014 - 768) / 2 + 101)$$

$$= 112.0632 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1014 - 768 - 45) / 2$$

$$= 100.4990 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 180, Bb = 178, Cc = 60 \text{ and } Ddg = 37 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 180 \cdot 178^3 \cdot (1/3 - 0.210 \cdot (178/180) \cdot (1 - 1/12 \cdot (178/180)^4))$$

$$= 144369504.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 60 \cdot 37^3 \cdot (1/3 - 0.105 \cdot (37/60) \cdot (1 - 1/192 \cdot (37/60)^4))$$

$$= 816420.562 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 8810 + 49.8$$



$$= 145185920.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 \left(\frac{1}{3} - 0.21 \left(\frac{t}{AR} \right) \left(1 - \frac{1}{12} \left(\frac{t}{AR} \right)^4 \right) \right)$$

$$= 180 \cdot 178^3 \left(\frac{1}{3} - 0.21 \left(\frac{178}{180} \right) \left(1 - \frac{1}{12} \left(\frac{178}{7.09} \right)^4 \right) \right)$$

$$= 144369504.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip, term1, Ip, term2)$$

$$= \max(145185920, 144369504)$$

$$= 145185920.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 3.73 \cdot 29^2 \cdot 149 \cdot 768 / 0.33$$

$$= 96253848.000 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	4355515.	100.4990	1.0000	437902336.
Face Pressure, Mt	510284.	112.0632	1.0000	57207280.
Gasket Load, Mg	1025362.	101.1274	1.0000	103734248.
Gasket Seating, Matm	7688536.	101.1274	1.0000	777837120.
Moment Load, Moe	...	...	...	1299901.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me \left[\frac{1}{(0.384 \cdot IP + I)} \right] \cdot \left(\frac{hD}{(C-2 \cdot hD)} \right) + FA \cdot hD$$

$$= 4 \cdot 1188450 \left[\frac{96253848}{(0.384 \cdot 145185920 + 96253848)} \right] \cdot \left(\frac{100}{(1014 - 2 \cdot 100)} \right) + 9228 \cdot 100$$

$$= 1299900.875 \text{ N-mm}$$

Total Moment for Operation, Mo 600143744. N-mm
 Total Moment for Gasket seating, Mg 777837120. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 149.238 mm
 Hub Ratio, h/h0 = HL / H0 0.402
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.552

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.860
 Factor V per 4.16.5 0.327
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1128/768$$

$$= 1.469$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 \cdot K^2 \cdot \log_{10}(K))) / (K^2 - 1) / (K - 1)$$

$$= (0.66845 + (5.7169 \cdot 1.47^2 \cdot \log_{10}(1.47))) / (1.47^2 - 1) / (1.47 - 1)$$

$$= 5.222$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.47^2(1 + 8.55246 \cdot \log_{10}(1.47)) - 1) / ((1.0472 + 1.9448 \cdot 1.47^2) \cdot (1.47 - 1))$$

$$= 1.724$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.47^2(1 + 8.55246 \cdot \log_{10}(1.47)) - 1) / (1.36136(1.47^2 - 1) \cdot (1.47 - 1))$$

$$= 5.738$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.47^2 + 1) / (1.47^2 - 1)$$

$$= 2.728$$



Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned} &= U \cdot g0^2 \cdot h0 / V \\ &= 5.74 \cdot 29^2 \cdot 149 / 0.33 \\ &= 2203228.000 \text{ mm}^3 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned} &= F / h0 \\ &= 0.86 / 149 \\ &= 0.006 \text{ 1/mm} \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned} &= (t \cdot e + 1) / T + t^3/d \\ &= (178 \cdot 0.0058 + 1) / 1.72 + 178^3 / 2203228 \\ &= 3.735 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned} &= (f \cdot Mop / Bcor) / (L \cdot g1^2) \\ &= (1 \cdot 600143744 / 768) / (3.73 \cdot 45^2) \\ &= 103.29 \text{ MPa} \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned} &= (f \cdot Matm / Bcor) / (L \cdot g1^2) \\ &= (1 \cdot 777837120 / 768) / (3.73 \cdot 45^2) \\ &= 133.88 \text{ MPa} \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned} &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\ &= (2.37 \cdot 600143744 / 768) / (3.73 \cdot 178^2) \\ &= 15.63 \text{ MPa} \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned} &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\ &= (2.37 \cdot 777837120 / 768) / (3.73 \cdot 178^2) \\ &= 20.26 \text{ MPa} \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned} &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\ &= (5.22 \cdot 600143744 / (178^2 \cdot 768)) - 2.73 \cdot 15.6 \\ &= 86.09 \text{ MPa} \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned} &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\ &= (5.22 \cdot 777837120 / (178^2 \cdot 768)) - 2.73 \cdot 20.3 \\ &= 111.59 \text{ MPa} \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned} &= (SHo + \max(SRo, STo)) / 2 \\ &= (103 + \max(15.6, 86.1)) / 2 \\ &= 94.69 \text{ MPa} \end{aligned}$$

Average Flange Stress, Seating [SAa]:

$$\begin{aligned} &= (SHa + \max(SRa, STa)) / 2 \\ &= (134 + \max(20.3, 112)) / 2 \\ &= 122.73 \text{ MPa} \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned} &= (Wm1 + Fa + 4Me/G) / Ab \\ &= 5906243 / 44705 \\ &= 132.13 \text{ MPa} \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned} &= (Wm2 / Ab) \\ &= (1253136 / 44705) \\ &= 28.03 \text{ MPa} \end{aligned}$$

Flange Stress Analysis Results: MPa



	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	103.29	207.00	133.88	207.00
Radial Flange	15.63	138.00	20.26	138.00
Tangential Flange	86.09	138.00	111.59	138.00
Maximum Average	94.69	138.00	122.73	138.00
Bolting	132.13	172.00	28.03	172.00

Minimum Required Flange Thickness 166.218 mm

Estimated M.A.W.P. (Operating) 9.97 MPa

Estimated Finished Weight of Flange at given Thk. 796.4 kgm

Estimated Unfinished Weight of Forging at given Thk 1002.2 kgm

Flange Stress results per WRC-538 at computed S_{fmax} (MPa):

		Actual	Operating Allowed	stress ratio	Actual	Gasket Seating Allowed	stress ratio
Longitudinal Hub	Sh	409.05	499.99	0.818	409.05	499.99	0.818
Tangential Flange	St	340.94	374.99	0.909	340.94	374.99	0.909
Tangential Flange	Sto	147.70	250.00	0.591	147.70	250.00	0.591
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	402.84	749.99	0.537	402.84	749.99	0.537
Combined	Sh + Sto	556.75	749.99	0.742	556.75	749.99	0.742
Radial Flange	Sr	61.90	499.99	0.124	61.90	499.99	0.124
Maximum Stress Ratio				1.000			1.000

Coefficients for *f* and *f2*

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:



$$= \max(1; F1num / F1den)$$

$$= \max(1; 1.37/1.44)$$

$$= 1.0000$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; fn2) \text{ if } h/ho < 0.35, \text{ else } \max(0.25; fn3)$$

$$= \max(0.25; 0.53/1.45) \text{ if } h/ho < 0.35, \text{ else } \max(0.25; 0.27/0.76)$$

$$= 0.3611$$

In this case, the hub ratio h/ho , 0.4020 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, $Sfmax$, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.4 MPa
Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	525.5 MPa
Target assembly gasket stress	SgT	172.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	1862.7250 mm ²
Max. permissible single flange rotation for gasket	Thetamax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	230.1 MPa
Single flange rotation at $Sfmax$	Thetaamax	0.334 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (826^2 - 794^2)$$

$$= 40714.9 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 172 \cdot 40715 / (24 \cdot 1863)$$

$$= 156.6 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(157; 507)$$

$$= 156.6 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(157; 290)$$

$$= 290.0 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(290; 526)$$

$$= 290.0 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$\therefore Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$\therefore 290 \geq 140 (40715 / (24 \cdot 1863))$$

$$\therefore 290.0 \geq 127.5 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$\therefore Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$\therefore 290 \geq (97 \cdot 40715 + \pi/4 \cdot 9.4 \cdot 794^2) / (0.7 \cdot 1863 \cdot 24)$$

$$\therefore 290.0 \geq 275.0 \text{ MPa} \text{ Check Passes}$$



Check if Gasket Maximum Stress is achieved. (Eq. O-9)

:: Sbsel $\leq$ Sgmax(Ag / (nb • Abs))
 :: 290 $\leq$ 380 (40715/(24 • 1863))
 :: 290.0 $\leq$ 346.1 MPa Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

:: Sbsel $\leq$ Sfmax(thetagmax / thetafmax)
 :: 290 $\leq$ 526 (1/0.33)
 :: 290.0 $\leq$ 1571.1 MPa Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

= Sbsel • K • Abs • ϕ b
 = 290 • 0.2 • 1863 • 56
 = 6052070.0 N-mm

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 MPa
 Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since Sb $\geq$ SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 = 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
 = 52.14•777837120/1•1•0.33/(3.22•202713•29²•149•0.3)
 = 0.539 (should be $\leq$ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 = 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
 = 52.14•600143744/1•1•0.33/(3.22•199226•29²•149•0.3)
 = 0.423 (should be $\leq$ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 = 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
 = 52.14•777837120/1•1•0.33/(3.73•202713•29²•149•0.3)
 = 0.464 (should be $\leq$ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 = 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
 = 52.14•600143744/1•1•0.33/(3.73•199226•29²•149•0.3)
 = 0.365 (should be $\leq$ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.87, Temperature Reduction per Fig. UCS 66.1 = 7 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -2 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	104.72	207.00	133.92	207.00
Radial Flange	17.64	138.00	22.55	138.00
Tangential Flange	81.74	138.00	104.53	138.00
Maximum Average	93.23	138.00	119.22	138.00
Bolting	132.13	172.00	28.03	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 162.636 mm

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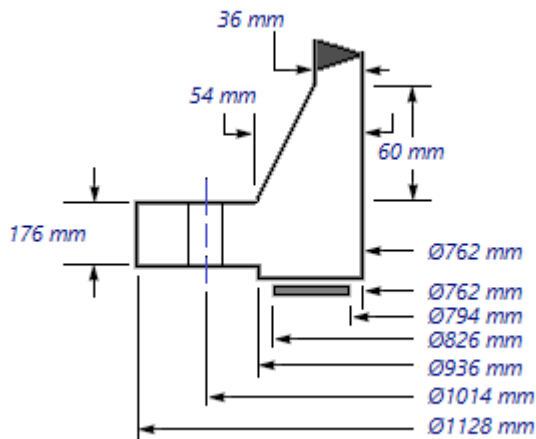


Flange Input Data Values

Description: FLANGE-8 :

Flange-8

Flange Type	Integral Weld Neck		
Design Pressure	P	9.40	MPa
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	762.002	mm
Flange Outside Diameter	A	1128.000	mm
Flange Thickness	t	176.0000	mm
Thickness of Hub at Small End	go	36.0000	mm
Thickness of Hub at Large End	gl	54.0000	mm
Length of Hub	h	60.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	1014.000	mm
Nominal Bolt Diameter	a	56.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	936.000	mm
Flange Face Inside Diameter	Fid	762.002	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	826.000	mm
Gasket Inside Diameter	Gi	794.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		2428.47	N
Bending Moment on Flange		555057.00	N-mm





Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (9.4 \cdot (762/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 9.4) + Ca$$

$$= 30.2804 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 33) / (384 + 0.6 \cdot 33)$$

$$= 11.278 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 36) / (381 + 0.6 \cdot 36)$$

$$= 12.340 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	768.002 mm
Corroded Large Hub,	g1Cor = g1-ci	51.000 mm
Corroded Small Hub,	g0Cor = go-ci	33.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	71.999 mm
Gasket Contact Width,	N = (Go - Gi)/2	16.000 mm
Basic Gasket Width,	bo = N/2	8.000 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	7.127 mm
Gasket Reaction Diameter,	G = Go - 2 * b	811.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 812^2 \cdot 9.4$$

$$= 4865799.500 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 812 \cdot 3 \cdot 9.4$$

$$= 1025361.688 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 768^2 \cdot 9.4$$

$$= 4355515.000 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 4865799 - 4355515$$

$$= 510284.281 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(4865799 + 1025361 + 0, 0)$$

$$= 5891161.500 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot Ip$$

$$= 68.9 \cdot 7.13 \cdot \pi \cdot 812 + 0 \cdot 0 \cdot 0$$

$$= 1253136.250 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((5891161 + 2428 + 4 \cdot 555057/812) / 172, 1253136/172)$$

$$= 34284.484 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 56 + 6 \cdot 176/(3 + 0.5)$$

$$= 413.714 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1014 \cdot \sin(\pi / 24)$$



= 132.354 mm

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{\frac{Bs}{(2a + t)}}, 1)$$

$$= \max(\sqrt{\frac{132}{(2 \cdot 56 + 176)}}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	34284.484	44705.398	
Radial Distance between Hub and Bolts:	71.500	71.999	
Radial Distance between Bolts and Edge:	56.500	57.000	
Circ. Spacing between the Bolts:	124.700	132.354	413.714

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 44705 \cdot 172 / (68.9 \cdot \pi(826 + 794))$$

$$= 21.912 \text{ mm [Note: Exceeds actual gasket width, 16.000]}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 44705$$

$$= 7688536.50 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 5891161 - 4865799$$

$$= 1025361.88 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1014 - 812) / 2$$

$$= 101.1274 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1014 - 768) / 2 + 101)$$

$$= 112.0632 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1014 - 768 - 51) / 2$$

$$= 97.4990 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$Aa = 180, Bb = 176, Cc = 60 \text{ and } Ddg = 42 \text{ mm}$$

Compute the value of [Kab]:

$$= Aa \cdot Bb^3 \cdot (1/3 - 0.210 \cdot (Bb/Aa) \cdot (1 - 1/12 \cdot (Bb/Aa)^4))$$

$$= 180 \cdot 176^3 \cdot (1/3 - 0.210 \cdot (176/180) \cdot (1 - 1/12 \cdot (176/180)^4))$$

$$= 140955200.000 \text{ mm}^4$$

Compute the value of [Kcd]:

$$= Cc \cdot Ddg^3 \cdot (1/3 - 0.105 \cdot (Ddg/Cc) \cdot (1 - 1/192 \cdot (Ddg/Cc)^4))$$

$$= 60 \cdot 42^3 \cdot (1/3 - 0.105 \cdot (42/60) \cdot (1 - 1/192 \cdot (42/60)^4))$$

$$= 1155439.500 \text{ mm}^4$$

Flange Inertia Term [Ip,term1]:

$$= Kab + Kcd = 8602 + 70.5$$



$$= 142110640.000 \text{ mm}^4$$

Flange Inertia Term [Ip,term2]:

$$= Ar \cdot t^3 \left(\frac{1}{3} - 0.21 \left(\frac{t}{AR} \right) \left(1 - \frac{1}{12} \left(\frac{t}{AR} \right)^4 \right) \right)$$

$$= 180 \cdot 176^3 \left(\frac{1}{3} - 0.21 \left(\frac{176}{180} \right) \left(1 - \frac{1}{12} \left(\frac{176}{7.09} \right)^4 \right) \right)$$

$$= 140955200.000 \text{ mm}^4$$

Flange Inertia [Ip]:

$$= \max(Ip,term1, Ip,term2)$$

$$= \max(142110640, 140955200)$$

$$= 142110640.0 \text{ mm}^4$$

Flange Moment of Inertia for Integral Types [I]:

$$= 0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$$

$$= 0.0874 \cdot 2.97 \cdot 33^2 \cdot 159 \cdot 768 / 0.34$$

$$= 103093784.000 \text{ mm}^4$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	4355515.	97.4990	1.0000	424830464.
Face Pressure, Mt	510284.	112.0632	1.0000	57207280.
Gasket Load, Mg	1025362.	101.1274	1.0000	103734248.
Gasket Seating, Matm	7688536.	101.1274	1.0000	777837120.
Moment Load, Moe	...	...	...	409696.

Moment Value, eq. 4.16.16, [Moe]:

$$= 4Me \left[\frac{1}{(0.384 \cdot IP + I)} \right] \cdot \left(\frac{hD}{(C-2 \cdot hD)} \right) + FA \cdot hD$$

$$= 4 \cdot 555057 \left[\frac{103093784}{(0.384 \cdot 142110640 + 103093784)} \right] \cdot \left(\frac{97.5}{(1014 - 2 \cdot 97.5)} \right) +$$

$$2428 \cdot 97.5$$

$$= 409696.500 \text{ N-mm}$$

Total Moment for Operation, Mo 586181696. N-mm
 Total Moment for Gasket seating, Mg 777837120. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 159.198 mm
 Hub Ratio, h/h0 = HL / H0 0.377
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.545

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.865
 Factor V per 4.16.5 0.336
 Factor f per 4.16.5 1.007

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1128/768$$

$$= 1.469$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 \cdot K^2 \cdot \log_{10}(K))) / (K^2 - 1) / (K - 1)$$

$$= (0.66845 + (5.7169 \cdot 1.47^2 \cdot \log_{10}(1.47))) / (1.47^2 - 1) / (1.47 - 1)$$

$$= 5.222$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.47^2(1 + 8.55246 \cdot \log_{10}(1.47)) - 1) / ((1.0472 + 1.9448 \cdot 1.47^2) \cdot (1.47 - 1))$$

$$= 1.724$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.47^2(1 + 8.55246 \cdot \log_{10}(1.47)) - 1) / (1.36136(1.47^2 - 1) \cdot (1.47 - 1))$$

$$= 5.738$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.47^2 + 1) / (1.47^2 - 1)$$

$$= 2.728$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned} &= U \cdot g_0^2 \cdot h_0 / V \\ &= 5.74 \cdot 33^2 \cdot 159 / 0.34 \\ &= 2962546.250 \text{ mm}^3 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned} &= F / h_0 \\ &= 0.87 / 159 \\ &= 0.005 \text{ 1/mm} \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned} &= (t \cdot e + 1) / T + t^3 / d \\ &= (176 \cdot 0.0054 + 1) / 1.72 + 176^3 / 2962546 \\ &= 2.975 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned} &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\ &= (1.01 \cdot 586181696 / 768) / (2.97 \cdot 51^2) \\ &= 99.28 \text{ MPa} \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned} &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\ &= (1.01 \cdot 777837120 / 768) / (2.97 \cdot 51^2) \\ &= 131.74 \text{ MPa} \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned} &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\ &= (2.28 \cdot 586181696 / 768) / (2.97 \cdot 176^2) \\ &= 18.84 \text{ MPa} \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned} &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\ &= (2.28 \cdot 777837120 / 768) / (2.97 \cdot 176^2) \\ &= 25.00 \text{ MPa} \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned} &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\ &= (5.22 \cdot 586181696 / (176^2 \cdot 768)) - 2.73 \cdot 18.8 \\ &= 77.22 \text{ MPa} \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned} &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\ &= (5.22 \cdot 777837120 / (176^2 \cdot 768)) - 2.73 \cdot 25 \\ &= 102.47 \text{ MPa} \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned} &= (SHo + \max(SRo, STo)) / 2 \\ &= (99.3 + \max(18.8, 77.2)) / 2 \\ &= 88.25 \text{ MPa} \end{aligned}$$

Average Flange Stress, Seating [SAa]:

$$\begin{aligned} &= (SHa + \max(SRa, STa)) / 2 \\ &= (132 + \max(25, 102)) / 2 \\ &= 117.11 \text{ MPa} \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned} &= (Wm_1 + Fa + 4Me/G) / Ab \\ &= 5896324 / 44705 \\ &= 131.90 \text{ MPa} \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned} &= (Wm_2 / Ab) \\ &= (1253136 / 44705) \\ &= 28.03 \text{ MPa} \end{aligned}$$

Flange Stress Analysis Results: MPa



	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Longitudinal Hub	99.28	207.00	131.74	207.00
Radial Flange	18.84	138.00	25.00	138.00
Tangential Flange	77.22	138.00	102.47	138.00
Maximum Average	88.25	138.00	117.11	138.00
Bolting	131.90	172.00	28.03	172.00

Minimum Required Flange Thickness 157.988 mm

Estimated M.A.W.P. (Operating) 11.3 MPa

Estimated Finished Weight of Flange at given Thk. 794.1 kgm

Estimated Unfinished Weight of Forging at given Thk 993.7 kgm

Flange Stress results per WRC-538 at computed S_{fmax} (MPa):

		Actual	Operating Allowed	stress ratio	Actual	Gasket Seating Allowed	stress ratio
Longitudinal Hub	Sh	425.21	499.99	0.850	425.21	499.99	0.850
Tangential Flange	St	329.38	374.99	0.878	329.38	374.99	0.878
Tangential Flange	Sto	151.84	250.00	0.607	151.84	250.00	0.607
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	409.73	749.99	0.546	409.73	749.99	0.546
Combined	Sh + Sto	577.04	749.99	0.769	577.04	749.99	0.769
Radial Flange	Sr	80.35	499.99	0.161	80.35	499.99	0.161
Maximum Stress Ratio				1.000			1.000

Coefficients for *f* and *f2*

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:



$$= \max(1; F1num / F1den)$$

$$= \max(1; 1.4/1.39)$$

$$= 1.0109$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; fn2) \text{ if } h/ho < 0.35, \text{ else } \max(0.25; fn3)$$

$$= \max(0.25; 0.5/1.38) \text{ if } h/ho < 0.35, \text{ else } \max(0.25; 0.27/0.76)$$

$$= 0.3610$$

In this case, the hub ratio h/ho , 0.3769 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, $Sfmax$, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 MPa
Minimum permissible bolt stress	Sbmin	290.0 MPa
Max. permissible bolt stress prior to flange damage	Sfmax	552.9 MPa
Target assembly gasket stress	SgT	172.0 MPa
Maximum permissible gasket stress	Sgmax	380.0 MPa
Minimum gasket operating stress	Sgmin-O	97.0 MPa
Minimum gasket seating stress	Sgmin-S	140.0 MPa
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	1862.7250 mm ²
Max. permissible single flange rotation for gasket	Thetamax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 MPa
Flange Material Operating Yield Stress	Syo	230.1 MPa
Single flange rotation at $Sfmax$	Thetaamax	0.338 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (826^2 - 794^2)$$

$$= 40714.9 \text{ mm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 172 \cdot 40715 / (24 \cdot 1863)$$

$$= 156.6 \text{ MPa}$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(157; 508)$$

$$= 156.6 \text{ MPa}$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(157; 290)$$

$$= 290.0 \text{ MPa}$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(290; 553)$$

$$= 290.0 \text{ MPa}$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$\therefore Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$\therefore 290 \geq 140 (40715 / (24 \cdot 1863))$$

$$\therefore 290.0 \geq 127.5 \text{ MPa} \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$\therefore Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$\therefore 290 \geq (97 \cdot 40715 + \pi/4 \cdot 9.4 \cdot 794^2) / (0.7 \cdot 1863 \cdot 24)$$

$$\therefore 290.0 \geq 275.0 \text{ MPa} \text{ Check Passes}$$



Check if Gasket Maximum Stress is achieved. (Eq. O-9)

:: Sbsel $\leq$ Sgmax(Ag / (nb • Abs))
 :: 290 $\leq$ 380 (40715/(24 • 1863))
 :: 290.0 $\leq$ 346.1 MPa Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

:: Sbsel $\leq$ Sfmax(thetagmax / thetafmax)
 :: 290 $\leq$ 553 (1/0.34)
 :: 290.0 $\leq$ 1635.3 MPa Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

= Sbsel • K • Abs • ϕ b
 = 290 • 0.2 • 1863 • 56
 = 6052070.0 N-mm

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 MPa
 Bolt Allowable Stress for: SA-193 B7 Sb 171 MPa

Note: Since Sb $\geq$ SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 = 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
 = 52.14•777837120/1•1•0.34/(2.41•202713•33²•159•0.3)
 = 0.536 (should be $\leq$ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 = 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
 = 52.14•586181696/1•1•0.34/(2.41•199226•33²•159•0.3)
 = 0.411 (should be $\leq$ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 = 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)
 = 52.14•777837120/1•1•0.34/(2.97•202713•33²•159•0.3)
 = 0.434 (should be $\leq$ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 = 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)
 = 52.14•586181696/1•1•0.34/(2.97•199226•33²•159•0.3)
 = 0.332 (should be $\leq$ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.78, Temperature Reduction per Fig. UCS 66.1 = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 9 °C
 Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -8 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -20 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	99.12	207.00	129.70	207.00
Radial Flange	20.83	138.00	27.26	138.00
Tangential Flange	73.14	138.00	95.70	138.00
Maximum Average	86.13	138.00	112.70	138.00
Bolting	131.90	172.00	28.03	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 152.629 mm

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
Skirt-2	...		16	1.6	813	...	1
Skirt-1	...		16	1.6	813	...	2
Bottom head	9.4123		36	3	762	138	3
Shell-1	9.4101		32	3	762	138	4
Flange-1	9.4029		178	3	762	138	5
Flange-2	9.4029		178	3	762	138	6
Shell-2	9.4029		32	3	762	138	7
Shell-3	9.4029		32	3	762	138	8
Flange-3	9.4029		178	3	762	138	9
Flange-4	9.4029		178	3	762	138	10
Shell-4	9.4029		32	3	826	138	11
Shell-5	9.4029		32	3	762	138	12
Flange-5	9.4029		178	3	762	138	13
Flange-6	9.4029		178	3	762	138	14
Shell-6	9.4029		32	3	762	138	15
Flange-7	9.4029		180	3	762	138	16
Flange-8	9.4029		178	3	762	138	17
Top head	9.4029		36	3	762	138	18

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
Skirt-2	...	No Calc	No Calc	No Calc	16	No Calc	1
Skirt-1	...	No Calc	No Calc	No Calc	16	No Calc	2
Bottom head	9.4029	9.73	10.8	30	29.0996		3
Shell-1	9.4029	9.96	11	32	30.3023		4
Flange-1	9.4029	9.97	10.7	176	161.468		5
Flange-2	9.4029	9.97	10.8	176	161.468		6
Shell-2	9.4029	9.97	11	32	30.2804		7
Shell-3	9.4029	9.97	11	32	30.2804		8
Flange-3	9.4029	9.97	11	176	161.468		9
Flange-4	9.4029	9.97	11	176	161.468		10
Shell-4	9.4029	9.97	11	32	30.3945		11
Shell-5	9.4029	9.97	11	32	30.2804		12
Flange-5	9.4029	9.97	11	176	161.468		13
Flange-6	9.4029	9.97	11	176	161.468		14
Shell-6	9.4029	9.97	11	32	30.2804		15
Flange-7	9.4029	9.97	11	178	166.218		16
Flange-8	9.4029	11.3	12.3	176	157.988		17
Top head	9.4029	9.56	10.6	29.5	29.0734		18

Minimum 9.56 10.6

MAWP: 9.56 MPa, limited by: Top head.

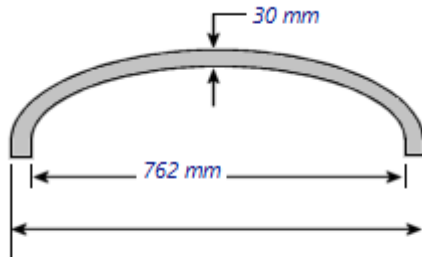
Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Bottom head

Elliptical Head Design Information:					
Design Pressure	P	9.412 MPa	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		82 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	762.002 mm	Actual Stress		133.4 MPa
Spec. Min. Thk t,e, or sa		30.0000 mm	Spec. Nominal Thk.		36.000 mm
Surface Area		884980.8 mm²	SG of Contents		0.920
Element Volume		80.7 ltr	Weight of Contents		727.9 N
Empty Weight		2428.3 N	Operating Weight		3156.2 N
			Comp. All. Ope. Stress		147.3 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 9.525 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 4.763 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (9.41 \cdot 768 \cdot 0.99) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 9.41)$$

$$= 26.0996 + 3.0000 = 29.0996 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.009 MPa

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 27) / (0.99 \cdot 768 + 0.2 \cdot 27)$$

$$= 9.73 - 0.0094 = 9.73 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 30) / (1 \cdot 762 + 0.2 \cdot 30)$$

$$= 10.8 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (9.41 \cdot (0.99 \cdot 768 + 0.2 \cdot 27)) / (2 \cdot 1 \cdot 27)$$

$$= 133.427 \text{ MPa}$$



Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (9.41 \cdot 384) / (138 \cdot 1 - 0.6 \cdot 9.41) + 3$
 $= 30.309 \text{ mm}$

Straight Flange Maximum Allowable Working Pressure:
 Less Operating Hydrostatic Head Pressure of 0.008 MPa
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138 \cdot 1 \cdot 33) / (384 + 0.6 \cdot 33)$
 $= 11.3 - 0.0077 = 11.3 \text{ MPa}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (768 / (2 \cdot 194))^2) / 6$
 $= 0.989704$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 18.300 %
 Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 30$, $t_r = 26.5$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.98$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 4 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 3 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 36$, $t_r = 27.8$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.84$, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 9 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 0 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	9.565 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-28 °C
Governing Thk.	36.000 mm	Basic MDMT	9 °C
Stress/Thk. Ratio	0.842	Temperature Difference	9 °C
UCS-66(c) red.	No	Computed Min. Temp.	0 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 9.525 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 4.763 mm

EXTERNAL PRESSURE RESULTS for: Bottom head

Elliptical Head From 30 to 40 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
27.000	822.00	30.44	0.0045620	118.28

MAEP $= B / (K_0 \cdot Do / t) = 118 / (0.9 \cdot 30.4) = 4.32 \text{ MPa}$

Results for Required Thickness	Tca
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PV Elite 28 U2 Licensee: SL User
FileName : 124-D_Ope_Rev 02
Element: Bottom head

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Step: 12 3:35pm Jul 30, 2026



HEXAGON

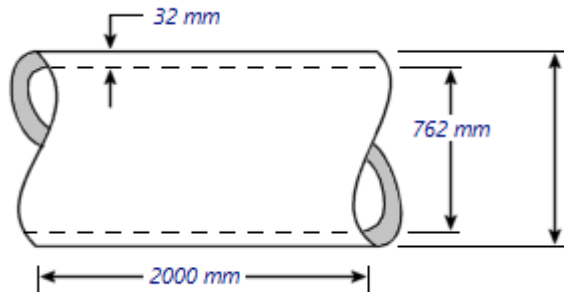
Tca	Outer Dia	Do/t	Factor A	Factor B
2.128	822.00	386.22	0.0003596	35.95

$$MAEP = B / (K0 * Do / t) = 36 / (0.9 * 386) = 0.1 \text{ MPa}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	9.410 MPa	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		82 °C
Length		2000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	762.002 mm	Actual Stress		130.2 MPa
Spec. Min. Thk t,e, or sa		32.0000 mm	Spec. Nominal Thk.		32.000 mm
Surface Area		5189924.0 mm²	SG of Contents		0.920
Element Volume		912.2 ltr	Weight of Contents		3289.8 N
Empty Weight		12133.1 N	Operating Weight		15422.9 N
			Comp. All. Ope. Stress		147.3 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (9.41 \cdot 384) / (138 \cdot 1 - 0.6 \cdot 9.41)$$

$$= 27.3023 + 3.0000 = 30.3023 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.007 MPa

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 29) / (384 + 0.6 \cdot 29)$$

$$= 9.97 - 0.0072 = 9.96 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 32) / (381 + 0.6 \cdot 32)$$

$$= 11 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (9.41 \cdot (384 + 0.6 \cdot 29)) / (1 \cdot 29)$$

$$= 130.249 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 4.030 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 32, tr = 27.8, c = 3 mm, E* = 1



Thickness Ratio = $tr * E^* / (tg - c) = 0.96$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 3 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	9.563 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-28 °C
Governing Thk.	32.000 mm	Basic MDMT	6 °C
Stress/Thk. Ratio	0.957	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	3 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-1

Cylindrical Shell From 40 to 50 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
29.000	826.00	2113.50	28.48	2.5587	0.0031987	113.02

$$MAEP = (4*B)/(3*(Do/t)) = (4*113)/(3*28.5) = 5.29 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
3.919	826.00	2113.50	210.78	2.5587	0.0001635	16.35

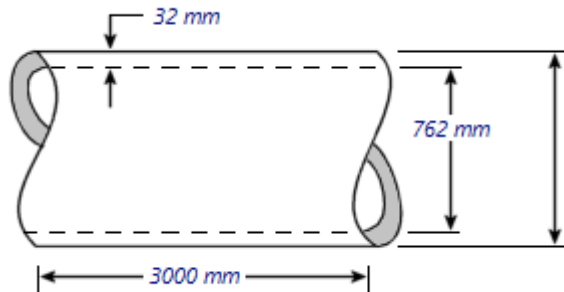
$$MAEP = (4*B)/(3*(Do/t)) = (4*16.4)/(3*211) = 0.1 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
29.000	826.00	98941.76	28.48	50.0000	0.0013690	94.37

$$MAEP = (4*B)/(3*(Do/t)) = (4*94.4)/(3*28.5) = 4.42 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	9.403 MPa	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		82 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	762.002 mm	Actual Stress		130.1 MPa
Spec. Min. Thk t _e or s _a		32.0000 mm	Spec. Nominal Thk.		32.000 mm
Surface Area		7784886.0 mm ²	SG of Contents		0.000
Element Volume		1368.4 ltr	Weight of Contents		0.0 N
Empty Weight		18199.7 N	Operating Weight		18199.7 N
			Comp. All. Ope. Stress		147.3 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (9.4 \cdot 384) / (138 \cdot 1 - 0.6 \cdot 9.4)$$

$$= 27.2804 + 3.0000 = 30.2804 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 29) / (384 + 0.6 \cdot 29)$$

$$= 9.97 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 32) / (381 + 0.6 \cdot 32)$$

$$= 11 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (9.4 \cdot (384 + 0.6 \cdot 29)) / (1 \cdot 29)$$

$$= 130.149 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 4.030 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 32, t_r = 27.7, c = 3 mm, E* = 1
 Thickness Ratio = t_r * E* / (t_g - c) = 0.96, Temp. Reduction = 2 °C



Min Metal Temp. w/o impact per UCS-66, Curve B 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 3 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	9.556 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-28 °C
Governing Thk.	32.000 mm	Basic MDMT	6 °C
Stress/Thk. Ratio	0.957	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	3 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-2

Cylindrical Shell From 70 to 80 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
29.000	826.00	4150.00	28.48	5.0242	0.0016252	98.83

$$MAEP = (4*B)/(3*(Do/t)) = (4*98.8)/(3*28.5) = 4.63 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.227	826.00	4150.00	158.03	5.0242	0.0001226	12.26

$$MAEP = (4*B)/(3*(Do/t)) = (4*12.3)/(3*158) = 0.1 \text{ MPa}$$

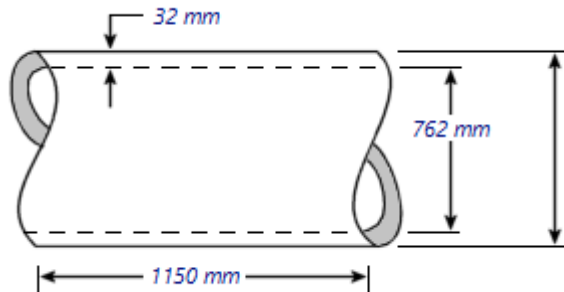
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
29.000	826.00	181399.97	28.48	50.0000	0.0013690	94.37

$$MAEP = (4*B)/(3*(Do/t)) = (4*94.4)/(3*28.5) = 4.42 \text{ MPa}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	9.403 MPa	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		82 °C
Length		1150.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	762.002 mm	Actual Stress		130.1 MPa
Spec. Min. Thk t _e , or s _a		32.0000 mm	Spec. Nominal Thk.		32.000 mm
Surface Area		2984206.2 mm ²	SG of Contents		0.000
Element Volume		524.5 ltr	Weight of Contents		0.0 N
Empty Weight		6976.5 N	Operating Weight		6976.5 N
			Comp. All. Ope. Stress		147.3 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (9.4 \cdot 384) / (138 \cdot 1 - 0.6 \cdot 9.4)$$

$$= 27.2804 + 3.0000 = 30.2804 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 29) / (384 + 0.6 \cdot 29)$$

$$= 9.97 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 32) / (381 + 0.6 \cdot 32)$$

$$= 11 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (9.4 \cdot (384 + 0.6 \cdot 29)) / (1 \cdot 29)$$

$$= 130.149 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 4.030 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 32, t_r = 27.7, c = 3 mm, E* = 1
 Thickness Ratio = t_r * E* / (t_g - c) = 0.96, Temp. Reduction = 2 °C



Min Metal Temp. w/o impact per UCS-66, Curve B 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 3 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	9.556 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-28 °C
Governing Thk.	32.000 mm	Basic MDMT	6 °C
Stress/Thk. Ratio	0.957	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	3 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-3

Cylindrical Shell From 80 to 90 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
29.000	826.00	4150.00	28.48	5.0242	0.0016252	98.83

$$MAEP = (4*B)/(3*(Do/t)) = (4*98.8)/(3*28.5) = 4.63 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.227	826.00	4150.00	158.03	5.0242	0.0001226	12.26

$$MAEP = (4*B)/(3*(Do/t)) = (4*12.3)/(3*158) = 0.1 \text{ MPa}$$

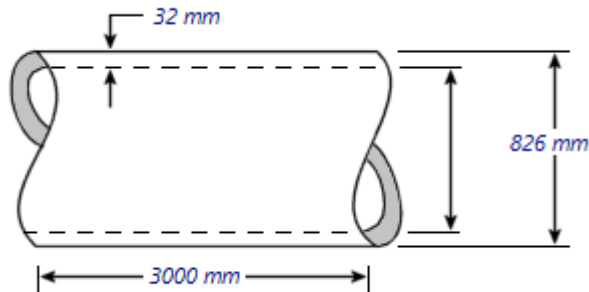
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
29.000	826.00	181399.97	28.48	50.0000	0.0013690	94.37

$$MAEP = (4*B)/(3*(Do/t)) = (4*94.4)/(3*28.5) = 4.42 \text{ MPa}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-4

Cylindrical Shell Design Information:					
Design Pressure	P	9.403 MPa	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		82 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Outside Diameter	Do	826.002 mm	Actual Stress		130.1 MPa
Spec. Min. Thk t _e or s _a		32.0000 mm	Spec. Nominal Thk.		32.000 mm
Surface Area		7784886.0 mm ²	SG of Contents		0.000
Element Volume		1368.4 ltr	Weight of Contents		0.0 N
Empty Weight		18199.7 N	Operating Weight		18199.7 N
			Comp. All. Ope. Stress		147.3 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R_o) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (9.4 \cdot 413) / (138 \cdot 1 + 0.4 \cdot 9.4)$$

$$= 27.3945 + 3.0000 = 30.3945 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (138 \cdot 1 \cdot 29) / (413 - 0.4 \cdot 29)$$

$$= 9.97 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (138 \cdot 1 \cdot 32) / (413 - 0.4 \cdot 32)$$

$$= 11 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R_o - 0.4 \cdot t)) / (E \cdot t)$$

$$= (9.4 \cdot (413 - 0.4 \cdot 29)) / (1 \cdot 29)$$

$$= 130.149 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 4.030 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 32, t_r = 27.8, c = 3 mm, E* = 1
 Thickness Ratio = t_r * E* / (t_g - c) = 0.96, Temp. Reduction = 2 °C



Min Metal Temp. w/o impact per UCS-66, Curve B 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 3 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	9.556 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-28 °C
Governing Thk.	32.000 mm	Basic MDMT	6 °C
Stress/Thk. Ratio	0.960	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	3 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-4

Cylindrical Shell From 110 to 120 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
29.000	826.00	4128.00	28.48	4.9976	0.0016289	98.89

$$MAEP = (4*B)/(3*(Do/t)) = (4*98.9)/(3*28.5) = 4.63 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.214	826.00	4128.00	158.43	4.9976	0.0001229	12.29

$$MAEP = (4*B)/(3*(Do/t)) = (4*12.3)/(3*158) = 0.1 \text{ MPa}$$

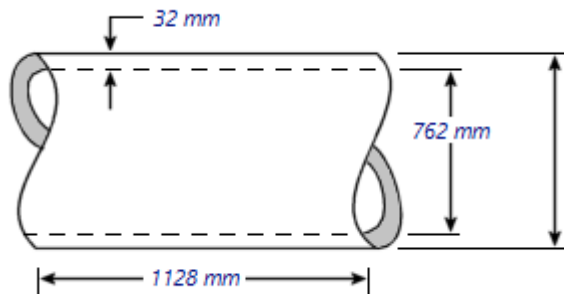
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
29.000	826.00	180490.12	28.48	50.0000	0.0013690	94.37

$$MAEP = (4*B)/(3*(Do/t)) = (4*94.4)/(3*28.5) = 4.42 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-5

Cylindrical Shell Design Information:					
Design Pressure	P	9.403 MPa	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		82 °C
Length		1128.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	762.002 mm	Actual Stress		130.1 MPa
Spec. Min. Thk t _e , or s _a		32.0000 mm	Spec. Nominal Thk.		32.000 mm
Surface Area		2927117.0 mm ²	SG of Contents		0.000
Element Volume		514.5 ltr	Weight of Contents		0.0 N
Empty Weight		6843.1 N	Operating Weight		6843.1 N
			Comp. All. Ope. Stress		147.3 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (9.4 \cdot 384) / (138 \cdot 1 - 0.6 \cdot 9.4)$$

$$= 27.2804 + 3.0000 = 30.2804 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 29) / (384 + 0.6 \cdot 29)$$

$$= 9.97 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 32) / (381 + 0.6 \cdot 32)$$

$$= 11 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (9.4 \cdot (384 + 0.6 \cdot 29)) / (1 \cdot 29)$$

$$= 130.149 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 4.030 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 32, t_r = 27.7, c = 3 mm, E* = 1
 Thickness Ratio = t_r * E* / (t_g - c) = 0.96, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 3 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	9.556 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-28 °C
Governing Thk.	32.000 mm	Basic MDMT	6 °C
Stress/Thk. Ratio	0.957	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	3 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-5

Cylindrical Shell From 120 to 130 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
29.000	826.00	4128.00	28.48	4.9976	0.0016289	98.89

$$MAEP = (4*B)/(3*(Do/t)) = (4*98.9)/(3*28.5) = 4.63 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.214	826.00	4128.00	158.43	4.9976	0.0001229	12.29

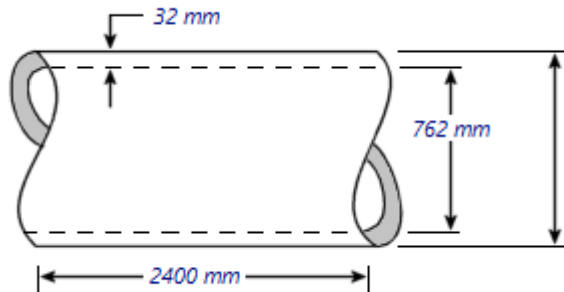
$$MAEP = (4*B)/(3*(Do/t)) = (4*12.3)/(3*158) = 0.1 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
29.000	826.00	180490.12	28.48	50.0000	0.0013690	94.37

$$MAEP = (4*B)/(3*(Do/t)) = (4*94.4)/(3*28.5) = 4.42 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-6

Cylindrical Shell Design Information:					
Design Pressure	P	9.403 MPa	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		82 °C
Length		2400.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	762.002 mm	Actual Stress		130.1 MPa
Spec. Min. Thk t _e or s _a		32.0000 mm	Spec. Nominal Thk.		32.000 mm
Surface Area		6227909.0 mm ²	SG of Contents		0.000
Element Volume		1094.7 ltr	Weight of Contents		0.0 N
Empty Weight		14559.8 N	Operating Weight		14559.8 N
			Comp. All. Ope. Stress		147.3 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (9.4 \cdot 384) / (138 \cdot 1 - 0.6 \cdot 9.4)$$

$$= 27.2804 + 3.0000 = 30.2804 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 29) / (384 + 0.6 \cdot 29)$$

$$= 9.97 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 32) / (381 + 0.6 \cdot 32)$$

$$= 11 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (9.4 \cdot (384 + 0.6 \cdot 29)) / (1 \cdot 29)$$

$$= 130.149 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 4.030 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 32, t_r = 27.7, c = 3 mm, E* = 1
 Thickness Ratio = t_r * E* / (t_g - c) = 0.96, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 3 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	9.556 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-28 °C
Governing Thk.	32.000 mm	Basic MDMT	6 °C
Stress/Thk. Ratio	0.957	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	3 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-6

Cylindrical Shell From 150 to 160 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
29.000	826.00	2400.00	28.48	2.9056	0.0027737	110.48

$$MAEP = (4*B)/(3*(Do/t)) = (4*110)/(3*28.5) = 5.17 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.145	826.00	2400.00	199.29	2.9056	0.0001546	15.46

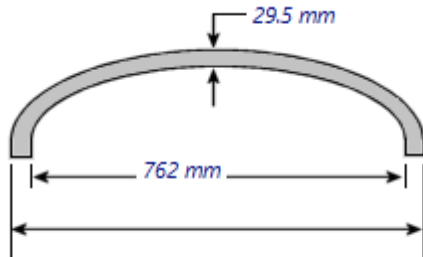
$$MAEP = (4*B)/(3*(Do/t)) = (4*15.5)/(3*199) = 0.1 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
29.000	826.00	111047.70	28.48	50.0000	0.0013690	94.37

$$MAEP = (4*B)/(3*(Do/t)) = (4*94.4)/(3*28.5) = 4.42 \text{ MPa}$$

Internal Pressure Calculation Results:

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Internal Pressure Results for: Top head

Elliptical Head Design Information:					
Design Pressure	P	9.403 MPa	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		82 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	762.002 mm	Actual Stress		135.8 MPa
Spec. Min. Thk t,e, or sa		29.5000 mm	Spec. Nominal Thk.		36.000 mm
Surface Area		884980.8 mm²	SG of Contents		0.000
Element Volume		80.7 ltr	Weight of Contents		0.0 N
Empty Weight		2428.3 N	Operating Weight		2428.3 N
			Comp. All. Ope. Stress		147.3 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 9.525 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 4.763 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (9.4 \cdot 768 \cdot 0.99) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 9.4)$$

$$= 26.0734 + 3.0000 = 29.0734 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 26.5) / (0.99 \cdot 768 + 0.2 \cdot 26.5)$$

$$= 9.56 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 29.5) / (1 \cdot 762 + 0.2 \cdot 29.5)$$

$$= 10.6 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (9.4 \cdot (0.99 \cdot 768 + 0.2 \cdot 26.5)) / (2 \cdot 1 \cdot 26.5)$$

$$= 135.791 \text{ MPa}$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (9.4 \cdot 384) / (138 \cdot 1 - 0.6 \cdot 9.4) + 3$$

$$= 30.280 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 33) / (384 + 0.6 \cdot 33)$$

$$= 11.3 \text{ MPa}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (768 / (2 \cdot 194))^2) / 6$$

$$= 0.989704$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 18.300 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 29.5, tr = 26.5, c = 3 mm, E* = 1
 Thickness Ratio = tr * E* / (tg - c) = 1, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 4 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 36, tr = 27.7, c = 3 mm, E* = 1
 Thickness Ratio = tr * E* / (tg - c) = 0.84, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 9 °C

Min Metal Temp. at Required thickness (UCS 66.1) 0 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	9.556 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-28 °C
Governing Thk.	36.000 mm	Basic MDMT	9 °C
Stress/Thk. Ratio	0.841	Temperature Difference	9 °C
UCS-66(c) red.	No	Computed Min. Temp.	0 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 9.525 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 4.763 mm

EXTERNAL PRESSURE RESULTS for: Top head

Elliptical Head From 180 to 190 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199955 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
26.500	821.00	30.98	0.0044830	118.05

$$MAEP = B / (K_0 \cdot Do / t) = 118 / (0.9 \cdot 31) = 4.23 \text{ MPa}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
2.126	821.00	386.22	0.0003596	35.95

PV Elite 28 U2 Licensee: SL User
FileName : 124-D_Ope_Rev 02
Element: Top head

Step: 19 3:35pm Jul 30, 2026

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HEXAGON

$MAEP = B / (K0 * Do / t) = 36 / (0.9 * 386) = 0.1 \text{ MPa}$

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	12.422 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	12.224 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	13.776 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	10.511 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	13.446 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	13.446 MPa
App 27-4	= M.A.W.P.	9.556 MPa

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	176.4	234.0	0.754	13.78
Shell-1	172.4	234.0	0.737	13.78
Shell-2	172.4	234.0	0.737	13.78
Shell-3	172.4	234.0	0.737	13.78
Shell-4	172.4	234.0	0.737	13.78
Shell-5	172.4	234.0	0.737	13.78
Shell-6	172.4	234.0	0.737	13.78
Top head	179.4	234.0	0.767	13.78

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N4_80	Nozzle	138.00	138.00	1.000
N1_150	Nozzle	138.00	138.00	1.000
K1_50	Nozzle	138.00	138.00	1.000
K2_50	Nozzle	138.00	138.00	1.000
K3_50	Nozzle	138.00	138.00	1.000
H2_200	Nozzle	138.00	138.00	1.000
H1_200	Nozzle	138.00	138.00	1.000
N2_80	Nozzle	138.00	138.00	1.000
K4_50	Nozzle	138.00	138.00	1.000
N3_150	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom head	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Flange-1	138.00	138.00	1.000
Flange-2	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Flange-3	138.00	138.00	1.000
Flange-4	138.00	138.00	1.000
Shell-4	138.00	138.00	1.000
Shell-5	138.00	138.00	1.000
Flange-5	138.00	138.00	1.000
Flange-6	138.00	138.00	1.000
Shell-6	138.00	138.00	1.000
Flange-7	138.00	138.00	1.000
Flange-8	138.00	138.00	1.000
Top head	138.00	138.00	1.000

Minimum 1.000



Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N4_80	49.56	225.00	0.220
N1_150	115.27	225.00	0.512
K1_50	21.27	225.00	0.095
K2_50	21.27	225.00	0.095
K3_50	21.27	225.00	0.095
H2_200	88.97	225.00	0.395
H1_200	77.16	225.00	0.343
N2_80	57.43	225.00	0.255
K4_50	21.27	225.00	0.095
N3_150	115.27	225.00	0.512

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	12.422 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	12.224 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	13.781 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	10.511 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	13.446 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	13.446 MPa
App 27-4	= M.A.W.P.	9.556 MPa

UG-99(b) Note 35, Test Pressure Calculation:
 = Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 9.4 • 1
 = 12.224 MPa

Vertical Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	175.4	234.0	0.749	12.37
Shell-1	171.2	234.0	0.732	12.37
Shell-2	170.9	234.0	0.730	12.34
Shell-3	170.5	234.0	0.728	12.32
Shell-4	170.2	234.0	0.728	12.30
Shell-5	169.8	234.0	0.726	12.27
Shell-6	169.6	234.0	0.725	12.25
Top head	176.6	234.0	0.755	12.23

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N4_80	Nozzle	138.00	138.00	1.000
N1_150	Nozzle	138.00	138.00	1.000
K1_50	Nozzle	138.00	138.00	1.000
K2_50	Nozzle	138.00	138.00	1.000
K3_50	Nozzle	138.00	138.00	1.000
H2_200	Nozzle	138.00	138.00	1.000
H1_200	Nozzle	138.00	138.00	1.000
N2_80	Nozzle	138.00	138.00	1.000
K4_50	Nozzle	138.00	138.00	1.000
N3_150	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom head	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Flange-1	138.00	138.00	1.000
Flange-2	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Flange-3	138.00	138.00	1.000
Flange-4	138.00	138.00	1.000
Shell-4	138.00	138.00	1.000
Shell-5	138.00	138.00	1.000
Flange-5	138.00	138.00	1.000
Flange-6	138.00	138.00	1.000
Shell-6	138.00	138.00	1.000



Flange-7	138.00	138.00	1.000
Flange-8	138.00	138.00	1.000
Top head	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N4_80	44.48	225.00	0.198
N1_150	103.44	225.00	0.460
K1_50	19.09	225.00	0.085
K2_50	19.09	225.00	0.085
K3_50	19.09	225.00	0.085
H2_200	79.68	225.00	0.354
H1_200	68.85	225.00	0.306
N2_80	51.05	225.00	0.227
K4_50	18.91	225.00	0.084
N3_150	102.25	225.00	0.454

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	3200	...	...	No Calc	No Calc
20	30	3200	...	...	No Calc	No Calc
30	40	No Calc	822.002	27	0.004562	118.277
40	50	2113.5	826.002	29	0.0031987	113.016
50	60	No Calc	...	173	No Calc	No Calc
60	70	No Calc	...	173	No Calc	No Calc
70	80	4150	826.002	29	0.0016252	98.8288
80	90	4150	826.002	29	0.0016252	98.8288
90	100	No Calc	...	173	No Calc	No Calc
100	110	No Calc	...	173	No Calc	No Calc
110	120	4128	826.002	29	0.0016289	98.8856
120	130	4128	826.002	29	0.0016289	98.8856
130	140	No Calc	...	173	No Calc	No Calc
140	150	No Calc	...	173	No Calc	No Calc
150	160	2400	826.002	29	0.0027737	110.482
160	170	No Calc	...	175	No Calc	No Calc
170	180	No Calc	...	173	No Calc	No Calc
180	190	No Calc	821.002	26.5	0.004483	118.053

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	30	5.12833	0.10343	4.32
40	50	32	6.91879	0.10343	5.29
50	60	176	161.468	0.10343	No Calc
60	70	176	161.468	0.10343	No Calc
70	80	32	8.22672	0.10343	4.63
80	90	32	8.22672	0.10343	4.63
90	100	176	161.468	0.10343	No Calc
100	110	176	161.468	0.10343	No Calc
110	120	32	8.21362	0.10343	4.63
120	130	32	8.21362	0.10343	4.63
130	140	176	161.468	0.10343	No Calc
140	150	176	161.468	0.10343	No Calc
150	160	32	7.14473	0.10343	5.17
160	170	178	166.218	0.10343	No Calc
170	180	176	157.988	0.10343	No Calc
180	190	29.5	5.12574	0.10343	4.23

Minimum

4.23

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	3200	No Calc	No Calc	No Calc
20	30	3200	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	2113.5	98941.8	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc



70	80	4150	181400	No Calc	No Calc
80	90	4150	181400	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc
100	110	No Calc	No Calc	No Calc	No Calc
110	120	4128	180490	No Calc	No Calc
120	130	4128	180490	No Calc	No Calc
130	140	No Calc	No Calc	No Calc	No Calc
140	150	No Calc	No Calc	No Calc	No Calc
150	160	2400	111048	No Calc	No Calc
160	170	No Calc	No Calc	No Calc	No Calc
170	180	No Calc	No Calc	No Calc	No Calc
180	190	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	1321.57	...	1253.27	...	...
20	30	310.497	...	279.447	...	...
30	40	247.633	80.7336	226.997	82.4733	...
40	50	1237.32	912.242	1125.56	926.664	...
50	60	791.498	109.84	778.198	110.273	...
60	70	791.498	109.84	778.198	110.273	...
70	80	1855.98	1368.36	1688.33	1390	...
80	90	711.457	524.539	647.194	532.832	...
90	100	791.498	109.84	778.198	110.273	...
100	110	791.498	109.84	778.198	110.273	...
110	120	1855.98	1368.36	1688.33	1390	...
120	130	697.847	514.504	634.814	522.639	...
130	140	791.498	109.84	778.198	110.273	...
140	150	791.498	109.84	778.198	110.273	...
150	160	1484.78	1094.69	1350.67	1112	...
160	170	804.804	110.767	791.393	111.2	...
170	180	802.57	109.84	789.27	110.273	...
180	190	247.633	80.7336	226.997	82.4733	...
Total		16327	6823.82	15371	6922.18	0

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Wght	200	...	...	...	Tailing lug
20	Insl	39.5469	...	750	...	FP
30	Liqd	74.2273	...	-95.2503	...	Liquid 30
30	Insl	34.1963	...	-70.2503	...	Ins: 30
30	Nozl	22.8393	...	-325.5	...	N4_80
30	Forc	...	...	-190	...	Guide sup later
40	Liqd	335.49	...	400	...	Liquid 40
40	Insl	160.491	...	1000	...	Ins: 40
40	Nozl	90.0621	-481.001	1175	...	N1_150
40	Nozl	23.5956	474.663	200	...	K1_50
40	Nozl	23.5956	474.663	1200	...	K2_50
40	Nozl	23.5956	-474.663	550	...	K3_50
40	Wght	259	-1649	1175	...	N1_FY
50	Insl	20.8233	...	103	...	Ins: 50
60	Insl	20.8233	...	103	...	Ins: 60
70	Pack	296.59	...	1550	...	Bed 2
70	Insl	240.736	...	1500	...	Ins: 70
70	Nozl	221.645	481.001	343.991	...	H2_200
70	Wght	50	350	343.991	...	H2_Davit
70	Wght	10	...	50.0004	...	Bed support 2
70	Wght	35	...	80.0003	...	Bed sup grid 2
80	Pack	81.8179	...	400	...	Bed2
80	Insl	92.2822	...	575	...	Ins: 80
80	Wght	10	...	820	...	Bed limiter sup
80	Wght	10	...	850	...	Bed limiter
90	Insl	20.8233	...	103	...	Ins: 90
100	Insl	20.8233	...	103	...	Ins: 100
110	Pack	298.84	...	1539	...	Bed-1
110	Insl	240.736	...	1500	...	Ins: 110
110	Nozl	221.93	481.001	673.2	...	H1_200
110	Wght	50	350	673.2	...	H1_Davit
110	Wght	400	...	665	...	Clip weight
110	Wght	10	...	49.9993	...	Bed support
110	Wght	35	...	59.9998	...	Bed sup grid
110	Wght	1300	...	607	...	Misc wt 7%
120	Pack	79.5679	...	389	...	Bed1



120	Ins1	90.5168	...	564	...	Ins: 120
120	Wght	10	...	800	...	Bed limiter sup
120	Wght	10	...	820	...	Bed limiter 1
130	Ins1	20.8233	...	103	...	Ins: 130
140	Ins1	20.8233	...	103	...	Ins: 140
150	Ins1	192.589	...	1200	...	Ins: 150
150	Noz1	34.9242	-481.001	744.701	...	N2_80
150	Noz1	23.6567	474.663	1506.5	...	K4_50
150	Wght	5	...	744.701	...	N2_Pipe
150	Wght	10	...	2250	...	Demister sup
150	Wght	20	...	2300	...	Demister
150	Wght	150	...	2300	...	Lifting lug1
150	Wght	150	...	2300	...	Lifting lug 2
160	Ins1	20.9609	...	103	...	Ins: 160
170	Ins1	21.1146	...	97	...	Ins: 170
180	Ins1	34.7592	...	120.25	...	Ins: 180
180	Noz1	89.804	...	325.75	...	N3_150
180	Wght	320	1649	...	...	N3_Wt
180	Forc	...	...	170	...	N3_F/M

Total Weight of Each Detail Type:

Packing	756.8
Liquid	409.7
Insulation	1292.9
Nozzles	775.6
Weights	3044.0

Sum of the Detail Weights 6279.0 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	16327.1	16327.1	16327.1	16327.1	16327.1	16327.1
Nozzles	775.6	775.6	775.6	775.6	775.6	775.6
Wld Weights	2350.0	2350.0	2350.0	2350.0	2350.0	2350.0
Packing	...	...	756.8	756.8	756.8	756.8
Insulation	...	...	1292.9	1292.9	1292.9	1292.9
Ope Weights	...	...	...	...	...	694.0
Ope. Liquid	...	...	...	...	...	409.7
Test Liquid	...	6819.7	...	...	...	...
Totals	19452.7	26272.4	21502.4	21502.4	21502.4	22606.1

Field Installation Options:

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	19452.7 kgm
Shop Test Wt.	- Fabricated weight + water (full)	26272.4 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	21502.4 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	21502.4 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	21502.4 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	22606.1 kgm
Field Test Wt.	- Empty weight + test medium (full)	27565.2 kgm

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	7719.1 kgm
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	9621.5 kgm
Mass of the Upper 1/3 of the Vertical Vessel (empty)	7715.1 kgm

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
------	----	---	--



10	20	5619052	5397884
20	30	2554114	2453583
30	40	884981	749108
40	50	5189924	4787800
50	60	1331141	569748
60	70	1331141	569748
70	80	7784886	7181699
80	90	2984206	2752985
90	100	1331141	569748
100	110	1331141	569748
110	120	7784886	7181699
120	130	2927117	2700319
130	140	1331141	569748
140	150	1331141	569748
150	160	6227909	5745360
160	170	1342887	574536
170	180	1338062	569748
180	190	884981	749108

Total 53509852.0 44262324.0 mm²
 Total 53.5 44.3 m²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kgm	Total. Ele. Oper. Wgt. kgm	Total. Ele. Hydro. Wgt. kgm	Total Dtl. Offset Mom. N-mm	Oper. Wgt. No Liquid kgm
10	20	1521.57	1521.57	1521.57	...	1521.57
20	30	350.044	350.044	350.044	...	350.044
30	40	304.668	378.895	385.352	...	304.668
40	50	1558.66	2153.15	2470.34	4504846	1817.66
50	60	812.321	812.321	922.095	...	812.321
60	70	812.321	812.321	922.095	...	812.321
70	80	2674.95	2709.95	3745.89	1217530	2709.95
80	90	895.557	905.557	1337.96	...	905.557
90	100	812.321	812.321	922.095	...	812.321
100	110	812.321	812.321	922.095	...	812.321
110	120	4377.48	4412.48	5446.17	1218875	4412.48
120	130	877.932	887.932	1312.55	...	887.932
130	140	812.321	812.321	922.095	...	812.321
140	150	812.321	812.321	922.095	...	812.321
150	160	2045.95	2070.95	3139.97	54637.4	2070.95
160	170	825.765	825.765	936.465	...	825.765
170	180	823.684	823.684	933.458	...	823.684
180	190	372.196	692.196	452.88	5176526	692.196

Empty Support Force + the Sum of the Y forces 225697.17 N
 Operating Support Force + the Sum of the Y forces 236520.20 N
 Hydro Support Force + the Sum of the Y forces 285149.28 N

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kgm	Cumulative Oper. Wgt. kgm	Cumulative Hydro. Wgt. kgm
10	20	22196.4	22606.1	27565.2
20	30	20674.8	21084.5	26043.7
30	40	20324.8	20734.5	25693.6
40	50	20020.1	20355.6	25308.3
50	60	18202.4	18202.4	22837.9
60	70	17390.1	17390.1	21915.8
70	80	16577.8	16577.8	20993.7
80	90	13867.9	13867.9	17247.8
90	100	12962.3	12962.3	15909.9
100	110	12150	12150	14987.8
110	120	11337.7	11337.7	14065.7
120	130	6925.17	6925.17	8619.52



130	140	6037.24	6037.24	7306.97
140	150	5224.92	5224.92	6384.87
150	160	4412.6	4412.6	5462.78
160	170	2341.65	2341.65	2322.8
170	180	1515.88	1515.88	1386.34
180	190	692.196	692.196	452.88

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. N-mm	Cumulative Oper. Mom. N-mm	Cumulative Hydro. Mom. N-mm
10	20	12172417	12172417	12172417
20	30	12172417	12172417	12172417
30	40	12172417	12172417	12172417
40	50	12172417	12172417	12172417
50	60	7667570	7667570	7667570
60	70	7667570	7667570	7667570
70	80	7667570	7667570	7667570
80	90	6450039	6450039	6450039
90	100	6450039	6450039	6450039
100	110	6450039	6450039	6450039
110	120	6450039	6450039	6450039
120	130	5231163	5231163	5231163
130	140	5231163	5231163	5231163
140	150	5231163	5231163	5231163
150	160	5231163	5231163	5231163
160	170	5176526	5176526	5176526
170	180	5176526	5176526	5176526
180	190	5176526	5176526	5176526



Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N4_80	14.34	15.31	82	900	GR 1.1	5.784	14.342	14.342	14.342
N1_150	14.34	15.31	82	900	GR 1.1	3.585	14.342	10.757	14.342
K1_50	14.34	15.31	82	900	GR 1.1	...	...	...	...
K2_50	14.34	15.31	82	900	GR 1.1	...	...	...	...
K3_50	14.34	15.31	82	900	GR 1.1	...	...	...	...
H2_200	14.34	15.31	82	900	GR 1.1	...	...	...	...
H1_200	14.34	15.31	82	900	GR 1.1	...	...	...	...
N2_80	14.34	15.31	82	900	GR 1.1	5.784	14.342	14.342	14.342
K4_50	14.34	15.31	82	900	GR 1.1	...	...	...	...
N3_150	14.34	15.31	82	900	GR 1.1	3.585	14.342	10.757	14.342
Min Rating	14.342	15.311, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
N4_80	107.950	1.40	1267201	5998	9.6	14.3	Pass
N1_150	192.151	1.00	4454994	11246	9.6	14.3	Pass
N2_80	107.950	1.40	1267201	5998	9.6	14.3	Pass
N3_150	192.151	1.00	4454994	11246	9.6	14.3	Pass

Evaluating Flange on Nozzle [N480]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1267201) + 4 \cdot \text{abs}(5998)107.950 \leq \pi \cdot 107.950^3 \cdot ((14.342 - 9.556) + 1.400 \cdot 14.342)$$

$$= 0.22866E+08 \leq 0.98301E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N480 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(14.342, 1.400 \cdot 14.342 + 14.342 - 22866090.0 / (\pi \cdot 107.950^3))$$

$$= \min(14.342, 28.637)$$

$$= 14.342 \text{ MPa}$$

Evaluating Flange on Nozzle [N1150]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(4454994) + 4 \cdot \text{abs}(11246)192.151 \leq \pi \cdot 192.151^3 \cdot ((14.342 - 9.556) + 1.000 \cdot 14.342)$$

$$= 0.79927E+08 \leq 0.42648E+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1150 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(14.342, 1.000 \cdot 14.342 + 14.342 - 79926880.0 / (\pi \cdot 192.151^3))$$

$$= \min(14.342, 25.100)$$

$$= 14.342 \text{ MPa}$$

Evaluating Flange on Nozzle [N280]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1267201) + 4 \cdot \text{abs}(5998)107.950 \leq \pi \cdot 107.950^3 \cdot ((14.342 - 9.556) + 1.400 \cdot 14.342)$$

$$= 0.22866E+08 \leq 0.98301E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N280 [Pmax]:



$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(14.342, 1.400 \cdot 14.342 + 14.342 - 22866090.0 / (\pi \cdot 107.950^3)) \\ &= \min(14.342, 28.637) \\ &= 14.342 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N3150]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(4454994) + 4 \cdot \text{abs}(11246)192.151 \leq \pi \cdot 192.151^3 [(14.342 - 9.556) + 1.000 \cdot 14.342] \\ &= 0.79927\text{E}+08 \leq 0.42648\text{E}+09 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N3150 [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(14.342, 1.000 \cdot 14.342 + 14.342 - 79926880.0 / (\pi \cdot 192.151^3)) \\ &= \min(14.342, 25.100) \\ &= 14.342 \text{ MPa} \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 1.20445 Hz.

The Natural Frequency for the Vessel (Ope...) is 1.20398 Hz.

The Natural Frequency for the Vessel (Filled) is 1.0942 Hz.

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Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	25228.9	-4795438
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	...	...
130	140	...	...
140	150	...	...
150	160	...	...
160	170	...	...
170	180	...	...
180	190	15161.8	9579076

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	25228.9	-4795438
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	...	...
130	140	...	...
140	150	...	...
150	160	...	...
160	170	...	...
170	180	...	...
180	190	15161.8	9579076

User Input Forces and Moments:

From Node	Distance From	- - - - Fx	Forces - - - - Fy	- - - - Fz	- - - - Mx	Moments - - - - My	- - - - Mz
30	-190.00	-20791.		14291.			
180	170.00	10721.	-14845.	10721.	0.495E+07	0.495E+07	



Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.2 \cdot 0.93)/(1 + 1.7 \cdot 3.4 \cdot 0.2)))$
 $= \min(0.85, 0.89)$
 $= 0.850$



Note: Vortex shedding loads have been applied at the analyst's request, even though the vessel was determined to be stable and not susceptible to vortex shedding.

Vortex Shedding Loads, per NBC 1990 Supplement, pages 149-150

Compute the critical wind speed using the weighted average diameter of the top one-third of the vessel [Vh(ope)]
 $= F_n(ope) \cdot D_{avg} / 0.2 \cdot 0.6818$
 $= 1.2 \cdot 3.39 / 0.2 \cdot 0.6818$
 $= 13.9 \text{ mile/hr ; } 22.4 \text{ km/hr}$

Compute the aspect ratio (γ) for the entire structure [γ]
 $= (\text{Total Height})^2 / \text{Sum of the Wind Area}$
 $= 17886^2 / 17164270 = 18.6$

Determine the Velocity Pressure Qh(ope) (English Units only)
 $= 0.5 \cdot 0.0668 \cdot \rho \cdot V_h(ope)^2$
 $= 0.5 \cdot 0.0668 \cdot 0.075 \cdot 13.9^2 = 0.48 \text{ psf ; } 0.023 \text{ kPa}$

Compute the equivalent force per unit area which will act on the top one-third of the vessel [F/area]:
 $= (q_h(ope) \cdot C_1) / ((\gamma \cdot (\beta(ope) - (C_2 \cdot \rho \cdot D_{avg}^2 / M_3))))^{1/2}$ [English Units]
 $= (0.48 \cdot 6) / (((18.6 \cdot (0.01 - (1.2 \cdot 0.075 \cdot 3.39^2 / 870))))^{1/2}$
 $= 7.156035 \text{ psf or } 0.342624 \text{ kPa}$

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
Tailing lug	0.00	300.00	0.00	0.00
N1FY	0.00	4725.00	0.00	0.00
H2Davit	0.00	6370.49	0.00	0.00



Bed support 2	0.00	6076.50	0.00	0.00
Bed sup grid	0.00	6106.50	0.00	0.00
Bed limiter s	0.00	9846.50	0.00	0.00
Bed limiter	0.00	9876.50	0.00	0.00
H1Davit	0.00	11326.20	0.25	0.00
Clip weight	0.00	11318.00	0.25	0.00
Bed support	0.00	10703.00	0.25	0.00
Bed sup grid	0.00	10713.00	0.25	0.00
Misc wt 7%	0.00	11260.00	0.25	0.00
Bed limiter s	0.00	14453.00	0.34	0.00
Bed limiter 1	0.00	14473.00	0.34	0.00
N2pipe	0.00	16002.20	0.34	0.00
Demister sup	0.00	17507.50	0.34	0.00
Demister	0.00	17557.50	0.34	0.00
Lifting lug1	0.00	17557.50	0.34	0.00
Lifting lug 2	0.00	17557.50	0.34	0.00
N3wt	0.00	18136.00	0.34	0.00

The following table contains results for Vortex Shedding Case (Ope).

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	1400	1333.32	2933303	...	...
20	30	3000	1464.52	1464519	...	...
30	40	3525	1610.48	80524.2	...	...
40	50	4550	1617.04	3234086	...	...
50	60	5672.5	1478.73	348980	...	...
60	70	5908.5	1478.73	348980	...	...
70	80	7526.5	1617.04	4851130	...	...
80	90	9601.5	1617.04	1859599	...	...
90	100	10299	1478.73	348980	...	...
100	110	10535	1478.73	348980	...	...
110	120	12153	1617.04	4851130	0.24867	805.992
120	130	14217	1617.04	1824024	0.34262	417.559
130	140	14903.5	1478.73	348980	0.34262	79.889
140	150	15139.5	1478.73	348980	0.34262	79.889
150	160	16457.5	1617.04	3880904	0.34262	888.423
160	170	17781	1476.8	351479	0.34262	80.4612
170	180	18018	1465.38	345831	0.34262	79.1682
180	190	18286.7	1608.84	459518	0.34262	105.194

End of Vortex Shedding Calculations

Natural Frequency of Vessel (Operating)	1.204 Hz
Natural Frequency of Vessel (Empty)	1.204 Hz
Natural Frequency of Vessel (Test)	1.094 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.668
Structure Height to Diameter ratio	19.269
Height to top of Structure	18185.998 mm

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:

Exposure Category: C from Table 26.11-1

α = 9.5 : Zg = 274320 mm

Effective Height [z]:

= Centroid Height + Vessel Base Elevation



= 1100 + 300 = 1400 mm
 = 4.59 ft. Imperial Units

Velocity Pressure coefficient evaluated at height z [Kz]:

Because z (4.59 ft.) < 15 ft.
 = $2.01(15 / Z_g)^{2/\alpha}$
 = $2.01(15/900)^{2/9.5}$
 = 0.85

Type of Hill: No Hill

Wind Directionality Factor [Kd]:

= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= $(1 + K1 \cdot K2 \cdot K3)^2$
 = $(1 + 0 \cdot 0 \cdot 0)^2$
 = 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= $\max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$
 = $\max(16, 0.00256 \cdot 0.85 \cdot 1 \cdot 1 \cdot 1 \cdot 60.4^2)$
 = 16 psf [0.77] kPa

Force on the first element [F]:

= $qz \cdot G \cdot Cf \cdot \text{WindArea}$
 = $16 \cdot 0.85 \cdot 0.67 \cdot 31.6$
 = 287 lbs. [1276] N

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz kPa
Skirt-2	1400.0	0.000	0.000	0.000	0.849	1.000	0.766
Skirt-1	3000.0	0.000	0.000	0.000	0.849	1.000	0.766
Bottom head	3525.0	0.000	0.000	0.000	0.849	1.000	0.766
Shell-1	4550.0	0.000	0.000	0.000	0.849	1.000	0.766
Flange-1	5672.5	0.000	0.000	0.000	0.888	1.000	0.766
Flange-2	5908.5	0.000	0.000	0.000	0.896	1.000	0.766
Shell-2	7526.5	0.000	0.000	0.000	0.943	1.000	0.766
Shell-3	9601.5	0.000	0.000	0.000	0.992	1.000	0.766
Flange-3	10299.0	0.000	0.000	0.000	1.007	1.000	0.766
Flange-4	10535.0	0.000	0.000	0.000	1.012	1.000	0.766
Shell-4	12153.0	0.000	0.000	0.000	1.043	1.000	0.766
Shell-5	14217.0	0.000	0.000	0.000	1.078	1.000	0.766
Flange-5	14903.5	0.000	0.000	0.000	1.089	1.000	0.766
Flange-6	15139.5	0.000	0.000	0.000	1.092	1.000	0.766
Shell-6	16457.5	0.000	0.000	0.000	1.112	1.000	0.766
Flange-7	17781.0	0.000	0.000	0.000	1.130	1.000	0.766
Flange-8	18018.0	0.000	0.000	0.000	1.133	1.000	0.766
Top head	18286.7	0.000	0.000	0.000	1.137	1.000	0.766

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
 D - Average outer insulated diameter of vessel, mm
 Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
 Dr - Average outer insulated of top half of vessel, mm
 f - Natural frequency of vibration (Hertz)
 f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(104)$
 L - Total height of structure mm
 Lc - Total length of conical section(s) of vessel, mm
 tb - Uncorroded plate thickness at bottom of vessel, mm



V30 - Design Wind Speed provided by user, km/hr
 Vc - Critical wind velocity, km/hr
 Vw - Maximum wind speed at top of structure, km/hr
 W - Total corroded weight of structure, N
 Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. N
 Z - Maximum amplitude of vibration at top of vessel, mm
 D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
 Vp - Vib. Chance, <= 0.31416E-05 (High); 0.31416E-05 < 0.39270E-05 (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: 0/17886 = 0

:: [D / L²] • 10⁴ < 8 (English Units, ft.)
 :: [3.12/58.7²] • 10⁴ = 9.05 [Geometry Violation]

Compute the vibration possibility. If Vp > 0.39270E-05 no chance. [Vp]:
 = W / (L • Dr²)
 = 212305/(17886 • 978²)
 = 0.12398E-04

Since Vp is > 0.39270E-05 no further vibration analysis is required !

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
Tailing lug	0.00	300.00	0.77	0.00
N1FY	0.00	4725.00	0.77	0.00
H2Davit	0.00	6370.49	0.77	0.00
Bed support 2	0.00	6076.50	0.77	0.00
Bed sup grid	0.00	6106.50	0.77	0.00
Bed limiter s	0.00	9846.50	0.77	0.00
Bed limiter	0.00	9876.50	0.77	0.00
H1Davit	0.00	11326.20	0.77	0.00
Clip weight	0.00	11318.00	0.77	0.00
Bed support	0.00	10703.00	0.77	0.00
Bed sup grid	0.00	10713.00	0.77	0.00
Misc wt 7%	0.00	11260.00	0.77	0.00
Bed limiter s	0.00	14453.00	0.77	0.00
Bed limiter 1	0.00	14473.00	0.77	0.00
N2Pipe	0.00	16002.20	0.77	0.00
Demister sup	0.00	17507.50	0.77	0.00
Demister	0.00	17557.50	0.77	0.00
Lifting lug1	0.00	17557.50	0.77	0.00
Lifting lug 2	0.00	17557.50	0.77	0.00
N3Wt	0.00	18136.00	0.77	0.00

The Natural Frequency for the Vessel (Ope...) is 1.20398 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	1400	1333.32	2933303	0.76606	1403.79
20	30	3000	1464.52	1464519	0.76606	700.876
30	40	3525	1610.48	80524.2	0.76606	38.5365
40	50	4550	1617.04	3234086	0.76606	1547.74
50	60	5672.5	1478.73	348980	0.76606	167.011
60	70	5908.5	1478.73	348980	0.76606	167.011
70	80	7526.5	1617.04	4851130	0.76606	2321.61
80	90	9601.5	1617.04	1859599	0.76606	889.95
90	100	10299	1478.73	348980	0.76606	167.011
100	110	10535	1478.73	348980	0.76606	167.011
110	120	12153	1617.04	4851130	0.76606	2321.61



120	130	14217	1617.04	1824024	0.76606	872.925
130	140	14903.5	1478.73	348980	0.76606	167.011
140	150	15139.5	1478.73	348980	0.76606	167.011
150	160	16457.5	1617.04	3880904	0.76606	1857.29
160	170	17781	1476.8	351479	0.76606	168.208
170	180	18018	1465.38	345831	0.76606	165.504
180	190	18286.7	1608.84	459518	0.76606	219.912

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1\end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
= $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
= $0.020(59.30.75)$
= 0.427 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
= (1/Natural Frequency) = (1/1.2)
= 0.831

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
= Minimum Value of ($1.7 \cdot 0.43$, 0.83) per 12.8.2
= 0.727

Seismic Response Coefficient per equation 12.8-2 [Cs]:
= $Sds / (R / I)$
= $0.16 / (2 / 1.25)$
= 0.098

Maximum value of C_s per equation 12.8-3 [Cs]:
= $Sd1 / (T(R / I_e))$
= $0.1 / (0.73 (2 / 1.25))$
= 0.089

Minimum value of C_s per equation 15.4-1 [Cs]:
= $\max((0.044 \cdot SDS \cdot I_e), 0.030)$
= $\max((0.044 \cdot 0.16 \cdot 1.25), 0.030)$
= 0.030

Total Base Shear [V]:
= $C_s \cdot W$ (Equation 12.8-1):
= $0.089 \cdot 221675$
= 19830.215 N

Vertical load per 12.4-4, [Ev]:
= $0.2 \cdot Sds \cdot W$
= $0.2 \cdot 0.16 \cdot 221675$



= 6951.7 N

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$) and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1.17

The Natural Frequency for the Vessel (Ope...) is 1.20398 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N	Element Emp Load N	Element Vertical Load N
10	20	1100	14920.6	97.5001	96.8857	-7893.06
20	30	2700	3432.53	63.8652	63.4534	-8360.97
30	40	3225	3715.45	85.0318	67.9308	-8468.61
40	50	4250	21113.8	666.51	479.338	-8585.13
50	60	5372.5	7965.62	330.425	328.257	-9247.26
60	70	5608.5	7965.62	347.399	345.118	-9497.06
70	80	7226.5	26573.7	1557.18	1526.91	-9746.86
80	90	9301.5	8879.89	698.295	685.991	-10580.2
90	100	9999	7965.62	681.467	676.927	-10858.7
100	110	10235	7965.62	700.246	695.579	-11108.5
110	120	11853	43268.8	4513.17	4447.42	-11358.3
120	130	13917	8707.06	1095.01	1075.41	-12715.2
130	140	14603.5	7965.62	1059.58	1052.46	-12988.3
140	150	14839.5	7965.62	1079.56	1072.3	-13238.1
150	160	16157.5	20307.7	3039.16	2982.24	-13487.9
160	170	17481	8097.45	1328.26	1319.29	-14124.7
170	180	17718	8077.05	1345.87	1336.78	-14378.6
180	190	17861	6787.67	1141.67	609.731	-14631.9



Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support mm	Cumulative Shr Wind Cas N	Cumulative Shr Eqk Cas N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1100	40390.7	40390.7	355082560	355082560
20	30	2700	40390.7	40390.7	266872384	266872384
30	40	3225	40390.7	40390.7	226863376	226863376
40	50	4250	15161.8	15161.8	230818496	230818496
50	60	5372.5	15161.8	15161.8	200414352	200414352
60	70	5608.5	15161.8	15161.8	196834720	196834720
70	80	7226.5	15161.8	15161.8	193255104	193255104
80	90	9301.5	15161.8	15161.8	147751312	147751312
90	100	9999	15161.8	15161.8	130239944	130239944
100	110	10235	15161.8	15161.8	126660304	126660304
110	120	11853	15161.8	15161.8	123080688	123080688
120	130	13917	15161.8	15161.8	77576896	77576896
130	140	14603.5	15161.8	15161.8	60399216	60399216
140	150	14839.5	15161.8	15161.8	56819584	56819584
150	160	16157.5	15161.8	15161.8	53239964	53239964
160	170	17481	15161.8	15161.8	16768661	16768661
170	180	17718	15161.8	15161.8	13158700	13158700
180	190	17861	15161.8	15161.8	9579076	9579076

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case + Vortex Shedding.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1100	16046.6	19830.2	161666480	255876320
20	30	2700	14642.8	19732.7	127894464	212339440
30	40	3225	13941.9	19668.9	113596320	192630704
40	50	4250	13903.4	19583.8	112899920	191648976
50	60	5372.5	12355.6	18917.3	86574608	153047072
60	70	5608.5	12188.6	18586.9	83677200	148619792
70	80	7226.5	12021.6	18239.5	80819240	144272512
80	90	9301.5	9700.01	16682.3	48223584	91868600
90	100	9999	8810.06	15984	37536312	73005888
100	110	10235	8643.05	15302.5	35476012	69312584
110	120	11853	8476.04	14602.3	33455146	65782384
120	130	13917	5348.44	10089.1	12710012	28730226
130	140	14603.5	4057.96	8994.11	7384389	17922432
140	150	14839.5	3811.06	7934.53	6455471	15924049
150	160	16157.5	3564.16	6854.96	5584844	14178185
160	170	17481	818.447	3815.8	319904	1350892
170	180	17718	569.778	2487.54	154639	600492
180	190	17861	325.105	1141.67	49000	172073

The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1100	13510	19830.2	124878792	255876320
20	30	2700	12106.2	19732.7	96689520	212339440
30	40	3225	11405.3	19668.9	84928960	192630704
40	50	4250	11366.8	19583.8	84359432	191648976
50	60	5372.5	9819.07	18917.3	63120756	153047072
60	70	5608.5	9652.06	18586.9	60822232	148619792
70	80	7226.5	9485.05	18239.5	58563152	144272512
80	90	9301.5	7163.44	16682.3	33580292	91868600
90	100	9999	6273.49	15984	25822684	73005888
100	110	10235	6106.48	15302.5	24361258	69312584
110	120	11853	5939.47	14602.3	22939264	65782384
120	130	13917	3617.86	10089.1	8597463	28730226
130	140	14603.5	2744.93	8994.11	4995039	17922432
140	150	14839.5	2577.92	7934.53	4366689	15924049
150	160	16157.5	2410.91	6854.96	3777769	14178185
160	170	17481	553.624	3815.8	216393	1350892
170	180	17718	385.416	2487.54	104603	600492
180	190	17861	219.912	1141.67	33145.2	172073

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	40390.7	0.3381	1.29362	0.0011226
20	30	40390.7	1.9083	2.62421	0.0015276
30	40	40390.7	2.66247	2.70086	0.0015383
40	50	15161.8	4.33601	6.15619	0.0019084
50	60	15161.8	6.38149	6.60703	0.0019123
60	70	15161.8	6.8328	7.0588	0.0019162
70	80	15161.8	10.1124	13.4953	0.0023552
80	90	15161.8	14.8701	16.2843	0.0024924
90	100	15161.8	16.5785	16.8728	0.0024949
100	110	15161.8	17.1673	17.4619	0.0024974
110	120	15161.8	21.3196	25.371	0.0027558
120	130	15161.8	26.9354	28.5187	0.0028225
130	140	15161.8	28.8518	29.185	0.0028237
140	150	15161.8	29.5182	29.8515	0.0028248
150	160	15161.8	33.2703	36.7324	0.0028968
160	170	15161.8	37.0771	37.4219	0.0028971
170	180	15161.8	37.7638	38.1056	0.0028973
180	190	15161.8	38.1781	38.2505	0.0028974

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)
 Allowable deflection : 89.430 Actual deflection : 38.251 mm

The following table is for Wind Deflection; Operating Case + Vortex Shedding.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	16046.6	0.1565	0.60315	0.00052776
20	30	14642.8	0.89307	1.23306	0.00072809
30	40	13941.9	1.25129	1.26959	0.00073342
40	50	13903.4	2.05006	2.91692	0.0009063
50	60	12355.6	3.02391	3.13101	0.00090802
60	70	12188.6	3.23821	3.3455	0.00090968
70	80	12021.6	4.78332	6.3459	0.0010764
80	90	9700.01	6.97149	7.60937	0.0011192
90	100	8810.06	7.74146	7.87358	0.0011199
100	110	8643.05	8.00575	8.13797	0.0011206
110	120	8476.04	9.84763	11.6005	0.0011788
120	130	5348.44	12.267	12.9361	0.0011885
130	140	4057.96	13.0764	13.2166	0.0011886
140	150	3811.06	13.3569	13.4972	0.0011887
150	160	3564.16	14.9263	16.3583	0.0011937
160	170	818.447	16.5004	16.6424	0.0011937
170	180	569.778	16.7833	16.9242	0.0011938
180	190	325.105	16.954	16.9838	0.0011938

Allowable deflection at the Tower Top (Emp)(6.000"/100ft. Criteria)
 Allowable deflection : 89.430 Actual deflection : 16.984 mm

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

		Cumulative	Centroid	Elem. End	Elem. Ang.
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From	To	Shear N	Deflection mm	Deflection mm	Rotation
10	20	13510	0.1205	0.46285	0.00040362
20	30	12106.2	0.68431	0.94341	0.00055425
30	40	11405.3	0.9573	0.97123	0.00055823
40	50	11366.8	1.56448	2.22154	0.00068593
50	60	9819.07	2.30251	2.38357	0.00068719
60	70	9652.06	2.46469	2.54588	0.0006884
70	80	9485.05	3.63124	4.80529	0.00080701
80	90	7163.44	5.27395	5.75111	0.0008366
90	100	6273.49	5.84984	5.94861	0.00083711
100	110	6106.48	6.0474	6.14622	0.00083759
110	120	5939.47	7.42227	8.72774	0.00087719
120	130	3617.86	9.22357	9.72125	0.00088377
130	140	2744.93	9.82554	9.92983	0.00088386
140	150	2577.92	10.0341	10.1384	0.00088394
150	160	2410.91	11.201	12.2655	0.00088733
160	170	553.624	12.371	12.4766	0.00088734
170	180	385.416	12.5813	12.6861	0.00088734
180	190	219.912	12.7082	12.7304	0.00088734

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	28.8173	180.108
20	30	31.653	197.831
30	40	34.8077	217.548
40	50	34.9495	218.434
50	60	31.96	199.75
60	70	31.96	199.75
70	80	34.9495	218.434
80	90	34.9495	218.434
90	100	31.96	199.75
100	110	31.96	199.75
110	120	34.9495	218.434
120	130	34.9495	218.434
130	140	31.96	199.75
140	150	31.96	199.75
150	160	34.9495	218.434
160	170	31.9184	199.49
170	180	31.6717	197.948
180	190	34.7723	217.327

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 89.430 Actual deflection : 12.730 mm

Total Deflection in the Operating Condition + Applied Forces:
 Allowable deflection : 89.430 Actual deflection : 50.981 mm

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Longitudinal Stress Constants:

From	To	Metal Area New mm ²	Metal Area Corroded mm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	40061.6	36055.4	7828326	7058846
20	30	40061.6	36055.4	7828326	7058846
30	40	74644.4	67434.4	14260671	12977350
40	50	79821.7	72611.7	15255562	13978457
50	60	79821.7	72611.7	15255562	13978457
60	70	79821.7	72611.7	15255562	13978457
70	80	79821.7	72611.7	15255562	13978457
80	90	79821.7	72611.7	15255562	13978457
90	100	79821.7	72611.7	15255562	13978457
100	110	79821.8	72611.8	15255572	13978468
110	120	79821.8	72611.8	15255572	13978468
120	130	79821.7	72611.7	15255562	13978457
130	140	79821.7	72611.7	15255562	13978457
140	150	79821.7	72611.7	15255562	13978457
150	160	79821.7	72611.7	15255562	13978457
160	170	79821.7	72611.7	15255562	13978457
170	180	73354	66144	14012853	12727970
180	190	73354	66144	14012853	12727970



Longitudinal Allowable Stresses:

From	To	Tensile MPa	Hydrotest Tensile MPa	Compressive MPa	Hydrotest Compressive MPa
10	20	115.918	196.556	-141.502	-141.502
20	30	82.7985	140.397	-141.502	-141.502
30	40	165.597	280.795	-147.277	-147.277
40	50	165.597	280.795	-147.277	-147.277
50	60	165.597	269.995	-147.277	-147.277
60	70	165.597	269.995	-147.277	-147.277
70	80	165.597	280.795	-147.277	-147.277
80	90	165.597	280.795	-147.277	-147.277
90	100	165.597	269.995	-147.277	-147.277
100	110	165.597	269.995	-147.277	-147.277
110	120	165.597	280.795	-147.277	-147.277
120	130	165.597	280.795	-147.277	-147.277
130	140	165.597	269.995	-147.277	-147.277
140	150	165.597	269.995	-147.277	-147.277
150	160	165.597	280.795	-147.277	-147.277
160	170	165.597	269.995	-147.277	-147.277
170	180	165.597	269.995	-147.277	-147.277
180	190	165.597	280.795	-147.277	-147.277



Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure MPa	Longitudinal Stress External Pressure MPa	Longitudinal Stress Hydrotest Pressure MPa
10	20	...	...	...
20	30	...	...	...
30	40	64.9847	-0.81392	84.4801
40	50	60.3733	-0.76326	78.4853
50	60	...	...	...
60	70	...	...	...
70	80	60.3733	-0.76326	78.4853
80	90	60.3733	-0.76326	78.4853
90	100	...	...	...
100	110	...	...	...
110	120	60.3733	-0.76326	78.4852
120	130	60.3733	-0.76326	78.4853
130	140	...	...	...
140	150	...	...	...
150	160	60.3733	-0.76326	78.4853
160	170	...	...	...
170	180	...	...	...
180	190	66.2463	-0.82778	86.1202

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty MPa	Wght. Str. Operating MPa	Wght. Str. Hydrotest MPa	Wght. Str. Emp. Mom. MPa	Wght. Str. Opr. Mom. MPa
10	20	-6.03727	-6.14871	-7.49755	1.72387	1.72387
20	30	-5.62341	-5.73485	-7.08369	1.72387	1.72387
30	40	-2.95578	-2.95578	-2.95578	0.93767	0.93767
40	50	-2.70388	-2.70388	-2.70388	0.87052	0.87052
50	60	-2.45839	-2.45839	-2.45839	0.54835	0.54835
60	70	-2.34868	-2.34868	-2.34868	0.54835	0.54835
70	80	-2.23897	-2.23897	-2.23897	0.54835	0.54835
80	90	-1.87297	-1.87297	-1.87297	0.46128	0.46128
90	100	-1.75067	-1.75067	-1.75067	0.46128	0.46128
100	110	-1.64096	-1.64096	-1.64096	0.46128	0.46128
110	120	-1.53125	-1.53125	-1.53125	0.46128	0.46128
120	130	-0.9353	-0.9353	-0.9353	0.37411	0.37411
130	140	-0.81538	-0.81538	-0.81538	0.37411	0.37411
140	150	-0.70567	-0.70567	-0.70567	0.37411	0.37411
150	160	-0.59596	-0.59596	-0.59596	0.37411	0.37411
160	170	-0.31626	-0.31626	-0.31626	0.3702	0.3702
170	180	-0.22475	-0.22475	-0.22475	0.40657	0.40657
180	190	-0.10263	-0.10263	-0.10263	0.40657	0.40657

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. MPa	Bend. Str. Oper. Wind MPa	Bend. Str. Oper. Equ. MPa	Bend. Str. Hyd. Wind MPa	Bend. Str. Hyd. Equ. MPa
10	20	1.72387	17.6854	36.2374	8.84272	...
20	30	1.72387	13.6932	30.0717	6.84662	...
30	40	0.93767	6.5423	14.8389	3.27115	...



40	50	0.87052	6.03303	13.7059	3.01651	...
50	60	0.54835	4.51413	10.9453	2.25706	...
60	70	0.54835	4.34975	10.6287	2.17487	...
70	80	0.54835	4.18819	10.3178	2.09409	...
80	90	0.46128	2.40152	6.57005	1.20076	...
90	100	0.46128	1.84673	5.22107	0.92336	...
100	110	0.46128	1.74221	4.95694	0.87111	...
110	120	0.46128	1.64052	4.70447	0.82026	...
120	130	0.37411	0.61485	2.05466	0.30743	...
130	140	0.37411	0.35722	1.28174	0.17861	...
140	150	0.37411	0.31229	1.13882	0.15614	...
150	160	0.37411	0.27017	1.01396	0.13508	...
160	170	0.3702	0.015476	0.09661	0.0077378	...
170	180	0.40657	0.0082157	0.047164	0.0041078	...
180	190	0.40657	0.0026033	0.013515	0.0013016	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case MPa	Vortex Shedding Empty Case MPa	Vortex Shedding Test Case MPa	Earthquake Empty Case MPa
10	20	22.8953	...	...	34.3428
20	30	18.1125	...	...	28.4788
30	40	8.75063	...	...	14.047
40	50	8.07412	...	...	12.9742
50	60	6.19145	...	...	10.3363
60	70	5.98424	...	...	10.0326
70	80	5.77985	...	...	9.73449
80	90	3.44875	...	...	6.14632
90	100	2.68444	...	...	4.85698
100	110	2.53709	...	...	4.60489
110	120	2.39257	...	...	4.3644
120	130	0.90897	...	...	1.85921
130	140	0.5281	...	...	1.13741
140	150	0.46167	...	...	1.00493
150	160	0.3994	...	...	0.89038
160	170	0.022878	...	...	0.072529
170	180	0.012146	...	...	0.030919
180	190	0.0038486	...	...	0.0072179

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind MPa	Longitudinal Stress Y Forces Seismic MPa
10	20	-0.41176	-0.21893
20	30	-0.41176	-0.23191
30	40	-0.22016	-0.12559
40	50	-0.20446	-0.11824
50	60	-0.20446	-0.12736
60	70	-0.20446	-0.1308
70	80	-0.20446	-0.13424
80	90	-0.20446	-0.14572
90	100	-0.20446	-0.14956
100	110	-0.20446	-0.153
110	120	-0.20446	-0.15644
120	130	-0.20446	-0.17513
130	140	-0.20446	-0.17889
140	150	-0.20446	-0.18233
150	160	-0.20446	-0.18577
160	170	-0.20446	-0.19454
170	180	-0.22445	-0.2174
180	190	-0.22445	-0.22123

Longitudinal Stresses due to User Forces and Moments:



From	To	Wind For/Mom Corroded MPa	Earthquake For/Mom Corroded MPa	Wind For/Mom No Corrosion MPa	Earthquake For/Mom No Corrosion MPa
10	20	50.2871	50.2871	45.3442	45.3442
20	30	37.7947	37.7947	34.0797	34.0797
30	40	17.4759	17.4759	15.9032	15.9032
40	50	16.5072	16.5072	15.1253	15.1253
50	60	14.3328	14.3328	13.1329	13.1329
60	70	14.0768	14.0768	12.8984	12.8984
70	80	13.8208	13.8208	12.6638	12.6638
80	90	10.5665	10.5665	9.68198	9.68198
90	100	9.31421	9.31421	8.53448	8.53448
100	110	9.0582	9.0582	8.2999	8.2999
110	120	8.8022	8.8022	8.06533	8.06533
120	130	5.54797	5.54797	5.08353	5.08353
130	140	4.31949	4.31949	3.95789	3.95789
140	150	4.06349	4.06349	3.72332	3.72332
150	160	3.8075	3.8075	3.48875	3.48875
160	170	1.19922	1.19922	1.09883	1.09883
170	180	1.03351	1.03351	0.93874	0.93874
180	190	0.75236	0.75236	0.68337	0.68337

Bending Stresses due to Blast Load:

From	To	Bending Stress MPa
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...
80	90	...
90	100	...
100	110	...
110	120	...
120	130	...
130	140	...
140	150	...
150	160	...
160	170	...
170	180	...
180	190	...



Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	63.25	115.92	-76.15	141.50	0.5456	0.5381
20	47.18	82.80	-59.25	141.50	0.5698	0.4187
30	21.78	165.60	-28.13	147.28	0.1315	0.1910
40	20.50	165.60	-26.32	147.28	0.1238	0.1787
70	16.11	165.60	-21.00	147.28	0.0973	0.1426
80	11.35	165.60	-15.51	147.28	0.0686	0.1053
110	9.17	165.60	-12.64	147.28	0.0554	0.0858
120	5.40	165.60	-7.68	147.28	0.0326	0.0521
150	3.65	165.60	-5.25	147.28	0.0220	0.0357
180	0.83	165.60	-1.49	147.28	0.0050	0.0101

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	80.10	115.92	-92.61	141.50	0.6910	0.6545
20	62.14	82.80	-73.85	141.50	0.7505	0.5219
30	29.38	165.60	-35.54	147.28	0.1774	0.2413
40	27.53	165.60	-33.17	147.28	0.1662	0.2252
70	21.73	165.60	-26.48	147.28	0.1312	0.1798
80	15.16	165.60	-19.19	147.28	0.0915	0.1303
110	11.94	165.60	-15.32	147.28	0.0721	0.1040
120	6.67	165.60	-8.89	147.28	0.0403	0.0604
150	4.29	165.60	-5.85	147.28	0.0259	0.0397



180 0.84 165.60 -1.49 147.28 0.0051 0.0101

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	63.14	115.92	-76.26	141.50	0.5447	0.5389
20	47.07	82.80	-59.36	141.50	0.5684	0.4195
30	21.78	165.60	-28.13	147.28	0.1315	0.1910
40	20.50	165.60	-26.32	147.28	0.1238	0.1787
70	16.11	165.60	-21.00	147.28	0.0973	0.1426
80	11.35	165.60	-15.51	147.28	0.0686	0.1053
110	9.17	165.60	-12.64	147.28	0.0554	0.0858
120	5.40	165.60	-7.68	147.28	0.0326	0.0521
150	3.65	165.60	-5.25	147.28	0.0220	0.0357
180	0.83	165.60	-1.49	147.28	0.0050	0.0101

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	81.88	115.92	-94.62	141.50	0.7064	0.6687
20	63.62	82.80	-75.56	141.50	0.7684	0.5340
30	30.17	165.60	-36.33	147.28	0.1822	0.2467
40	28.26	165.60	-33.91	147.28	0.1707	0.2302
70	22.31	165.60	-27.06	147.28	0.1347	0.1837
80	15.58	165.60	-19.62	147.28	0.0941	0.1332
110	12.28	165.60	-15.66	147.28	0.0742	0.1063
120	6.87	165.60	-9.09	147.28	0.0415	0.0617
150	4.41	165.60	-5.98	147.28	0.0267	0.0406
180	0.85	165.60	-1.50	147.28	0.0051	0.0102

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	3.07	196.56	-18.06	141.50	0.0156	0.1277
20	1.49	140.40	-15.65	141.50	0.0106	0.1106
30	1.25	280.79	-7.16	147.28	0.0045	0.0486
40	1.18	280.79	-6.59	147.28	0.0042	0.0448
70	0.40	280.79	-4.88	147.28	0.0014	0.0331
80		280.79	-3.54	147.28		0.0240
110		280.79	-2.81	147.28		0.0191
120		280.79	-1.62	147.28		0.0110
150		280.79	-1.11	147.28		0.0075
180	0.31	280.79	-0.51	147.28	0.0011	0.0035

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		196.56	-9.22	141.50		0.0652
20		140.40	-8.81	141.50		0.0622
30		280.79	-3.89	147.28		0.0264
40		280.79	-3.57	147.28		0.0243
70		280.79	-2.79	147.28		0.0189
80		280.79	-2.33	147.28		0.0158
110		280.79	-1.99	147.28		0.0135
120		280.79	-1.31	147.28		0.0089
150		280.79	-0.97	147.28		0.0066
180	0.30	280.79	-0.51	147.28	0.0011	0.0035

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	63.14	115.92	-76.26	141.50	0.5447	0.5389
20	47.07	82.80	-59.36	141.50	0.5684	0.4195
30	86.76	165.60		147.28	0.5240	
40	80.88	165.60		147.28	0.4884	



70	76.49	165.60	147.28	0.4619
80	71.73	165.60	147.28	0.4331
110	69.54	165.60	147.28	0.4199
120	65.77	165.60	147.28	0.3972
150	64.02	165.60	147.28	0.3866
180	67.08	165.60	147.28	0.4051

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	81.88	115.92	-94.62	141.50	0.7064	0.6687
20	63.62	82.80	-75.56	141.50	0.7684	0.5340
30	95.16	165.60		147.28	0.5746	
40	88.63	165.60		147.28	0.5352	
70	82.69	165.60		147.28	0.4993	
80	75.95	165.60		147.28	0.4587	
110	72.65	165.60		147.28	0.4387	
120	67.24	165.60		147.28	0.4060	
150	64.79	165.60		147.28	0.3912	
180	67.09	165.60		147.28	0.4052	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	63.14	115.92	-76.26	141.50	0.5447	0.5389
20	47.07	82.80	-59.36	141.50	0.5684	0.4195
30	20.97	165.60	-28.95	147.28	0.1266	0.1965
40	19.74	165.60	-27.08	147.28	0.1192	0.1839
70	15.35	165.60	-21.76	147.28	0.0927	0.1478
80	10.59	165.60	-16.27	147.28	0.0639	0.1105
110	8.41	165.60	-13.40	147.28	0.0508	0.0910
120	4.63	165.60	-8.44	147.28	0.0280	0.0573
150	2.89	165.60	-6.02	147.28	0.0174	0.0408
180	0.01	165.60	-2.32	147.28	0.0000	0.0157

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	81.88	115.92	-94.62	141.50	0.7064	0.6687
20	63.62	82.80	-75.56	141.50	0.7684	0.5340
30	29.36	165.60	-37.15	147.28	0.1773	0.2522
40	27.50	165.60	-34.67	147.28	0.1661	0.2354
70	21.55	165.60	-27.82	147.28	0.1301	0.1889
80	14.82	165.60	-20.38	147.28	0.0895	0.1384
110	11.52	165.60	-16.42	147.28	0.0695	0.1115
120	6.10	165.60	-9.85	147.28	0.0369	0.0669
150	3.65	165.60	-6.74	147.28	0.0220	0.0458
180	0.02	165.60	-2.32	147.28	0.0001	0.0158

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	3.07	196.56	-18.06	141.50	0.0156	0.1277
20	1.49	140.40	-15.65	141.50	0.0106	0.1106
30	85.73	280.79		147.28	0.3053	
40	79.67	280.79		147.28	0.2837	
70	78.89	280.79		147.28	0.2809	
80	78.27	280.79		147.28	0.2788	
110	78.24	280.79		147.28	0.2786	
120	78.23	280.79		147.28	0.2786	
150	78.40	280.79		147.28	0.2792	
180	86.43	280.79		147.28	0.3078	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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10		196.56	-9.22	141.50		0.0652
20		140.40	-8.81	141.50		0.0622
30	82.46	280.79		147.28	0.2937	
40	76.65	280.79		147.28	0.2730	
70	76.79	280.79		147.28	0.2735	
80	77.07	280.79		147.28	0.2745	
110	77.42	280.79		147.28	0.2757	
120	77.92	280.79		147.28	0.2775	
150	78.26	280.79		147.28	0.2787	
180	86.42	280.79		147.28	0.3078	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-7.76	141.50		0.0548
20		82.80	-7.35	141.50		0.0519
30	62.97	165.60		147.28	0.3802	
40	58.54	165.60		147.28	0.3535	
70	58.68	165.60		147.28	0.3544	
80	58.96	165.60		147.28	0.3561	
110	59.30	165.60		147.28	0.3581	
120	59.81	165.60		147.28	0.3612	
150	60.15	165.60		147.28	0.3632	
180	66.55	165.60		147.28	0.4019	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-5.53	141.50		0.0391
20		82.80	-5.16	141.50		0.0365
30	62.31	165.60		147.28	0.3763	
40	57.91	165.60		147.28	0.3497	
70	58.34	165.60		147.28	0.3523	
80	58.67	165.60		147.28	0.3543	
110	58.98	165.60		147.28	0.3562	
120	59.52	165.60		147.28	0.3594	
150	59.83	165.60		147.28	0.3613	
180	66.15	165.60		147.28	0.3995	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	18.47	115.92	-30.77	141.50	0.1593	0.2174
20	14.10	82.80	-25.57	141.50	0.1703	0.1807
30	71.72	165.60		147.28	0.4331	
40	66.61	165.60		147.28	0.4023	
70	64.46	165.60		147.28	0.3893	
80	62.41	165.60		147.28	0.3769	
110	61.70	165.60		147.28	0.3726	
120	60.72	165.60		147.28	0.3667	
150	60.55	165.60		147.28	0.3657	
180	66.55	165.60		147.28	0.4019	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-7.76	141.50		0.0548
20		82.80	-7.35	141.50		0.0519
30	62.97	165.60		147.28	0.3802	
40	58.54	165.60		147.28	0.3535	
70	58.68	165.60		147.28	0.3544	
80	58.96	165.60		147.28	0.3561	
110	59.30	165.60		147.28	0.3581	
120	59.81	165.60		147.28	0.3612	
150	60.15	165.60		147.28	0.3632	



180 66.55 165.60 147.28 0.4019

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	18.47	115.92	-30.77	141.50	0.1593	0.2174
20	14.10	82.80	-25.57	141.50	0.1703	0.1807
30	6.73	165.60	-12.64	147.28	0.0407	0.0859
40	6.24	165.60	-11.65	147.28	0.0377	0.0791
70	4.09	165.60	-8.57	147.28	0.0247	0.0582
80	2.04	165.60	-5.78	147.28	0.0123	0.0393
110	1.32	165.60	-4.39	147.28	0.0080	0.0298
120	0.35	165.60	-2.22	147.28	0.0021	0.0151
150	0.18	165.60	-1.37	147.28	0.0011	0.0093
180	0.31	165.60	-0.51	147.28	0.0019	0.0035

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	45.64	96.60	-58.38	117.92	0.4725	0.4951
20	33.55	69.00	-45.49	117.92	0.4863	0.3857
30	80.32	138.00		122.73	0.5820	
40	74.93	138.00		122.73	0.5430	
70	72.37	138.00		122.73	0.5244	
80	69.38	138.00		122.73	0.5028	
110	67.95	138.00		122.73	0.4924	
120	65.18	138.00		122.73	0.4724	
150	63.77	138.00		122.73	0.4621	
180	67.08	138.00		122.73	0.4861	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	45.64	96.60	-58.38	117.92	0.4725	0.4951
20	33.55	69.00	-45.49	117.92	0.4863	0.3857
30	14.52	138.00	-22.31	122.73	0.1052	0.1818
40	13.79	138.00	-20.96	122.73	0.0999	0.1708
70	11.23	138.00	-17.51	122.73	0.0814	0.1426
80	8.25	138.00	-13.81	122.73	0.0598	0.1125
110	6.81	138.00	-11.71	122.73	0.0494	0.0954
120	4.05	138.00	-7.80	122.73	0.0293	0.0635
150	2.64	138.00	-5.73	122.73	0.0191	0.0467
180	0.01	138.00	-2.31	122.73	0.0001	0.0188

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		115.92	-7.87	141.50		0.0556
20		82.80	-7.46	141.50		0.0527
30	62.97	165.60		147.28	0.3802	
40	58.54	165.60		147.28	0.3535	
70	58.68	165.60		147.28	0.3544	
80	58.96	165.60		147.28	0.3561	
110	59.30	165.60		147.28	0.3581	
120	59.81	165.60		147.28	0.3612	
150	60.15	165.60		147.28	0.3632	
180	66.55	165.60		147.28	0.4019	

Absolute Maximum of the all of the Stress Ratio's 0.7684

Governing Element: Skirt-1

Governing Load Case 4 : NP+OW+EQ+FS+BS

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
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Skirt-2	...	-6.4	1.7	86.5	81.9	+ Tens
Skirt-2	...	-6.4	-1.7	-86.5	-94.6	- Comp
Skirt-1	...	-6.0	1.7	67.9	63.6	+ Tens
Skirt-1	...	-6.0	-1.7	-67.9	-75.6	- Comp
Bottom head	...	-3.1	0.9	32.3	30.2	+ Tens
Bottom head	...	-3.1	-0.9	-32.3	-36.3	- Comp
Shell-1	...	-2.8	0.9	30.2	28.3	+ Tens
Shell-1	...	-2.8	-0.9	-30.2	-33.9	- Comp
Shell-2	...	-2.4	0.5	24.1	22.3	+ Tens
Shell-2	...	-2.4	-0.5	-24.1	-27.1	- Comp
Shell-3	...	-2.0	0.5	17.1	15.6	+ Tens
Shell-3	...	-2.0	-0.5	-17.1	-19.6	- Comp
Shell-4	...	-1.7	0.5	13.5	12.3	+ Tens
Shell-4	...	-1.7	-0.5	-13.5	-15.7	- Comp
Shell-5	...	-1.1	0.4	7.6	6.9	+ Tens
Shell-5	...	-1.1	-0.4	-7.6	-9.1	- Comp
Shell-6	...	-0.8	0.4	4.8	4.4	+ Tens
Shell-6	...	-0.8	-0.4	-4.8	-6.0	- Comp
Top head	...	-0.3	0.4	0.8	0.8	+ Tens
Top head	...	-0.3	-0.4	-0.8	-1.5	- Comp

Fatigue Stress Calculations due to Wind Induced Vortex Shedding Loads

From	Elevation	Alternating Stress	Sif	Number of Cycles	Hours of Total Vibration
10	2200.00	47.54	2.0	1000000	230.72
20	3200.00	56.42	3.0	1000000	230.72
30	3250.00	18.17	2.0	1000000	230.72
40	5250.00	16.77	2.0	1000000	230.72
50	5490.50	12.86	2.0	1000000	230.72
60	5726.50	12.43	2.0	1000000	230.72
70	8726.50	12.00	2.0	1000000	230.72
80	9876.50	7.16	2.0	1000000	230.72
90	10117.00	5.57	2.0	1000000	230.72
100	10353.00	5.27	2.0	1000000	230.72
110	13353.00	4.97	2.0	1000000	230.72
120	14481.00	1.89	2.0	1000000	230.72
130	14721.50	1.10	2.0	1000000	230.72
140	14957.50	0.96	2.0	1000000	230.72
150	17357.50	0.83	2.0	1000000	230.72
160	17600.00	0.05	2.0	1000000	230.72
170	17836.00	0.03	2.0	1000000	230.72
180	17886.00	0.01	2.0	1000000	230.72

Cycle Life Determination based on ASME Div. II, Figure 3-F.1

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Skirt Support Analysis Per EN 13445-3 Section 16.12:

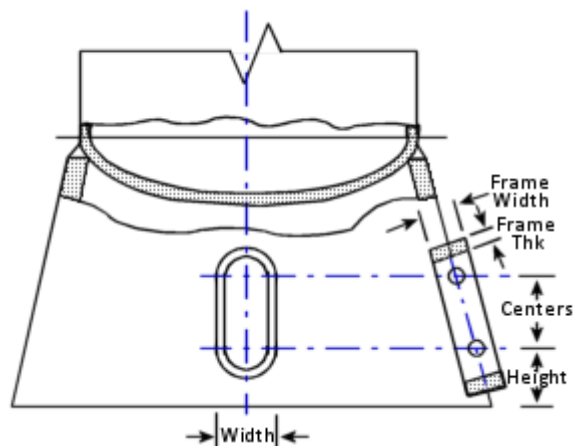
The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

The skirt comprises more than one piece. If this is the case, the Openings will only be considered in this Section of the Skirt! Otherwise, make the whole skirt all one piece.

Item: Skirt-2

Total vessel (empty) weight	Fg	201653.641 N
Weight of vessel contents (liquid)	Ff	4017.686 N
Vessel Gross Weight	Fg + Ff	205671.328 N
Skirt Mean Diameter	Dz	797.000 mm
Skirt Analysis Thickness (Corroded)	ez	14.400 mm
Skirt Height	H	2200.000 mm
Additional Axial Force at section 1-1	F1	-14844.800 N
Moment at section 1-1	M1	479211840.000 N-mm
Applied Moment - Bottom of Skirt	Mb	610958976.000 N-mm
Weight at Bottom of Skirt	W	221675.406 N



Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Obround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	300.0000	0.0	700.00	0.00	80.000	18.000

Section properties at least metal area [A]:

Number of Openings At Critical Height = 1

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
1	-22.1116	22.1116

The points for integrating Ixx, Iyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
-337.888	-22.112



Height of Weakest Section	H	700.030	mm
Rotation of Principle Axes	a	-0.000	deg.
Centroid Position in X direction	g	-19.076	mm
Centroid Position in Y direction	h	0.000	mm
Principle Moment of Inertia	Ixx	2886281216.000	mm ⁴
Principle Moment of Inertia	Iyy	2574906112.000	mm ⁴
Distance to Extreme Fiber	ybar	405.700	mm
Distance to Extreme Fiber	xbar	424.776	mm
Minimum Section Modulus	W4	6061800.500	mm ³
Metal Area at the Cutting Plane	A4	34506.309	mm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:

$$= \sin^{-1}(\text{Largest Opening Width} / D_z)$$

$$= \sin^{-1}(300/797)$$

$$= 0.39 \text{ rad (22.1 deg)}$$

Skirt Mean Radius at Section 4-4 [r]:

$$= D_z / 2$$

$$= 797/2$$

$$= 398.5 \text{ mm}$$

This calculation of ratio below is per an earlier version of EN 13445.

Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:

$$= \delta \cdot \sqrt{(r / e_z)}$$

$$= 0.39 \cdot \sqrt{(398/14.4)}$$

$$= 2.030$$

WARNING: ratio Exceeds 2 - See EN 13445 16.12.76(a).

This equation is a precaution against radial buckling at the opening.
The frame (if there is one), may prevent this. The code does not
say what size this frame should be. Your judgement is required.

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:

$$= e_x(\text{new}) - c_i - c_{ext}$$

$$= 16 - 0.8 - 0.8$$

$$= 14.4 \text{ mm}$$

Find the moment acting at the Critical Height [M4]:

$$= (M_{\text{bottom}} - M_{\text{top}}) \cdot (H - \text{CutHeight}) / H + M_{\text{top}}$$

$$= (610958912 - 479211808) \cdot (2200 - 700) / 2200 + 479211808$$

$$= 569037568 \text{ N-mm}$$

Compute the section modulus [Zxx and Zyy]:

$$Z_{xx} = I_{xx} / y_{\text{bar}} = 2.89\text{E}+09 / 406 = 7114331 \text{ mm}^3$$

$$Z_{yy} = I_{yy} / x_{\text{bar}} = 2.57\text{E}+09 / 425 = 6061800 \text{ mm}^3$$

$$Z = \min(Z_{xx}, Z_{yy}) = \min(7114331, 6061800) = 6061800 \text{ mm}^3$$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area}(A_4))}$$

$$= \sqrt{(\min(2.89\text{E}+09, 2.57\text{E}+09) / 34506)}$$

$$= 273 \text{ mm}$$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:

$$= r_g / L$$

$$= 273/2200$$

$$= 0.124$$

Location of the Skirt Centroid at critical height [e]:

$$= \sqrt{(e_x^2 + e_y^2)}$$

$$= \sqrt{(-19.1^2 + 0^2)}$$

$$= 24.625 \text{ mm}$$



Where $e_x = -19.1$: $e_y = 0$

Moment Increment at Centroid [ΔM_4]:
 $= \text{abs}(F_1 - F_f - F_g) \cdot e = F_4 \cdot e$
 $= \text{abs}(-14845 - 201654 - 4018) \cdot 19.1$
 $= 4208314.000 \text{ N-mm}$

Check of the Skirt in Regions with Openings, per 16.12.4.3:

Protrusion, Figure 16.12-5 [ht]:
 $= \text{abs}(l_t - ea_3) / 2$
 $= \text{abs}(80 - 14.4) / 2$
 $= 32.800 \text{ mm}$

Mean Diameter of Opening, Figure 16.12-5 [d]:
 $= \text{opening width} - e_t$
 $= 300 - 18$
 $= 282.000 \text{ mm}$

Angle, 16.12-72 [δ]:
 $= \sin^{-1}(d / D_3)$
 $= \sin^{-1}(282/797)$
 $= 0.362 \text{ radians}$

Area in the Frame [At]:
 $= 2 \cdot l_t \cdot e_t$
 $= 2 \cdot 80 \cdot 18$
 $= 2880.000 \text{ mm}^2$

Cross Sectional Shell Area at Opening, 16.12-73(b) [AS]:
 $= (\pi - \delta) \cdot D_3 \cdot ea_3$
 $= (\pi - 0.36) \cdot 797 \cdot 14.4$
 $= 31904.738 \text{ mm}^2$

Combined Area at Opening, 16.12-73 [A4]:
 $= AS + AT$
 $= 31905 + 2880$
 $= 34784.738 \text{ mm}^2$

Computed Offset, 16.12-74(b) [yt]:
 $= 0.5 \cdot D_3 \cdot \cos(\delta) + h_t - 0.5 \cdot l_t$
 $= 0.5 \cdot 797 \cdot \cos(0.36) + 32.8 - 0.5 \cdot 80$
 $= 365.521 \text{ mm}$

Distance to Neutral Axis, 16.12-74 [yG]:
 $= (0.5 \cdot D_3 \cdot ea_3 \cdot d - 2 \cdot l_t \cdot e_t \cdot y_t) / A_4$
 $= (0.5 \cdot 797 \cdot 14.4 \cdot 282 - 2 \cdot 80 \cdot 18 \cdot 366) / 34785$
 $= 16.258 \text{ mm}$

Computed Offset [ymax]:
 $= \max[\text{abs}(0.5 \cdot d_3 \cdot \cos(\delta) + h_t + y_G); \text{abs}(0.5 \cdot D_3 - y_G)]$
 $= \max[\text{abs}(0.5 \cdot 797 \cdot \cos(0.36) + 32.8 + 16.3); \text{abs}(0.5 \cdot 797 - 16.3)]$
 $= 421.779 \text{ mm}$

Inertia Value, 16.12-75(b) [IS]:
 $= (\pi - \delta - \sin(\delta) \cdot \cos(\delta)) \cdot ea_3 \cdot (D_3/2)^3$
 $= (\pi - 0.36 - \sin(0.36) \cdot \cos(0.36)) \cdot 14.4 \cdot (797/2)^3$
 $= 2231695616.0 \text{ mm}^4$

Section Modulus [W4]:
 $= (I_s + A_t(y_t^2 + l_t^2/12) - A_4 \cdot y_G^2) / y_{\max}$
 $= (2.23E+09 + 2880(366^2 + 80^2/12) - 34785 \cdot 16.3^2) / 422$
 $= 6185283.000 \text{ mm}^3$

Weakening Factor, 16.12-70(b) [psi1]:
 $= \min(1; A_4 / (\pi \cdot D_3 \cdot ea_3))$
 $= \min(1; 34785 / (\pi \cdot 797 \cdot 14.4))$
 $= 0.965$

Weakening Factor, 16.12-70(c) [psi2]:



$$\begin{aligned} &= \min(1; 4 \cdot W4 / (\pi \cdot D^2 / 3 / (\pi \cdot D3 \cdot ea3))) \\ &= \min(1; 4.0 \cdot 6185283 / (\pi \cdot 797^2 \cdot 14.4)) \\ &= 0.861 \end{aligned}$$

Maximum Bending Moment [Mmax]:

$$\begin{aligned} &= \pi/4 \cdot d3^2 \cdot ea3 \cdot \sigma_{call} \\ &= \pi/4 \cdot 797^2 \cdot 14.4 \cdot 142 \\ &= 1016884352.0 \text{ N-mm} \end{aligned}$$

Maximum Compressive Force [Fc,max]:

$$\begin{aligned} &= \pi \cdot d3 \cdot ea3 \cdot \sigma_{call} \\ &= \pi \cdot 797 \cdot 14.4 \cdot 142 \\ &= 5101493.0 \text{ N} \end{aligned}$$

Unity Check of Weakened Section, 16.12-70 [UC1]:

$$\begin{aligned} &= F4 / (\psi_{i1} \cdot F_{c,max}) + (\text{abs}(M4) + F4 \cdot \text{abs}(y_q)) / (\psi_{i2} \cdot M_{max}) \\ &= 220516 / (0.96 \cdot 5101493) + (\text{abs}(569037568) + 220516 \cdot \text{abs}(16.3)) / \\ &\quad (0.86 \cdot 1016884352) \\ &= 0.699 \text{ must be } \leq \text{ to } 1, \text{ Passed} \end{aligned}$$

The stresses are satisfactory in the region with the opening(s).

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Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

Item: Skirt-1

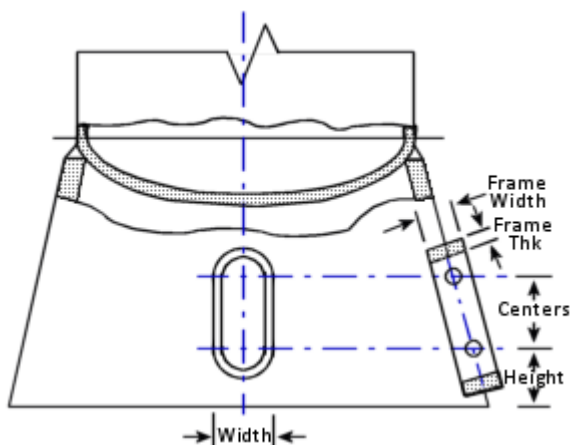
Total vessel (empty) weight	Fg	201653.641 N
Head Weight below section 2-2	DeltaFg	2987.574 N
Weight of vessel contents (liquid)	Ff	4017.686 N
Vessel Gross Weight	Fg + Ff	205671.328 N
Skirt Mean Diameter	Dz	797.000 mm
Skirt Analysis Thickness (Corroded)	ez	14.400 mm
Skirt Height	H	1000.000 mm
Additional Axial Force at section 1-1	F1	-14844.800 N
Moment at section 1-1	M1	419494080.000 N-mm
Applied Moment - Bottom of Skirt	Mb	479211840.000 N-mm
Weight at Bottom of Skirt	W	206754.859 N

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:	Elliptical
Head Mean Diameter at Skirt Top	Db 795.002 mm
Head Corroded Thickness	eb 27.000 mm

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 194 \text{ mm} \\
 K &= D_i / (2 * h_i) = 768 / (2 * 194) = 1.98 \\
 R &= D_i (0.44 * K + 0.02) = 768 * (0.44 * 1.98 + 0.02) = 686 \text{ mm} \\
 r &= D_i ((0.5 / K) - 0.08) = 768 * ((0.5 / 1.98) - 0.08) = 132 \text{ mm}
 \end{aligned}$$



Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Obround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	150.0000	0.0	450.00	0.00	80.000	16.000
2	80.0000	0.0	850.00	0.00	80.000	16.000
3	80.0000	180.0	850.00	0.00	80.000	16.000

Interference check:

Access	Access	Pitch	Ligament
--------	--------	-------	----------



		mm	mm
1	2	400.0000	285.0000
2	3	1251.9036	1171.9037
1	3	1314.2535	1199.2537

Interference Check: No Overlap Detected between Openings

Minimum Pitch between openings is: 400.0000 mm
 Minimum Ligament between Openings is: 285.0000 mm

Recommended minimum Ligament [IMin]:

$$= 2 \cdot \sqrt{(D \cdot t)}$$

$$= 2 \cdot \sqrt{(797 \cdot 14.4)} \text{ mm}$$

$$= 214.260 \text{ mm}$$

Carefully evaluate this acceptance criteria.

Section properties at least metal area [A]:

Number of Openings At Critical Height = 3

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
2	-5.7608	5.7608
3	174.2392	185.7608

The points for integrating Ixx, Iyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
0.000	-5.761
5.761	174.239
-174.239	0.000

Height of Weakest Section	H	849.967	mm
Rotation of Principle Axes	a	0.000	deg.
Centroid Position in X direction	g	24.625	mm
Centroid Position in Y direction	h	-0.000	mm
Principle Moment of Inertia	Ixx	2867176960.000	mm ⁴
Principle Moment of Inertia	Iyy	3687956736.000	mm ⁴
Distance to Extreme Fiber	ybar	405.700	mm
Distance to Extreme Fiber	xbar	430.325	mm
Minimum Section Modulus	W4	7067241.000	mm ³
Metal Area at the Cutting Plane	A4	41427.543	mm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:

$$= \sin^{-1}(\text{Largest Opening Width} / D_z)$$

$$= \sin^{-1}(150/797)$$

$$= 0.19 \text{ rad (10.8 deg)}$$

Skirt Mean Radius at Section 4-4 [r]:

$$= D_z / 2$$

$$= 797/2$$

$$= 398.5 \text{ mm}$$

Because the Code uses the words 'may be' in the evaluation below, the results should be considered optional.

Check areas for openings per EN 13445 Equation 16.12-76 [A,check]:

At,i	Adelta,i	Status	lt,i <= 8*eat,i
2560.0	1938.8	...	...
2560.0	922.6	...	...
2560.0	922.6	...	...

This calculation of ratio below is per an earlier version of EN 13445.



Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:

$$\begin{aligned} &= \delta \cdot \sqrt{(r / e_z)} \\ &= 0.19 \cdot \sqrt{(398 / 14.4)} \\ &= 0.996 \end{aligned}$$

Ok as ratio is < 2 - See EN 13445 16.12.76(a).

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:

$$\begin{aligned} &= e_{x(new)} - c_i - c_{ext} \\ &= 16 - 0.8 - 0.8 \\ &= 14.4 \text{ mm} \end{aligned}$$

Find the moment acting at the Critical Height [M4]:

$$\begin{aligned} &= (M_{bottom} - M_{top}) \cdot (H - CutHeight) / H + M_{top} \\ &= (479211808 - 419494048) \cdot (1000 - 850) / 1000 + 419494048 \\ &= 428453696 \text{ N-mm} \end{aligned}$$

Compute the section modulus [Zxx and Zyy]:

$$\begin{aligned} Z_{xx} &= I_{xx} / y_{bar} = 2.87E+09 / 406 = 7067241 \text{ mm}^3 \\ Z_{yy} &= I_{yy} / x_{bar} = 3.69E+09 / 430 = 8570171 \text{ mm}^3 \\ Z &= \min(Z_{xx}, Z_{yy}) = \min(7067241, 8570171) = 7067241 \text{ mm}^3 \end{aligned}$$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:

$$\begin{aligned} &= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area}(A_4))} \\ &= \sqrt{(\min(2.87E+09, 3.69E+09) / 41428)} \\ &= 263 \text{ mm} \end{aligned}$$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:

$$\begin{aligned} &= r_g / L \\ &= 263 / 1000 \\ &= 0.263 \end{aligned}$$

Location of the Skirt Centroid at critical height [e]:

$$\begin{aligned} &= \sqrt{(e_x^2 + e_y^2)} \\ &= \sqrt{(24.6^2 + -0.21691E-05^2)} \\ &= 19.076 \text{ mm} \end{aligned}$$

$$\text{Where } e_x = 24.6 : e_y = -0.21691E-05$$

Moment Increment at Centroid [ΔM_4]:

$$\begin{aligned} &= \text{abs}(F_1 - F_f - F_g) \cdot e = F_4 \cdot e \\ &= \text{abs}(-14845 - 201654 - 4018) \cdot 24.6 \\ &= 5432448.500 \text{ N-mm} \end{aligned}$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$\begin{aligned} &= F_1 - F_g - F_f + 4 \cdot M_1 / D_z \\ &= -14845 - 201654 - 4018 + 4 \cdot 419494048 / 797 \\ &= 1883996 \text{ N} \end{aligned}$$

Force at the joint, 16.12-2 [Fzq]:

$$\begin{aligned} &= F_1 - F_g - F_f - 4 \cdot M_1 / D_z \\ &= -14845 - 201654 - 4018 - 4 \cdot 419494048 / 797 \\ &= -2325028 \text{ N} \end{aligned}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$\begin{aligned} &= (F_{zp} + \Delta F_g) + F_f / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\ &= (1883996 + 2988 + 4018) / (\pi \cdot 795 \cdot 27) + 9.4 \cdot 795 / (4 \cdot 27) \end{aligned}$$



= 97.26 MPa

Stress [σ_{1qm}]:

$$= (Fzq + \Delta Fq + Ff) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (-2325028 + 2988 + 4018) / (\pi \cdot 795 \cdot 27) + 9.4 \cdot 795 / (4 \cdot 27)$$

$$= 34.84 \text{ MPa}$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$= (Ff + \Delta Fq) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (4018 + 2988) / (\pi \cdot 795 \cdot 27) + 9.4 \cdot 795 / (4 \cdot 27)$$

$$= 69.32 \text{ MPa}$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$= Fzp / (\pi \cdot Dz \cdot ez)$$

$$= 1883996 / (\pi \cdot 797 \cdot 14.4)$$

$$= 52.26 \text{ MPa}$$

Stress [σ_{3qm}]:

$$= Fzq / (\pi \cdot Dz \cdot ez)$$

$$= -2325028 / (\pi \cdot 797 \cdot 14.4)$$

$$= -64.49 \text{ MPa}$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [$\cos(\gamma)$]:

$$= 1 - 0.5(Db + eb - Dz + ez) / (r + eb)$$

$$= 1 - 0.5(795 + 27 - 797 + 14.4) / (132 + 27)$$

$$= 0.88$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.88) = 28.8 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma)}$$

$$= 0.5 \cdot \sqrt{27^2 + 14.4^2 + 2 \cdot 27 \cdot 14.4 \cdot 0.88}$$

$$= 20.1 \text{ mm}$$

Bending moment at verification points p and q [M_p & M_q]:

$$M_p = a \cdot Fzp = 20.11 \cdot 1883996.0 = 37.9025 \times 10^6 \text{ N-mm}$$

$$M_q = a \cdot Fzq = 20.11 \cdot -2325028.2 = -46.7753 \times 10^6 \text{ N-mm}$$

Correction factor [C]:

$$= 0.63 - 0.057(eb/ez)^2$$

$$= 0.63 - 0.057(27/14.4)^2$$

$$= 0.430$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot Db \cdot eb^2)$$

$$= 0.43 \cdot 6 \cdot 37902512 / (\pi \cdot 795 \cdot 27^2)$$

$$= 53.64 \text{ MPa}$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot Db \cdot eb^2)$$

$$= 0.43 \cdot 6 \cdot -46775260 / (\pi \cdot 795 \cdot 27^2)$$

$$= -66.20 \text{ MPa}$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot Dz \cdot ez^2)$$

$$= 0.43 \cdot 6 \cdot 37902512 / (\pi \cdot 797 \cdot 14.4^2)$$

$$= 188.11 \text{ MPa}$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot Dz \cdot ez^2)$$

$$= 0.43 \cdot 6 \cdot -46775260 / (\pi \cdot 797 \cdot 14.4^2)$$

$$= -232.15 \text{ MPa}$$



Geometry variables, 16.12-32 to 36:

$$eb/Db = 1.06/31.3 = 0.034$$

$$y = 125 * eb/Db = 125 * 27/795 = 4.25$$

As this an elliptical head, it will be treated a a Korbogen type head

Geometric value [α]:

$$= 4.2 - 0.2 * y = 4.2 - 0.2 * 4.25 = 3.35$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= (P + Ph) * Db ((y / yA) * \alpha - 1) / (4 * eb)$$

$$= (9.4 + 0) * 795 ((28.8/40) * 3.35 - 1) / (4 * 27)$$

$$= 97.89 \text{ MPa}$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$$

$$= 97.3 - 53.6 + 97.9$$

$$= 141.51 \text{ MPa}$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)}$$

$$= 97.3 + 53.6 - 97.9$$

$$= 53.01 \text{ MPa}$$

Stress [σ_{1qitot}]:

$$= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$$

$$= 34.8 - 66.2 + 97.9$$

$$= 198.93 \text{ MPa}$$

Stress [σ_{1qotot}]:

$$= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$$

$$= 34.8 + 66.2 - 97.9$$

$$= -129.26 \text{ MPa}$$

Stress [σ_{2pitot}]:

$$= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$$

$$= 69.3 + 53.6 + 97.9$$

$$= 220.86 \text{ MPa}$$

Stress [σ_{2potot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 69.3 - 53.6 - 97.9$$

$$= -82.22 \text{ MPa}$$

Stress [σ_{2qitot}]:

$$= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$$

$$= 69.3 + 66.2 + 97.9$$

$$= 101.01 \text{ MPa}$$

Stress [σ_{2qotot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 69.3 - 66.2 - 97.9$$

$$= 37.63 \text{ MPa}$$

Stress [σ_{3pitot}]:

$$= \sigma_{3pm} - \sigma_{3pb}$$

$$= 52.3 - 188$$

$$= -135.86 \text{ MPa}$$

Stress [σ_{3potot}]:

$$= \sigma_{3pm} + \sigma_{3pb}$$

$$= 52.3 + 188$$

$$= 240.37 \text{ MPa}$$

Stress [σ_{3qitot}]:



$$\begin{aligned}
 &= \sigma_{3qm} - \sigma_{3qb} \\
 &= -64.5 - -232 \\
 &= 167.66 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{Stress } [\sigma_{3qotot}] &: \\
 &= \sigma_{3qm} + \sigma_{3qb} \\
 &= -64.5 + -232 \\
 &= -296.64 \text{ MPa}
 \end{aligned}$$

Allowable stresses:

$$\begin{aligned}
 \text{For } \sigma_{1pitot} \text{ and } \sigma_{1potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2 \\
 &= 138 (3 - (1 / 1.5)) \cdot (97.3 / 138)^2 \\
 &= 368.29 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{1qitot} \text{ and } \sigma_{1qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2 \\
 &= 138 (3 - (1 / 1.5)) \cdot (34.8 / 138)^2 \\
 &= 408.13 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{2pitot} \text{ and } \sigma_{2potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2 \\
 &= 138 (3 - (1 / 1.5)) \cdot (69.3 / 138)^2 \\
 &= 390.78 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{2qitot} \text{ and } \sigma_{2qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2 \\
 &= 138 (3 - (1 / 1.5)) \cdot (69.3 / 138)^2 \\
 &= 390.78 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{3pitot} \text{ \& } \sigma_{3potot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{3pm} / f)^2 \\
 &= 138 (3 - (1 / 1.5)) \cdot (52.3 / 138)^2 \\
 &= 400.80 \text{ MPa}
 \end{aligned}$$

$$\begin{aligned}
 \text{For } \sigma_{3qitot} \text{ \& } \sigma_{3qotot}, \text{ the stress must not exceed:} \\
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{3qm} / f)^2 \\
 &= 138 (3 - (1 / 1.5)) \cdot (-64.5 / 138)^2 \\
 &= 393.90 \text{ MPa}
 \end{aligned}$$

Check of the Skirt in Regions with Openings, per 16.12.4.3:

$$\begin{aligned}
 \text{Weakening Factor, 16.12-70(b) [psi1]:} \\
 &= \min(1; A4 / (\pi \cdot D3 \cdot ea3)) \\
 &= \min(1; 41428 / (\pi \cdot 797 \cdot 14.4)) \\
 &= 1.000
 \end{aligned}$$

$$\begin{aligned}
 \text{Weakening Factor, 16.12-70(c) [psi2]:} \\
 &= \min(1; 4 \cdot W4 / (\pi \cdot D^2 / 3 / (\pi \cdot D3 \cdot ea3))) \\
 &= \min(1; 4.0 \cdot 8570171 / (\pi \cdot 797^2 \cdot 14.4)) \\
 &= 1.000
 \end{aligned}$$

$$\begin{aligned}
 \text{Maximum Bending Moment [Mmax]:} \\
 &= \pi / 4 \cdot d3^2 \cdot ea3 \cdot \sigma_{call} \\
 &= \pi / 4 \cdot 797^2 \cdot 14.4 \cdot 142 \\
 &= 1016884352.0 \text{ N-mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Maximum Compressive Force [Fc,max]:} \\
 &= \pi \cdot d3 \cdot ea3 \cdot \sigma_{call} \\
 &= \pi \cdot 797 \cdot 14.4 \cdot 142 \\
 &= 5101493.0 \text{ N}
 \end{aligned}$$

$$\begin{aligned}
 \text{Unity Check of Weakened Section, 16.12-70 [UC1]:} \\
 &= F4 / (\psi1 \cdot Fc,max) + (abs(M4) + F4 \cdot abs(yq)) / (\psi2 \cdot Mmax) \\
 &= 220516 / (1 \cdot 5101493) + (abs(428453696) + 220516 \cdot abs(24.6)) / \\
 &\quad (1 \cdot 1016884352) \\
 &= 0.470 \text{ must be } \leq \text{ to } 1, \text{ Passed}
 \end{aligned}$$

Please note: None of the following stresses in the table below are in the region where there is



an access opening.

Summary of Stresses: (MPa)

Stress	Actual	Allowed	Result	Equation
Stot1pi	141.51	368.29	Pass	46
Stot1po	53.01	368.29	Pass	47
Stot1qi	198.93	408.13	Pass	48
Stot1qo	-129.26	408.13	Pass	49
Stot2pi	220.86	390.78	Pass	50
Stot2po	-82.22	390.78	Pass	51
Stot2qi	101.01	390.78	Pass	52
Stot2qo	37.63	390.78	Pass	53
Stot3pi	-135.86	400.80	Pass	54
Stot3po	240.37	400.80	Pass	55
Stot3qi	167.66	393.90	Pass	56
Stot3qo	-296.64	393.90	Pass	57

The stresses are satisfactory in the region without openings.
The stresses are satisfactory in the region with the opening(s).

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Shop/Field Installation Options :

Packing is installed in the Shop.
Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Packing	9978.749 mm
Center of Gravity of the Liquid	3551.218 mm
Center of Gravity of the Insulation	10328.857 mm
Center of Gravity of the Nozzles	9155.763 mm
Center of Gravity of the Added Weights (Operating)	10952.885 mm
Center of Gravity of the Added Weights (Empty)	10748.147 mm
Center of Gravity of Bare Shell New and Cold	10093.839 mm
Center of Gravity of Bare Shell Corroded	10123.719 mm
Vessel CG in the Operating Condition	10088.422 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	10141.589 mm
Vessel CG in the Test Condition	10238.794 mm

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	813.0000	mm
Skirt Thickness	STHK	16.0000	mm
Skirt Internal Corrosion Allowance	SCA	0.8000	mm
Skirt External Corrosion Allowance		0.8000	mm
Skirt Material		SA-516 70	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	54.0000	mm
Design Temperature of the Basing		38.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	138.00	MPa
Basing Yield Stress		262.00	MPa
Inside Diameter of Basing	DI, D4	629.0000	mm
Outside Diameter of Basing	DOU, D5	1149.0000	mm
Nominal Diameter of Bolts	BND, d(b0)	48.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Bolt Material		SA-193 B7	
Bolt Material Class/Thickness Range		<= 64	
Bolt Operating Allowable Stress	SA, fB	138.00	MPa
Number of Bolts	RN, nb	16	
Diameter of Bolt Circle	DC, DBC	981.0000	mm
Bolt Allowable Shear Stress		82.800	MPa
Ultimate Comp. Strength of Concrete	FPC	41.0	MPa
Allowable Comp. Strength of Concrete	FC	22.5	MPa
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	16.0000	mm
Width of Gussets at Top Plate	TWDT	158.0000	mm
Width of Gussets at Base Plate	BWDT	158.0000	mm
Gusset Plate Elastic Modulus	E	201746.0	MPa
Gusset Plate Yield Stress	SY	262.0	MPa
Height of Chair	h1	350.0000	mm
Distance between Gussets	RG, b6	96.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	48.0000	mm
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	44.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	168.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	64.0000	mm
External Corrosion Allowance	CA	3.0000	mm
Dead Weight of Vessel	DW	225697.2	N
Operating Weight of Vessel	ROW	236520.2	N
Earthquake Moment on Basing	EQMOM	257206448.0	N-mm
Wind Moment on Basing	WIMOM	126208888.0	N-mm
Vortex Shedding Moment on Basing	Vormom	161666480.0	N-mm
User Moment at the Base [Earthquake Case]		355082560.0	N-mm (UEQMOM)
User Moment at the Base [WindLoad Case]		355082560.0	N-mm (UWIMOM)
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:



$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 612289024/981) - 229568) / 16$$

$$= 141625.7 \text{ N}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 481291456/981) - 225697) / 16$$

$$= 108497.5 \text{ N}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 161666480/981) - 236520) / 16$$

$$= 26400.2 \text{ N}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

$$\text{Area Available in 1 Bolt (corroded)} \quad \text{Abss} : 1155.1653 \text{ mm}^2$$

$$\text{Area Available in all the Bolts} \quad \text{Abss} \cdot \text{RN} : 18482.6445 \text{ mm}^2$$

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.300	0.150	1.511	2.441	0.438	0.781	670064.6	899633.0	60202.7
4	0.225	0.263	1.297	2.605	0.454	0.777	668772.6	898341.1	56308.2
6	0.282	0.291	1.459	2.482	0.442	0.780	669610.9	899179.4	59172.0
8	0.296	0.293	1.498	2.451	0.439	0.781	669943.3	899511.8	59941.2
10	0.295	0.294	1.495	2.454	0.439	0.781	669913.9	899482.5	59876.2
12	0.294	0.294	1.492	2.456	0.439	0.781	669892.0	899460.5	59827.6
13	0.294	0.294	1.493	2.455	0.439	0.781	669895.7	899464.1	59835.7

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 669896 / (6 \cdot 981 \cdot 2.46)$$

$$= 92.8 \text{ MPa, Should be } \leq 138 \text{ MPa}$$

Thickness of the Band of Bolting Steel [T1]

$$= \text{RN} \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 16 \cdot 1155 / (\pi \cdot 981)$$

$$= 6 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 3.93[(2 \cdot 0.29 \cdot 981 + 260) / (2 \cdot 0.29 \cdot 981)]$$

$$= 5.7 \text{ MPa, Should be } \leq 22.5 \text{ MPa}$$

Values for table 10.3, l = 168, b = 96, l/b = 1.74437714

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(6705, 20012) = 20012 \text{ N}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 20012 / 173)} + 3$$

$$= 29.4 \text{ mm}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 168, b = 96, b/l = 0.5714286

As b/l (0.57) is less than 1, inverting b/l = 1.75 0.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2l \sin(\pi \cdot a/l)) / (\pi \cdot g))) + 1] - [(0.7 - g2)P / (4\pi)]$$

$$= 159399 / (4 \cdot \pi) [1.3(\ln((2 \cdot 168 \cdot \sin(\pi \cdot 84/168))) / (\pi \cdot 40)) + 1] - [(0.7 - 0.044) \cdot 159399 / (4 \cdot \pi)]$$



$$= 20587.5703 \text{ N}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot \text{Abs}(\text{Mo}) / \text{Fb}) + \text{Ca}}$$

$$= \sqrt{(6 \cdot \text{Abs}(20588) / 175) + 3}$$

$$= 29.5947 \text{ mm}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot \text{lavg}) + \text{Ca}$$

$$= 159399 / (124 \cdot 158) + 3$$

$$= 12.5 \text{ (not less than } 9.52 + 3 \text{) mm}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 981 / 16 = 193 \text{ mm}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76 (P \cdot a / (m \cdot (h + tba) \cdot 1.5 \cdot \text{Sktope}))^{2/3} \cdot r^{1/3} + \text{Ca}$$

$$= 1.76 (159399 \cdot 84 / (193 \cdot 306 \cdot 207))^{2/3} \cdot 4061^{1/3} + \text{Ca}$$

$$= 13.9 + 1.6 = 15.5 \text{ mm}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 60221 / (1155 \cdot 16)$$

$$= 3.3 \text{ MPa. Must be less than } 82.8 \text{ MPa.}$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	29.3522	mm
Actual Basing Thickness as entered by user	54.0000	mm
Required Thickness of Chair Cap	29.5947	mm
Actual Top Ring Thickness as entered by user	44.0000	mm
Required Gusset thickness, + CA	12.5250	mm
Actual Gusset Thickness as entered by user	16.0000	mm
Required Thickness of Skirt for Local Stress	15.4722	mm
Given Thickness of Skirt	16.0000	mm
Required Gusset Height to meet local stress	283.3070	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - \text{CA})$$

$$= 94.6 \cdot (16 - 1.6)$$

$$= 1362 \text{ N/mm}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 1362 / [(0.4 \cdot 243) \cdot 2 \cdot 0.707]$$

$$= 9.93 \text{ mm}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 159399 / (156 + 2 \cdot (252 + 44))$$

$$= 213 \text{ N/mm}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 159399 \cdot 84 / (156 \cdot (296) + 0.6667 \cdot (296)^2)$$

$$= 128 \text{ N/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{213^2 + 128^2}$$

$$= 249 \text{ N/mm}$$



Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$\begin{aligned} &= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707] \\ &= 249 / [(0.4 \cdot 243) \cdot 2 \cdot 0.707] \\ &= 1.81 \text{ mm} \end{aligned}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$\begin{aligned} &= \text{Bolt Load} / (2 \cdot \text{TopWth}) \\ &= 159399 / (2 \cdot 158) \\ &= 504 \text{ N/mm} \end{aligned}$$

Weld Load [Wh]:

$$\begin{aligned} &= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth}) \\ &= 159399 \cdot 84 / (2 \cdot 296 \cdot 158) \\ &= 143 \text{ N/mm} \end{aligned}$$

Resultant Weld Load [Wr]:

$$\begin{aligned} &= \sqrt{W_v^2 + W_h^2} \\ &= \sqrt{504^2 + 143^2} \\ &= 524 \text{ N/mm} \end{aligned}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$\begin{aligned} &= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707] \\ &= 524 / [(0.4 \cdot 243) \cdot 2 \cdot 0.707] \\ &= 3.82 \text{ mm} \end{aligned}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	9.9296	mm
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm
Required Top Plate to Skirt Weld Size	6.3500	mm
Required Gusset to Top Plate Double Fillet Weld Size	3.8217	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N480

From: 30

Pressure for Reinforcement Calculations	P	9.4123	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	82	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	762.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	30.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
-------------------------------	----	--------	----

User Entered Minimum Design Metal Temperature		-28.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		80.0000	mm

Size and Thickness Basis		Nominal	
Nominal Thickness		160	

Flange Material		SA-105	
-----------------	--	--------	--

Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	13.8750	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		25.0000	mm

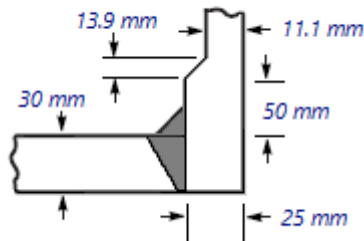
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		900	
Flange Grade		GR 1.1	

Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		600.0	mm
Centerline Distance to Tangent		600.0	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N480

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	88.900 mm.
Actual Thickness Used in Calculation	11.125 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 9.41 \cdot 768 \cdot 0.89 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 9.41)$
 $= 23.5584 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 9.41 \cdot 44.5 / (138 \cdot 1 + 0.4 \cdot 9.41)$
 $= 2.9512 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3989 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 13.9 / 0.5773$
 $= 24.0319 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

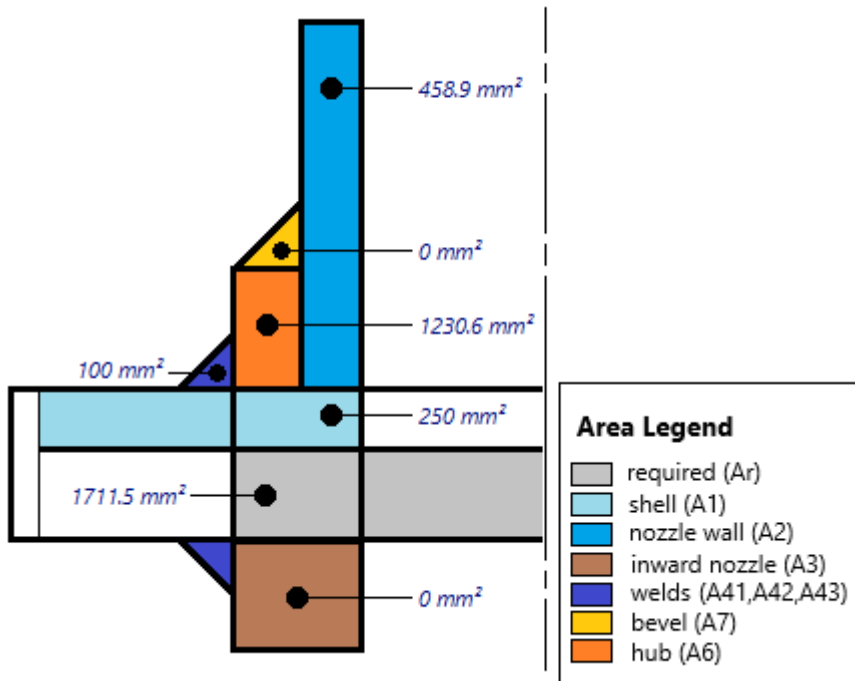
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	145.2992 mm
Parallel to Vessel Wall, opening length	d	72.6496 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	44.3449 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	1711.507	77.311	NA
Area in Shell	A1	250.032	1806.916	NA
Area in Nozzle Wall	A2	458.881	685.240	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	1230.552	1230.552	NA
TOTAL AREA AVAILABLE	Atot	2039.466	3822.709	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (72.6 \cdot 23.6 \cdot 1 + 2 \cdot 22 \cdot 23.6 \cdot 1 \cdot (1 - 1))$
 $= 1711.507 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 72.6(1 \cdot 27 - 1 \cdot 23.6) - 2 \cdot 22$
 $(1 \cdot 27 - 1 \cdot 23.6) \cdot (1 - 1)$
 $= 250.032 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 44.3)(8.13 - 2.95)1$
 $= 458.881 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 100.000 \text{ mm}^2$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2} \\ &= (2 \cdot \min(44.3, 200, 50)) \cdot (25 - 11.1) \cdot 1 \\ &= 1230.552 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.9512 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 29.0996 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 29.0996 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.2849 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[7.8, \max(29.1, 4.5)] \\ &= 7.8000 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(5.95, 7.8) \\ &= 7.8000 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = $0.875 \cdot 11.125 = 9.735 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 49.2,	Allowable	: 138.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 295.8 MPa	Passed
Occasional	: 18.9,	Allowable	: 183.5 MPa	Passed
Shear	: 25.1,	Allowable	: 96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 9.73, tr = 2.95, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.44$, Temp. Reduction = 42 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 25, tr = 2.95, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.13$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-39 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $9.41/15.31 = 0.615$

Weld Size Calculations, Description: N480

Intermediate Calc. for nozzle/shell welds Tmin 19.0000 mm



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (1712 - 250 + 2 \cdot 22 \cdot 1 \cdot (1 \cdot 27 - 23.6)) \cdot 138)$
 $= 222558.09 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (1689 + 100 - 0 \cdot 1) \cdot 138$
 $= 246916.34 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (1689 + 0 + 100 + (1188)) \cdot 138$
 $= 410843.44 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (1689 + 0 + 100 + 0 + (1188)) \cdot 138$
 $= 410843.44 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 88.9 \cdot 10 \cdot 0.49 \cdot 138$
 $= 94417. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 40.4) \cdot (25 - 3) \cdot 0.7 \cdot 138$
 $= 269624. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 88.9 \cdot (30 - 3) \cdot 0.74 \cdot 138$
 $= 384999. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (94417 + 269624) = 364041 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (94417 + 0 + 384999 + 0) = 479416 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (94417 + 384999 + 0) = 479416 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 364041 N	, must exceed W = 222558 N	or W1 = 246916 N
Path 2-2 = 479416 N	, must exceed W = 222558 N	or W2 = 410843 N
Path 3-3 = 479416 N	, must exceed W = 222558 N	or W3 = 410843 N

Maximum Allowable Pressure for this Nozzle at this Location:



Converged Maximum Allowable Pressure in the Operating case: 9.73 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 1.3931 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 231.3931 mm

Input Echo, WRC107/537 Item 1, Description: N480 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	1371.604	mm
Vessel Thickness	Tv	30.000	mm
Design Temperature	T1	82.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	116.650	mm
Nozzle Thickness	Tn	25.000	mm
Nozzle Material		SA-105	
Nozzle UNS Number		K03504	
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	138.00	MPa
Design Internal Pressure	Dp	9.412	MPa
Include Pressure Thrust		No	

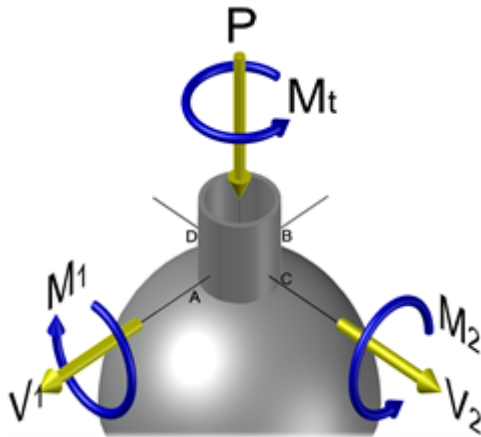
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-5997.7	N
Longitudinal Shear (SUS)	(V1) V1	3998.5	N
Circumferential Shear (SUS)	(Vc) V2	3998.5	N
Circumferential Moment (SUS)	(Mc) M1	896046.4	N-mm
Longitudinal Moment (SUS)	(Ml) M2	896046.4	N-mm
Torsional Moment (SUS)	Mt	1280070.8	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 116.65 + 2 \cdot 1.65 \cdot \sqrt{(702.302 (30.0 - 3.0))}$$

$$= 571.070 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load		P	-5997.7	N
Circumferential Shear	(VC)	V2	3998.5	N
Longitudinal Shear	(VL)	V1	3998.5	N
Circumferential Moment	(MC)	M1	896046.4	N-mm
Longitudinal Moment	(ML)	M2	896046.4	N-mm
Torsional Moment		MT	1280070.8	N-mm

Dimensionless Param: U = 0.42 TAU = 5.00 (2.15) RHO = 1.23

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 2	0.08312	(A,B,C,D)
$M(x) / P$	SP 2	0.09214	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 2	0.17564	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 2	0.25353	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 2	0.17564	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 2	0.25353	(A,B,C,D)
$N(y) * T / P$	SP 2	0.16411	(A,B,C,D)
$M(y) / P$	SP 2	0.04349	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 2	0.08855	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 2	0.14753	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 2	0.08855	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 2	0.14753	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7
Rad. Bend.	P	4.5	-4.5	4.5	-4.5	4.5	-4.5	4.5	-4.5
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-1.6	-1.6	1.6	1.6
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-13.6	13.6	13.6	-13.6
Rad. Memb.	ML	-1.6	-1.6	1.6	1.6	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-13.6	13.6	13.6	-13.6	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-9.9	8.1	20.4	-15.9	-9.9	8.1	20.4	-15.9



Tang. Memb. P	1.4	1.4	1.4	1.4	1.4	1.4	1.4	1.4
Tang. Bend. P	2.1	-2.1	2.1	-2.1	2.1	-2.1	2.1	-2.1
Tang. Memb. MC	0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Tang. Bend. MC	0.0	0.0	0.0	0.0	-7.9	7.9	7.9	-7.9
Tang. Memb. ML	-0.8	-0.8	0.8	0.8	0.0	0.0	0.0	0.0
Tang. Bend. ML	-7.9	7.9	7.9	-7.9	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-5.2	6.3	12.2	-7.9	-5.2	6.3	12.2	-7.9
Shear VC	0.8	0.8	-0.8	-0.8	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Shear MT	2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2
Tot. Shear	3.0	3.0	1.4	1.4	1.4	1.4	3.0	3.0
Str. Int.	11.4	10.4	20.6	16.1	10.3	8.9	21.4	16.9

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		120.2	120.2	120.2	120.2	120.2	120.2	120.2	120.2
Rad. Pl (SUS)		-0.9	-0.9	2.3	2.3	-0.9	-0.9	2.3	2.3
Rad. Q (SUS)		-9.0	9.0	18.1	-18.1	-9.0	9.0	18.1	-18.1
Long. Pm (SUS)		120.2	120.2	120.2	120.2	120.2	120.2	120.2	120.2
Long. Pl (SUS)		0.6	0.6	2.1	2.1	0.6	0.6	2.1	2.1
Long. Q (SUS)		-5.8	5.8	10.0	-10.0	-5.8	5.8	10.0	-10.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.8	0.8	-0.8	-0.8	-0.8	-0.8	0.8	0.8
Shear Q (SUS)		2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2
Pm (SUS)		120.2	120.2	120.2	120.2	120.2	120.2	120.2	120.2
Pm+Pl (SUS)		121.1	121.1	123.2	123.2	121.1	121.1	123.2	123.2
Pm+Pl+Q (Total)		116.5	130.6	140.8	112.5	115.4	129.1	141.6	113.3

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	120.18	138.00	Passed
Pm+Pl (SUS)	123.19	207.00	Passed
Pm+Pl+Q (TOTAL)	141.55	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N1150

From: 40

Pressure for Reinforcement Calculations	P	9.4029	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	82	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	762.00	mm
Design Length of Section	L	2113.5002	mm
Shell Finished (Minimum) Thickness	t	32.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1225.00	mm
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User Entered Minimum Design Metal Temperature		-28.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		150.0000	mm

Size and Thickness Basis		Minimum	
Nominal Thickness		80	

Flange Material		SA-105 +IT	
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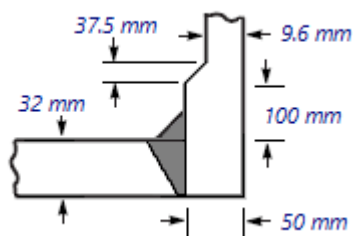
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	37.5095	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.8000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.





Reinforcement CALCULATION, Description: N1150

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 168.275 mm.
Actual Thickness Used in Calculation 9.601 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $9.4 \cdot 384 / (138 \cdot 1 - 0.6 \cdot 9.4)$
= 27.2804 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $9.4 \cdot 84.1 / (138 \cdot 1 + 0.4 \cdot 9.4)$
= 5.5808 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.5805 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= $40.4 / 0.5773$
= 69.9728 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

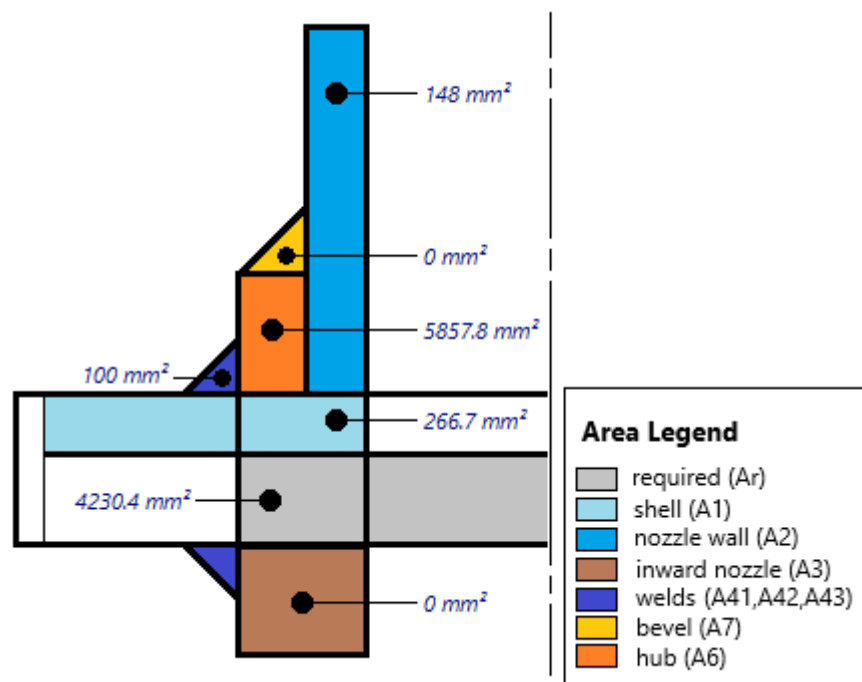
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	310.1452	mm
Parallel to Vessel Wall, opening length	d	155.0726	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	72.5000	mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 138 / 138)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 138 / 138)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1, 1)$
= 1.000



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	4230.448	303.848	NA
Area in Shell	A1	266.657	3889.408	NA
Area in Nozzle Wall	A2	147.962	873.005	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	5857.826	5857.826	NA
TOTAL AREA AVAILABLE	Atot	6372.444	10720.238	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (155 \cdot 27.3 \cdot 1 + 2 \cdot 47 \cdot 27.3 \cdot 1 \cdot (1 - 1))$
 $= 4230.448 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 155(1 \cdot 29 - 1 \cdot 27.3) - 2 \cdot 47(1 \cdot 29 - 1 \cdot 27.3) \cdot (1 - 1)$
 $= 266.657 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 72.5)(6.6 - 5.58)1$
 $= 147.962 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 100.000 \text{ mm}^2$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2} \\ &= (2 \cdot \min(72.5, 200, 100)) \cdot (50 - 9.6) \cdot 1 \\ &= 5857.826 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 8.5808 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 30.2804 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 30.2804 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.2879 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 9.2200 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[9.22, \max(30.3, 4.5)] \\ &= 9.2200 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(8.58, 9.22) \\ &= 9.2200 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 9.6012 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 83.8,	Allowable	: 138.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 261.2 MPa	Passed
Occasional	: 53.0,	Allowable	: 183.5 MPa	Passed
Shear	: 24.6,	Allowable	: 96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 32, tr = 27.3, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.94, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	6 °C
Min Metal Temp. at Required thickness (UCS 66.1)	2 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 2 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-18 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-39 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 9.40/15.31 = 0.614$$

Weld Size Calculations, Description: N1150

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm



Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v) \\ &= \max(0, (4230 - 267 + 2 \cdot 47 \cdot 1 \cdot (1 \cdot 29 - 27.3)) \cdot 138) \\ &= 569250.62 \text{ N} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v \\ &= (6006 + 100 - 0 \cdot 1) \cdot 138 \\ &= 842511.81 \text{ N} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v \\ &= (6006 + 0 + 100 + (2726)) \cdot 138 \\ &= 1218661.12 \text{ N} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S \\ &= (6006 + 0 + 100 + 0 + (2726)) \cdot 138 \\ &= 1218661.12 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.14/2.0) \cdot 168 \cdot 10 \cdot 0.49 \cdot 138 \\ &= 178719. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\ &= (3.14 \cdot 80.8) \cdot (50 - 3) \cdot 0.7 \cdot 138 \\ &= 1152916. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng} \\ &= (3.14/2.0) \cdot 168 \cdot (32 - 3) \cdot 0.74 \cdot 138 \\ &= 782729. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (178719 + 1152915) = 1331634 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ &= (178719 + 0 + 782729 + 0) = 961447 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ &= (178719 + 782729 + 0) = 961447 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 1331634 N	, must exceed W = 569250 N	or W1 = 842511 N
Path 2-2 = 961447 N	, must exceed W = 569250 N	or W2 = 1218661 N
Path 3-3 = 961447 N	, must exceed W = 569250 N	or W3 = 1218661 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 9.96 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa



The Drop for this Nozzle is : 9.4063 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 241.4062 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 155.07 mm

Input Echo, WRC297 Item 1, Description: N1150 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	762.002	mm
Shell Thickness	32.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	249.073	mm
Nozzle Thickness	50.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-105	
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa
Nozzle Hot S.I. Allowable	SNmh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	9.40	MPa
Radial Load	P	-11245.70	N
Circumferential Shear	Vc	7497.10	N
Longitudinal Shear	Vl	7497.10	N
Circumferential Moment	Mc	0.315E+07	N-mm
Longitudinal Moment	Ml	0.315E+07	N-mm
Torsional Moment	Mt	0.450E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (4.981) is < 10
 check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 249.073 + 2 \cdot 1.65 \cdot \sqrt{(398.501(32.0 - 3.0))}$$

$$= 603.827 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	797.002 mm
Nozzle Outside Diameter	(d) =	249.073 mm
Vessel Thickness used	(T) =	29.000 mm
Nozzle Thickness used	(t) =	47.000 mm
T / t	=	0.617
d / t	=	10.000



$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.638$$

Nr/P = 0.082
 Mr/P = 0.156
 M0/P = 0.047
 N0/P = 0.107
 MrD/Mc = 0.285
 NrDL/Mc = 0.109
 M0d/Mc = 0.092
 N0DL/Mc = 0.059
 MrD/Ml = 0.134
 NrDL/Ml = 0.076
 M0D/Ml = 0.045
 N0DL/Ml = 0.131

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-1	-1	1	1
Outplane Bending (ML)	-4	4	4	-4
Normal Pressure Stress	119	129	119	129
Outplane Stress Summary	119	129	131	124

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	12	-12	12	-12
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-12	12	12	-12
Inplane Pressure Stress	59	59	59	59
Inplane Stress Summary	60	59	86	37

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	119	129	131	125

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	3	-3	3	-3



Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-8	8	8	-8
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	59	59	59	59

Outplane Stress Summary	55	65	74	50
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	12	-12	12	-12
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-25	25	25	-25
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	119	129	119	129

Inplane Stress Summary	106	141	160	93
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	2	2
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	106	141	160	94
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-1	-1	1	1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	14	25	14	25

Hoop Stress Summary	13	25	17	28
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-3	3	3	-3
Axial Pressure Stress	7	7	7	7

Axial Stress Summary	5	4	16	3
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	1	1	0	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	13	25	17	28

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	14	25	14	25
Hoop Stress Summary	14	26	16	27

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-7	7	7	-7
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	7	7	7	7
Axial Stress Summary	0	9	21	-1

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	1	1

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	14	26	21	29

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl

Circ. Pm (SUS)	119	129	119	129	119	129	119	129
Circ. Pl (SUS)	0	0	2	2	0	0	2	2
Circ. Q (SUS)	-1	1	7	-7	-13	13	37	-37
Long. Pm (SUS)	59	59	59	59	59	59	59	59
Long. Pl (SUS)	0	0	2	2	1	1	1	1
Long. Q (SUS)	0	0	24	-24	-5	5	11	-11
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	119.0	129.0	119.0	129.0	119.0	129.0	119.0	129.0
Pm+Pl (SUS)	119.0	129.0	121.0	131.0	119.0	129.0	121.0	131.0
Pm+Pl+Q (Total)	118.0	130.0	128.0	124.0	106.0	142.0	158.0	94.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	129.00	138.00	Passed
Pm+Pl (SUS)	131.00	207.00	Passed
Pm+Pl+Q (TOTAL)	158.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	14	25	14	25	14	25	14	25
Circ. Pl (SUS)	0	0	2	2	1	1	1	1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	0	0	6	-6	-4	4	10	-10
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	14.0	25.0	14.0	25.0	14.0	25.0	14.0	25.0
Pm+Pl (SUS)	14.0	25.0	16.0	27.0	15.0	26.0	15.0	26.0
Pm+Pl+Q (Total)	14.0	25.0	16.0	27.0	15.0	26.0	19.0	27.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	25.00	138.00	Passed
Pm+Pl (SUS)	27.00	207.00	Passed
Pm+Pl+Q (TOTAL)	27.00	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: K150

From: 40

Pressure for Reinforcement Calculations	P	9.4083	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	82	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	762.00	mm
Design Length of Section	L	2113.5002	mm
Shell Finished (Minimum) Thickness	t	32.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent 250.00 mm

User Entered Minimum Design Metal Temperature -28.00 °C

Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm

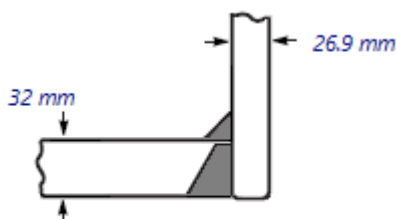
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	187.3250	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.8000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K150

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 50.800 mm.
 Actual Thickness Used in Calculation 26.924 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 9.41 \cdot 384 / (138 \cdot 1 - 0.6 \cdot 9.41)$
 $= 27.2968$ mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 28.4 \cdot (\exp([9.41 / (138 \cdot 1)] - 1))$
 $= 2.0037$ mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4278 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	162.6480	mm
Parallel to Vessel Wall	Rn+tn+t	81.3240	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	59.8100	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K150.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.0037	mm
Wall Thickness per UG16(b),	tr16b = 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1 = 30.2968	mm
Wall Thickness	tb1 = max(trb1, tr16b) = 30.2968	mm
Wall Thickness, shell/head, external pressure	trb2 = 3.2879	mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000	mm
Wall Thickness per table UG-45	tb3 = 8.2578	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.26, \max(30.3, 4.5)]$
 $= 8.2578$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(5, 8.26)$
 $= 8.2578$ mm

Available Nozzle Neck Thickness = 26.9240 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 26.9, tr = 2, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.084$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-39 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $9.41 / 15.31 = 0.614$



Weld Size Calculations, Description: K150

Intermediate Calc. for nozzle/shell Welds T_{min} 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.6100 mm

The Cut Length for this Nozzle is, Drop + H_o + H + T : 222.9350 mm

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Input, Nozzle Desc: K250

From: 40

Pressure for Reinforcement Calculations	P	9.4029	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	82	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	762.00	mm
Design Length of Section	L	2113.5002	mm
Shell Finished (Minimum) Thickness	t	32.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1250.00	mm
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User Entered Minimum Design Metal Temperature		-28.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm

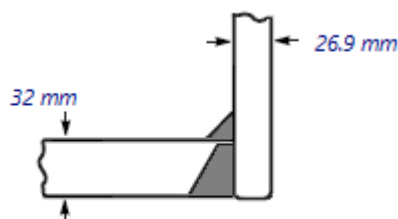
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	187.3250	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	30.8000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K250

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 50.800 mm.
Actual Thickness Used in Calculation 26.924 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $9.4 \cdot 384 / (138 \cdot 1 - 0.6 \cdot 9.4)$
= 27.2804 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $28.4 \cdot (\exp([9.4 / (138 \cdot 1)] - 1))$
= 2.0025 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4278 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	162.6480 mm
Parallel to Vessel Wall	Rn+tn+t	81.3240 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	59.8100 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K250.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.0025 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 30.2804 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 30.2804 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.2879 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 8.2578 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[8.26, max(30.3, 4.5)]
= 8.2578 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(5, 8.26)
= 8.2578 mm

Available Nozzle Neck Thickness = 26.9240 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 26.9, tr = 2, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.084$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-39 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $9.40 / 15.31 = 0.614$



Weld Size Calculations, Description: K250

Intermediate Calc. for nozzle/shell Welds T_{min} 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.6100 mm

The Cut Length for this Nozzle is, Drop + H_o + H + T : 222.9350 mm

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Input, Nozzle Desc: K350

From: 40

Pressure for Reinforcement Calculations	P	9.4052	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	82	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	762.00	mm
Design Length of Section	L	2113.5002	mm
Shell Finished (Minimum) Thickness	t	32.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		600.00	mm
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User Entered Minimum Design Metal Temperature		-28.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm

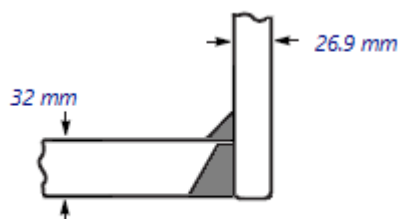
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	187.3250	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		30.8000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K350

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 50.800 mm.
 Actual Thickness Used in Calculation 26.924 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 9.41 \cdot 384 / (138 \cdot 1 - 0.6 \cdot 9.41)$
 $= 27.2873$ mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 28.4 \cdot (\exp([9.41 / (138 \cdot 1)] - 1))$
 $= 2.0030$ mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4278 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	162.6480	mm
Parallel to Vessel Wall	Rn+tn+t	81.3240	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	59.8100	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K350.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.0030	mm
Wall Thickness per UG16(b),	tr16b = 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1 = 30.2873	mm
Wall Thickness	tb1 = max(trb1, tr16b) = 30.2873	mm
Wall Thickness, shell/head, external pressure	trb2 = 3.2879	mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000	mm
Wall Thickness per table UG-45	tb3 = 8.2578	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[8.26, \max(30.3, 4.5)]$
 $= 8.2578$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(5, 8.26)$
 $= 8.2578$ mm

Available Nozzle Neck Thickness = 26.9240 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 26.9, tr = 2, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.084$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-39 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $9.41 / 15.31 = 0.614$



Weld Size Calculations, Description: K350

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.6100 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.9350 mm

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 81.32 mm

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Input, Nozzle Desc: H2200

From: 70

Pressure for Reinforcement Calculations	P	9.4029	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	82	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	762.00	mm
Design Length of Section	L	4150.0000	mm
Shell Finished (Minimum) Thickness	t	32.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		2870.49	mm
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User Entered Minimum Design Metal Temperature		-28.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		200.0000	mm

Size and Thickness Basis		Minimum	
Nominal Thickness		120	

Flange Material		SA-105	
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Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	34.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm

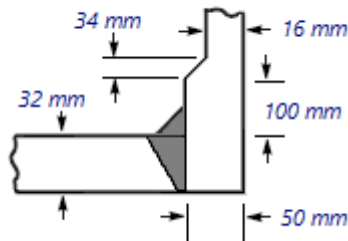
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

This is a Manway, Coupling or Access Opening.

Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: H2200

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	219.075 mm.
Actual Thickness Used in Calculation	15.980 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 9.4 \cdot 384 / (138 \cdot 1 - 0.6 \cdot 9.4)$
 $= 27.2804 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 9.4 \cdot 110 / (138 \cdot 1 + 0.4 \cdot 9.4)$
 $= 7.2655 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.6751 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 34 / 0.5773$
 $= 58.9248 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

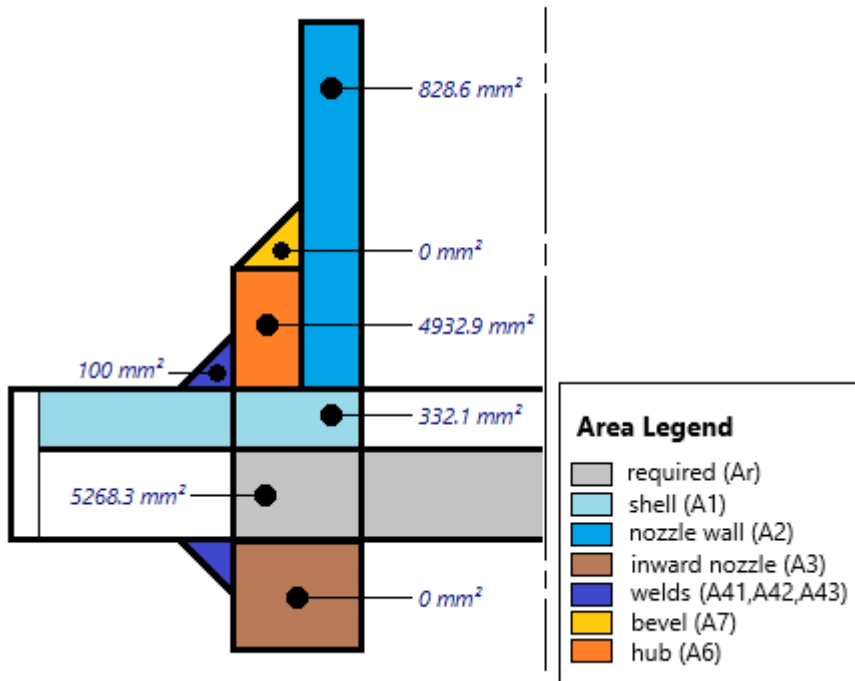
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	386.2309 mm
Parallel to Vessel Wall, opening length	d	193.1154 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	72.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	5268.274	504.680	NA
Area in Shell	A1	332.073	4590.988	NA
Area in Nozzle Wall	A2	828.565	1784.177	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	4932.932	4932.932	NA
TOTAL AREA AVAILABLE	Atot	6193.571	11408.097	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (193 \cdot 27.3 \cdot 1 + 2 \cdot 47 \cdot 27.3 \cdot 1 \cdot (1 - 1))$
 $= 5268.274 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 193(1 \cdot 29 - 1 \cdot 27.3) - 2 \cdot 47(1 \cdot 29 - 1 \cdot 27.3) \cdot (1 - 1)$
 $= 332.073 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 72.5)(13 - 7.27)1$
 $= 828.565 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 100.000 \text{ mm}^2$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(72.5, 200, 100)) \cdot (50 - 16) \cdot 1$$

$$= 4932.932 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 10.2655 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 10.2655 \text{ mm}$$

Available Nozzle Neck Thickness = 15.9798 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 16$, $t_r = 7.27$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.56$, Temp. Reduction = $26 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A $6 \text{ }^\circ\text{C}$
 Min Metal Temp. at Required thickness (UCS 66.1) $-20 \text{ }^\circ\text{C}$

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $t_g = 32$, $t_r = 27.3$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.94$, Temp. Reduction = $3 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B $6 \text{ }^\circ\text{C}$
 Min Metal Temp. at Required thickness (UCS 66.1) $2 \text{ }^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : $2 \text{ }^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) $-18 \text{ }^\circ\text{C}$
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) $-39 \text{ }^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $9.40/15.31 = 0.614$

Weld Size Calculations, Description: H2200

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (5268 - 332 + 2 \cdot 47 \cdot 1 \cdot (1 \cdot 29 - 27.3)) \cdot 138)$$

$$= 703429.25 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_4 - (W_i - C_{an} / 707)^2 \cdot fr_2) \cdot S_v$$

$$= (5761 + 100 - 0 \cdot 1) \cdot 138$$

$$= 808803.25 \text{ N}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$$

$$= (5761 + 0 + 100 + (2726)) \cdot 138$$



= 1184952.50 N

Weld Load [W3]:

= $(A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$
= $(5761 + 0 + 100 + 0 + (2726)) \cdot 138$
= 1184952.50 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

= $(\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
= $(3.14/2.0) \cdot 219 \cdot 10 \cdot 0.49 \cdot 138$
= 232671. N

Shear, Nozzle Wall [Snw]:

= $(\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
= $(3.14 \cdot 103) \cdot (50 - 3) \cdot 0.7 \cdot 138$
= 1469690. N

Tension, Shell Groove Weld [Tngw]:

= $(\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$
= $(3.14/2.0) \cdot 219 \cdot (32 - 3) \cdot 0.74 \cdot 138$
= 1019024. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

= $(SONW + SNW) = (232671 + 1469690) = 1702361 \text{ N}$

Failure Path [Path 2-2]:

= $(Sonw + Tpgw + Tngw + Sinw)$
= $(232671 + 0 + 1019024 + 0) = 1251695 \text{ N}$

Failure Path [Path 3-3]:

= $(Sonw + Tngw + Sinw)$
= $(232671 + 1019024 + 0) = 1251695 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 1702361 N	, must exceed W = 703429 N	or W1 = 808803 N
Path 2-2 = 1251695 N	, must exceed W = 703429 N	or W2 = 1184952 N
Path 3-3 = 1251695 N	, must exceed W = 703429 N	or W3 = 1184952 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 9.97 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 16.0855 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 248.0855 mm

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Input, Nozzle Desc: H1200

From: 110

Pressure for Reinforcement Calculations	P	9.4029	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	82	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	762.00	mm
Design Length of Section	L	4128.0000	mm
Shell Finished (Minimum) Thickness	t	32.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		7826.20	mm
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User Entered Minimum Design Metal Temperature		-28.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		200.0000	mm

Size and Thickness Basis		Nominal	
Nominal Thickness		120	

Flange Material		SA-105	
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Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	32.7400	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm

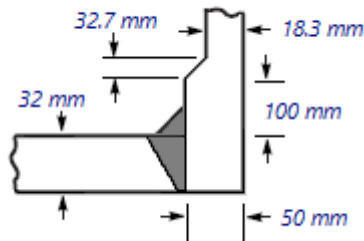
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

This is a Manway, Coupling or Access Opening.

Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: H1200

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	219.075 mm.
Actual Thickness Used in Calculation	18.263 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 9.4 \cdot 384 / (138 \cdot 1 - 0.6 \cdot 9.4)$
 $= 27.2804 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 9.4 \cdot 110 / (138 \cdot 1 + 0.4 \cdot 9.4)$
 $= 7.2655 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.6751 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 31.7 / 0.5773$
 $= 54.9708 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

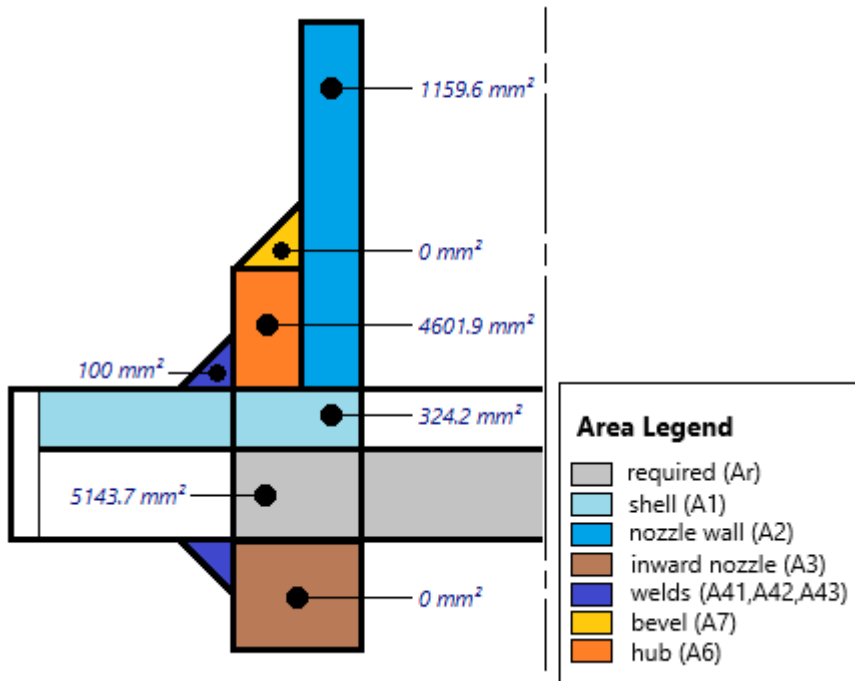
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	377.0996 mm
Parallel to Vessel Wall, opening length	d	188.5498 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	72.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	5143.721	491.514	NA
Area in Shell	A1	324.223	4484.917	NA
Area in Nozzle Wall	A2	1159.575	2115.187	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	4601.923	4601.923	NA
TOTAL AREA AVAILABLE	Atot	6185.721	11302.026	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (189 \cdot 27.3 \cdot 1 + 2 \cdot 47 \cdot 27.3 \cdot 1 \cdot (1 - 1))$$

$$= 5143.721 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 189(1 \cdot 29 - 1 \cdot 27.3) - 2 \cdot 47(1 \cdot 29 - 1 \cdot 27.3) \cdot (1 - 1)$$

$$= 324.223 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 72.5)(15.3 - 7.27)1$$

$$= 1159.575 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 100.000 \text{ mm}^2$$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(72.5, 200, 100)) \cdot (50 - 18.3) \cdot 1$$

$$= 4601.923 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 10.2655 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 10.2655 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 18.263 = 15.980 \text{ mm} \rightarrow \text{OK}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 16$, $t_r = 7.27$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.56$, Temp. Reduction = 26°C

Min Metal Temp. w/o impact per UCS-66, Curve A 6°C
 Min Metal Temp. at Required thickness (UCS 66.1) -20°C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $t_g = 32$, $t_r = 27.3$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.94$, Temp. Reduction = 3°C

Min Metal Temp. w/o impact per UCS-66, Curve B 6°C
 Min Metal Temp. at Required thickness (UCS 66.1) 2°C

Gov. MDMT of the nozzle to shell joint welded assembly : 2°C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18°C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -39°C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 9.40/15.31 = 0.614$$

Weld Size Calculations, Description: H1200

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm}$

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld $6.0000 = \text{Min per Code}$ $7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (5144 - 324 + 2 \cdot 47 \cdot 1 \cdot (1 \cdot 29 - 27.3)) \cdot 138)$$

$$= 687325.94 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_4 - (W_i - C_{an} / 707)^2 \cdot fr_2) \cdot S_v$$

$$= (5761 + 100 - 0 \cdot 1) \cdot 138$$

$$= 808803.25 \text{ N}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$$

$$= (5761 + 0 + 100 + (2726)) \cdot 138$$



= 1184952.50 N

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r 1)) \cdot S \\ &= (5761 + 0 + 100 + 0 + (2726)) \cdot 138 \\ &= 1184952.50 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.14/2.0) \cdot 219 \cdot 10 \cdot 0.49 \cdot 138 \\ &= 232671. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.14 \cdot 102) \cdot (50 - 3) \cdot 0.7 \cdot 138 \\ &= 1453411. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\ &= (3.14/2.0) \cdot 219 \cdot (32 - 3) \cdot 0.74 \cdot 138 \\ &= 1019024. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (232671 + 1453411) = 1686082 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (232671 + 0 + 1019024 + 0) = 1251695 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Sonw + Tngw + Sinw) \\ &= (232671 + 1019024 + 0) = 1251695 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 1686082 N	, must exceed W = 687325 N	or W1 = 808803 N
Path 2-2 = 1251695 N	, must exceed W = 687325 N	or W2 = 1184952 N
Path 3-3 = 1251695 N	, must exceed W = 687325 N	or W3 = 1184952 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 9.97 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 16.0855 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 248.0855 mm

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Input, Nozzle Desc: N280

From: 150

Pressure for Reinforcement Calculations	P	9.4029	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	82	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	762.00	mm
Design Length of Section	L	2400.0002	mm
Shell Finished (Minimum) Thickness	t	32.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		12502.20	mm
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User Entered Minimum Design Metal Temperature		-28.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		80.0000	mm

Size and Thickness Basis		Minimum	
Nominal Thickness		160	

Flange Material		SA-105	
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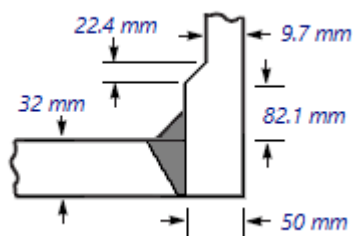
Hub Height of Integral Nozzle	h	82.1444	mm
Height of Beveled Transition	L'	22.4444	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	32.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.





Reinforcement CALCULATION, Description: N280

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 88.900 mm.
Actual Thickness Used in Calculation 9.735 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $9.4 \cdot 384 / (138 \cdot 1 - 0.6 \cdot 9.4)$
= 27.2804 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $9.4 \cdot 44.5 / (138 \cdot 1 + 0.4 \cdot 9.4)$
= 2.9483 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3989 mm

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / $\tan(30)$
= $31 / 0.5773$
= 53.6605 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

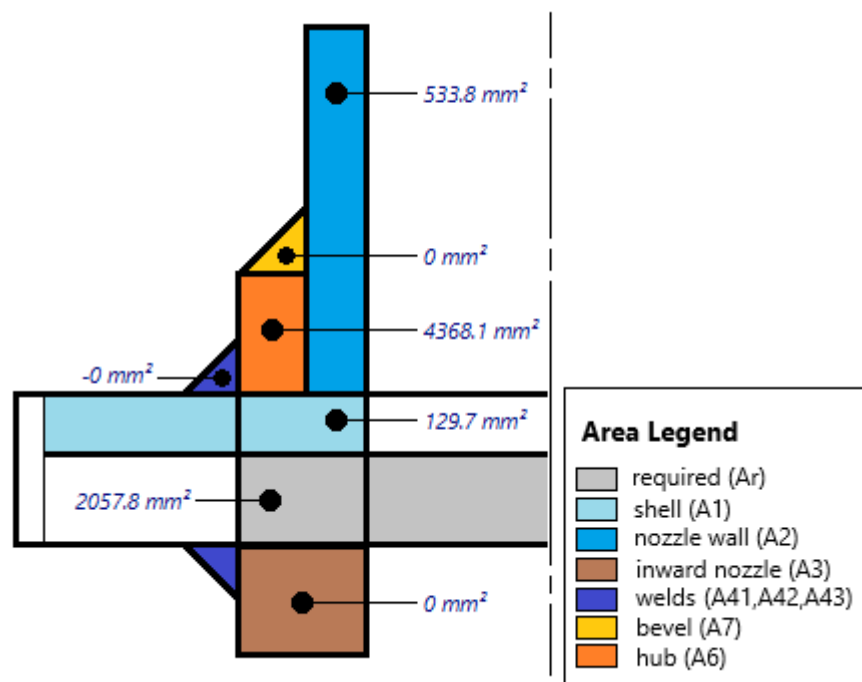
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	150.8618	mm
Parallel to Vessel Wall, opening length	d	75.4309	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	70.4969	mm

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 138 / 138)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 138 / 138)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1, 1)$
= 1.000



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	2057.788	156.320	NA
Area in Shell	A1	129.708	1874.855	NA
Area in Nozzle Wall	A2	533.833	893.283	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	-0.000	-0.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	4368.114	4368.114	NA
TOTAL AREA AVAILABLE	Atot	5031.655	7136.251	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (75.4 \cdot 27.3 \cdot 1 + 2 \cdot 37.7 \cdot 27.3 \cdot 1 \cdot (1 - 1))$
 $= 2057.788 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 75.4(1 \cdot 29 - 1 \cdot 27.3) - 2 \cdot 37.7$
 $(1 \cdot 29 - 1 \cdot 27.3) \cdot (1 - 1)$
 $= 129.708 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 70.5)(6.73 - 2.95)1$
 $= 533.833 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= (Wo^2 - \text{Area Lost}) \cdot fr2 + (Wi - can/0.707)^2 - \text{Area Lost} \cdot fr2$
 $= (10^2 - 100) \cdot 1 + (0^2 - 0) \cdot 1$
 $= -0.000 \text{ mm}^2$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lp}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2} \\ &= (2 \cdot \min(70.5, 200, 82.1)) \cdot (40.7 - 9.73) \cdot 1 \\ &= 4368.114 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.9483 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 30.2804 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 30.2804 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.2879 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[7.8, \max(30.3, 4.5)] \\ &= 7.8000 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(5.95, 7.8) \\ &= 7.8000 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 9.7346 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	58.8,	Allowable	:	138.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	286.2 MPa	Passed
Occasional	:	24.2,	Allowable	:	183.5 MPa	Passed
Shear	:	28.3,	Allowable	:	96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 9.73, tr = 2.95, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.44, Temp. Reduction = 42 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 32, tr = 27.3, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.94, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	6 °C
Min Metal Temp. at Required thickness (UCS 66.1)	2 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 2 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-39 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 9.40/15.31 = 0.614$$

Weld Size Calculations, Description: N280

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (2058 - 130 + 2 \cdot 37.7 \cdot 1 \cdot (1 \cdot 29 - 27.3)) \cdot 138)$$

$$= 283945.47 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (4902 + -0.28841\text{E-}04 - 0 \cdot 1) \cdot 138$$

$$= 676398.88 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (4902 + 0 + -0.28841\text{E-}04 + (2187)) \cdot 138$$

$$= 978242.25 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (4902 + 0 + -0.28841\text{E-}04 + 0 + (2187)) \cdot 138$$

$$= 978242.25 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 88.9 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 94417. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 41.1) \cdot (40.7 - 3) \cdot 0.7 \cdot 138$$

$$= 470185. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 88.9 \cdot (32 - 3) \cdot 0.74 \cdot 138$$

$$= 413517. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (94417 + 470185) = 564602 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (94417 + 0 + 413517 + 0) = 507934 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (94417 + 413517 + 0) = 507934 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 564602 N	, must exceed W = 283945 N	or W1 = 676398 N
Path 2-2 = 507934 N	, must exceed W = 283945 N	or W2 = 978242 N
Path 3-3 = 507934 N	, must exceed W = 283945 N	or W3 = 978242 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 9.97 MPa



Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 2.6018 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 234.6018 mm

Input Echo, WRC297 Item 1, Description: N280 :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	3.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	762.002	mm	
Shell Thickness	32.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	169.431	mm	
Nozzle Thickness	50.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-105		
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	9.40	MPa
Radial Load	P	-5997.68	N
Circumferential Shear	Vc	3998.47	N
Longitudinal Shear	Vl	3998.47	N
Circumferential Moment	Mc	896046.44	N-mm
Longitudinal Moment	Ml	896046.44	N-mm
Torsional Moment	Mt	0.128E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (3.389) is < 10
 check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))} \\
 &= 169.431 + 2 \cdot 1.65 \cdot \sqrt{(398.501(32.0 - 3.0))} \\
 &= 524.185 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	797.002 mm
Nozzle Outside Diameter	(d) =	169.431 mm
Vessel Thickness used	(T) =	29.000 mm
Nozzle Thickness used	(t) =	47.000 mm
T / t	=	0.617
d / t	=	10.000



$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.114$$

Nr/P	=	0.104
Mr/P	=	0.201
M0/P	=	0.061
N0/P	=	0.125
MrD/Mc	=	0.299
NrDL/Mc	=	0.105
M0d/Mc	=	0.098
N0DL/Mc	=	0.043
MrD/M1	=	0.183
NrDL/M1	=	0.090
M0D/M1	=	0.061
N0DL/M1	=	0.125

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-2	2	2	-2
Normal Pressure Stress	119	129	119	129
Outplane Stress Summary	120	129	126	126

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-6	6	6	-6
Inplane Pressure Stress	59	59	59	59
Inplane Stress Summary	61	58	76	45

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	1	1	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	120	129	126	126

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2



Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-3	3	3	-3
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	59	59	59	59

Outplane Stress Summary	59	61	67	54
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-11	11	11	-11
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	119	129	119	129

Inplane Stress Summary	117	132	141	110
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	1	1
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	117	132	141	110
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	16	5	16

Hoop Stress Summary	5	16	7	18
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	1	0	9	0
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	5	17	9	18

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	16	5	16
Hoop Stress Summary	6	17	6	18

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-3	3	3	-3
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	0	2	11	0

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	6	17	11	18

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl

Circ. Pm (SUS)	119	129	119	129	119	129	119	129
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	4	-4	-3	3	19	-19
Long. Pm (SUS)	59	59	59	59	59	59	59	59
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	2	-2	14	-14	-1	1	5	-5
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	119.0	129.0	119.0	129.0	119.0	129.0	119.0	129.0
Pm+Pl (SUS)	119.0	129.0	119.0	129.0	119.0	129.0	119.0	129.0
Pm+Pl+Q (Total)	119.0	129.0	123.0	125.0	116.0	132.0	138.0	110.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	129.00	138.00	Passed
Pm+Pl (SUS)	129.00	207.00	Passed
Pm+Pl+Q (TOTAL)	138.00	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	5	16	5	16	5	16	5	16
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	1	-1	3	-3	-1	1	5	-5
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	5.0	16.0	5.0	16.0	5.0	16.0	5.0	16.0
Pm+Pl (SUS)	5.0	16.0	5.0	16.0	5.0	16.0	5.0	16.0
Pm+Pl+Q (Total)	5.0	16.0	6.0	16.0	5.0	16.0	8.0	18.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	16.00	138.00	Passed
Pm+Pl (SUS)	16.00	207.00	Passed
Pm+Pl+Q (TOTAL)	18.00	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: K450

From: 150

Pressure for Reinforcement Calculations	P	9.4029	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	82	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	762.00	mm
Design Length of Section	L	2400.0002	mm
Shell Finished (Minimum) Thickness	t	32.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		13264.00	mm
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User Entered Minimum Design Metal Temperature		-28.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm

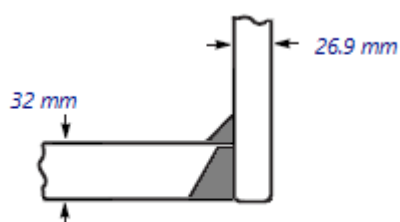
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	187.3250	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		32.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K450

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 50.800 mm.
Actual Thickness Used in Calculation 26.924 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $9.4 \cdot 384 / (138 \cdot 1 - 0.6 \cdot 9.4)$
= 27.2804 mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
= $28.4 \cdot (\exp([9.4 / (138 \cdot 1)] - 1))$
= 2.0025 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4278 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	162.6480 mm
Parallel to Vessel Wall	Rn+tn+t	81.3240 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	59.8100 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K450.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 5.0025 mm
Wall Thickness per UG16(b),	tr16b	= 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 30.2804 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 30.2804 mm
Wall Thickness, shell/head, external pressure	trb2	= 3.2879 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000 mm
Wall Thickness per table UG-45	tb3	= 8.2578 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[8.26, max(30.3, 4.5)]
= 8.2578 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(5, 8.26)
= 8.2578 mm

Available Nozzle Neck Thickness = 26.9240 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 26.9, tr = 2, c = 3 mm, E* = 1
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.084$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-39 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $9.40 / 15.31 = 0.614$



Weld Size Calculations, Description: K450

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.6100 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.9350 mm

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Input, Nozzle Desc: N3150

From: 180

Pressure for Reinforcement Calculations	P	9.4029	MPa
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	82	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	762.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	29.5000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
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User Entered Minimum Design Metal Temperature		-28.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		150.0000	mm

Size and Thickness Basis		Minimum	
Nominal Thickness		80	

Flange Material		SA-105	
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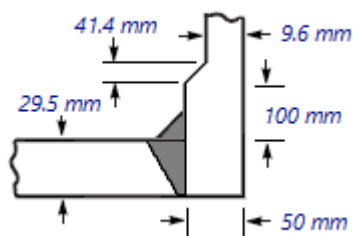
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	41.4000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	29.4000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		900	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.





Reinforcement CALCULATION, Description: N3150

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	168.275	mm.
Actual Thickness Used in Calculation	9.601	mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $9.4 \cdot 768 \cdot 0.89 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 9.4)$
= 23.5347 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot Ro / (Sn \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $9.4 \cdot 84.1 / (138 \cdot 1 + 0.4 \cdot 9.4)$
= 5.5808 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.5805 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= 40.4/0.5773
= 69.9728 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

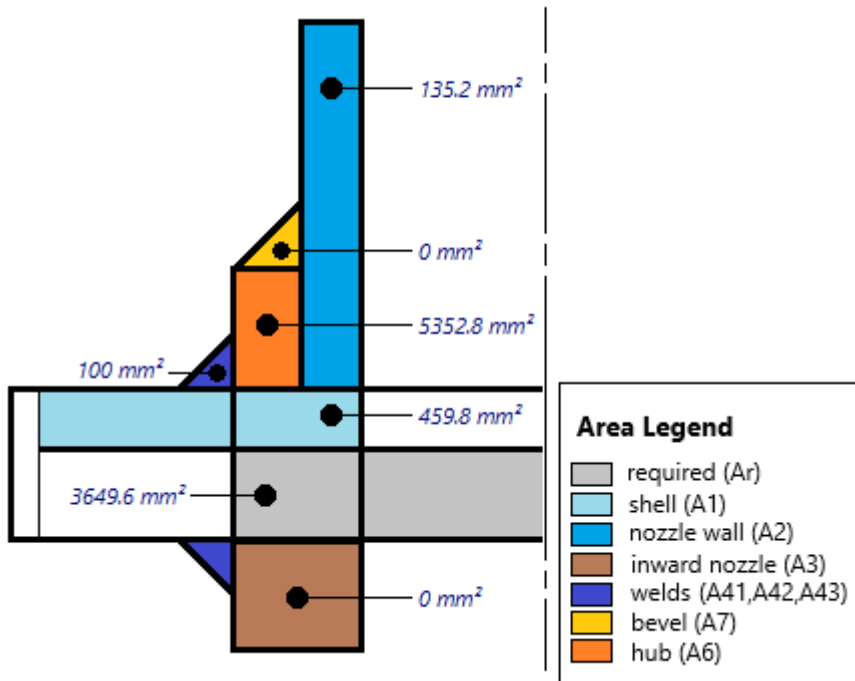
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	310.1452	mm
Parallel to Vessel Wall, opening length	d	155.0726	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	66.2500	mm

Weld Strength Reduction Factor [fr1]:
= min(1, Sn/Sv)
= min(1, 138/138)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, Sn/Sv)
= min(1, 138/138)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1, 1)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	3649.593	164.822	NA
Area in Shell	A1	459.831	3779.779	NA
Area in Nozzle Wall	A2	135.207	797.746	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	5352.841	5352.841	NA
TOTAL AREA AVAILABLE	Atot	6047.879	10030.365	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (155 \cdot 23.5 \cdot 1 + 2 \cdot 47 \cdot 23.5 \cdot 1 \cdot (1 - 1))$
 $= 3649.593 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 155(1 \cdot 26.5 - 1 \cdot 23.5) - 2 \cdot 47(1 \cdot 26.5 - 1 \cdot 23.5) \cdot (1 - 1)$
 $= 459.831 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 66.2)(6.6 - 5.58)1$
 $= 135.207 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 100.000 \text{ mm}^2$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lp}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2} \\ &= (2 \cdot \min(66.2, 200, 100)) \cdot (50 - 9.6) \cdot 1 \\ &= 5352.841 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 8.5808 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 29.0734 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 29.0734 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.2849 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 9.2200 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[9.22, \max(29.1, 4.5)] \\ &= 9.2200 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(8.58, 9.22) \\ &= 9.2200 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 9.6012 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 83.8,	Allowable	: 138.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 261.2 MPa	Passed
Occasional	: 53.0,	Allowable	: 183.5 MPa	Passed
Shear	: 24.6,	Allowable	: 96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 9.6, tr = 5.58, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.85, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-16 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 29.5, tr = 23.5, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.89, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-3 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -3 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-39 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 9.40/15.31 = 0.614$$

Weld Size Calculations, Description: N3150

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (3650 - 460 + 2 \cdot 47 \cdot 1 \cdot (1 \cdot 26.5 - 23.5)) \cdot 138)$
 $= 478603.12 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (5488 + 100 - 0 \cdot 1) \cdot 138$
 $= 771071.12 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (5488 + 0 + 100 + (2491)) \cdot 138$
 $= 1114793.75 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (5488 + 0 + 100 + 0 + (2491)) \cdot 138$
 $= 1114793.75 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 168 \cdot 10 \cdot 0.49 \cdot 138$
 $= 178719. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 80.8) \cdot (50 - 3) \cdot 0.7 \cdot 138$
 $= 1152916. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 168 \cdot (29.5 - 3) \cdot 0.74 \cdot 138$
 $= 715252. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (178719 + 1152915) = 1331634 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (178719 + 0 + 715252 + 0) = 893971 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (178719 + 715252 + 0) = 893971 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 1331634 N	, must exceed W = 478603 N	or W1 = 771071 N
Path 2-2 = 893970 N	, must exceed W = 478603 N	or W2 = 1114793 N
Path 3-3 = 893970 N	, must exceed W = 478603 N	or W3 = 1114793 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 9.56 MPa



Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0.1 MPa

The Drop for this Nozzle is : 5.0072 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 234.5072 mm

Input Echo, WRC107/537 Item 1, Description: N3150 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	1371.604	mm
Vessel Thickness	Tv	29.500	mm
Design Temperature	T1	82.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	249.073	mm
Nozzle Thickness	Tn	50.000	mm
Nozzle Material		SA-105	
Nozzle UNS Number		K03504	
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	138.00	MPa
Design Internal Pressure	Dp	9.403	MPa
Include Pressure Thrust		No	

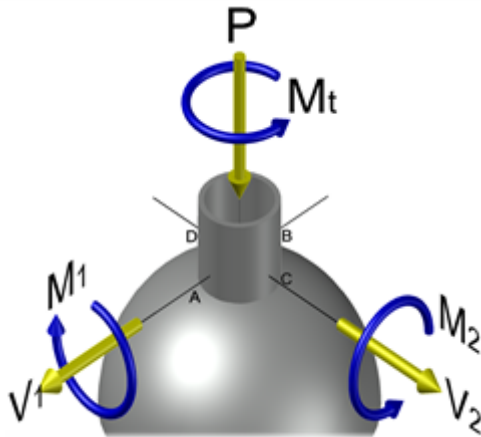
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-11245.7	N
Longitudinal Shear (SUS)	(V1) V1	7497.1	N
Circumferential Shear (SUS)	(Vc) V2	7497.1	N
Circumferential Moment (SUS)	(Mc) M1	3150156.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	3150156.0	N-mm
Torsional Moment (SUS)	Mt	4500244.5	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 249.073 + 2 \cdot 1.65 \cdot \sqrt{(702.052 (29.5 - 3.0))}$$

$$= 699.185 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load		P	-11245.7	N
Circumferential Shear	(VC)	V2	7497.1	N
Longitudinal Shear	(VL)	V1	7497.1	N
Circumferential Moment	(MC)	M1	3150156.0	N-mm
Longitudinal Moment	(ML)	M2	3150156.0	N-mm
Torsional Moment		MT	4500244.5	N-mm

Dimensionless Param: U = 0.91 TAU = 5.00 (2.15) RHO = 0.56

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.04456	(A,B,C,D)
$M(x) / P$	SP 1	0.04720	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.05655	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.09040	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.05655	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.09040	(A,B,C,D)
$N(y) * T / P$	SP 1	0.07488	(A,B,C,D)
$M(y) / P$	SP 1	0.01436	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.07449	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.03124	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.07449	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.03124	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7
Rad. Bend.	P	4.5	-4.5	4.5	-4.5	4.5	-4.5	4.5	-4.5
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-1.9	-1.9	1.9	1.9
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-17.8	17.8	17.8	-17.8
Rad. Memb.	ML	-1.9	-1.9	1.9	1.9	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-17.8	17.8	17.8	-17.8	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-14.4	12.2	24.9	-19.8	-14.4	12.2	24.9	-19.8



Tang. Memb. P	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2
Tang. Bend. P	1.4	-1.4	1.4	-1.4	1.4	-1.4	1.4	-1.4
Tang. Memb. MC	0.0	0.0	0.0	0.0	-2.4	-2.4	2.4	2.4
Tang. Bend. MC	0.0	0.0	0.0	0.0	-6.2	6.2	6.2	-6.2
Tang. Memb. ML	-2.4	-2.4	2.4	2.4	0.0	0.0	0.0	0.0
Tang. Bend. ML	-6.2	6.2	6.2	-6.2	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-6.0	3.5	11.2	-3.9	-6.0	3.5	11.2	-3.9
Shear VC	0.7	0.7	-0.7	-0.7	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.7	-0.7	0.7	0.7
Shear MT	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7
Tot. Shear	2.5	2.5	1.0	1.0	1.0	1.0	2.5	2.5
Str. Int.	15.1	12.8	25.0	19.9	14.6	12.3	25.4	20.2

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	122.3	122.3	122.3	122.3	122.3	122.3	122.3	122.3
Rad.	Pl (SUS)	-1.1	-1.1	2.6	2.6	-1.1	-1.1	2.6	2.6
Rad.	Q (SUS)	-13.3	13.3	22.4	-22.4	-13.3	13.3	22.4	-22.4
Long.	Pm (SUS)	122.3	122.3	122.3	122.3	122.3	122.3	122.3	122.3
Long.	Pl (SUS)	-1.2	-1.2	3.6	3.6	-1.2	-1.2	3.6	3.6
Long.	Q (SUS)	-4.8	4.8	7.5	-7.5	-4.8	4.8	7.5	-7.5
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.7	0.7	-0.7	-0.7	-0.7	-0.7	0.7	0.7
Shear	Q (SUS)	1.7	1.7	1.7	1.7	1.7	1.7	1.7	1.7
Pm (SUS)		122.3	122.3	122.3	122.3	122.3	122.3	122.3	122.3
Pm+Pl (SUS)		121.8	121.8	126.3	126.3	121.8	121.8	126.3	126.3
Pm+Pl+Q (Total)		117.0	135.1	147.3	118.5	116.4	134.6	147.7	118.8

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	122.32	138.00	Passed
Pm+Pl (SUS)	126.33	207.00	Passed
Pm+Pl+Q (TOTAL)	147.69	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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